Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()



    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```python
    nan_ratio = df[cols_to_check].isna().mean()
    valid_features = nan_ratio[nan_ratio <= 0.15].index
    valid_features_list = list(valid_features)

    #df = df[valid_features]
    # Removing of last np.nan
    mask = np.isfinite(df[valid_features]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(valid_features_list + #target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    new_feats_X = [x for x in features_X if x in valid_features_list]
    new_feats_add = [x for x in feats_eng if x in valid_features_list]

    data["df"]=df
    data["features_X"] = new_feats_X
    data["feats_added"] = new_feats_add

    column_removed = [x for x in cols_to_check if x not in valid_features_list]
    print(f'Column removed because np.isnan > 15%: {column_removed}')

    return data
```

```python
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
```

```
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r"
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
```

```
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	8028
Nombre de lignes df clean	8028
Nombre de features	77
% de NaN	0.0
Taille dataset entraînement	5619
Taille dataset test	2408
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_mean, r_NH3_st_2h_std, r_NH3_st_2h_rms, r_NH3_st_2h_median, r_NH3_st_2h_min, r_NH3_st_2h_max, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_NH3_st_3h_mean, r_NH3_st_3h_std, r_NH3_st_3h_rms, r_NH3_st_3h_median, r_NH3_st_3h_min, r_NH3_st_3h_max, r_NH3_st_3h_diff_last, r_NH3_st_3h_slope, r_NH3_st_4h_mean, r_NH3_st_4h_std, r_NH3_st_4h_rms, r_NH3_st_4h_median, r_NH3_st_4h_min, r_NH3_st_4h_max, r_NH3_st_4h_diff_last, r_NH3_st_4h_slope, r_temperature_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_rms, r_temperature_st_2h_median, r_temperature_st_2h_min, r_temperature_st_2h_max, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_temperature_st_3h_mean, r_temperature_st_3h_std, r_temperature_st_3h_rms, r_temperature_st_3h_median, r_temperature_st_3h_min, r_temperature_st_3h_max, r_temperature_st_3h_diff_last, r_temperature_st_3h_slope, r_temperature_st_4h_mean, r_temperature_st_4h_std, r_temperature_st_4h_rms, r_temperature_st_4h_median, r_temperature_st_4h_min, r_temperature_st_4h_max, r_temperature_st_4h_diff_last, r_temperature_st_4h_slope, r_umidity_st_2h_mean, r_umidity_st_2h_std, r_umidity_st_2h_rms, r_umidity_st_2h_median, r_umidity_st_2h_min, r_umidity_st_2h_max, r_umidity_st_2h_diff_last, r_umidity_st_2h_slope, r_umidity_st_3h_mean, r_umidity_st_3h_std, r_umidity_st_3h_rms, r_umidity_st_3h_median, r_umidity_st_3h_min, r_umidity_st_3h_max, r_umidity_st_3h_diff_last, r_umidity_st_3h_slope, r_umidity_st_4h_mean, r_umidity_st_4h_std, r_umidity_st_4h_rms, r_umidity_st_4h_median, r_umidity_st_4h_min, r_umidity_st_4h_max, r_umidity_st_4h_diff_last, r_umidity_st_4h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

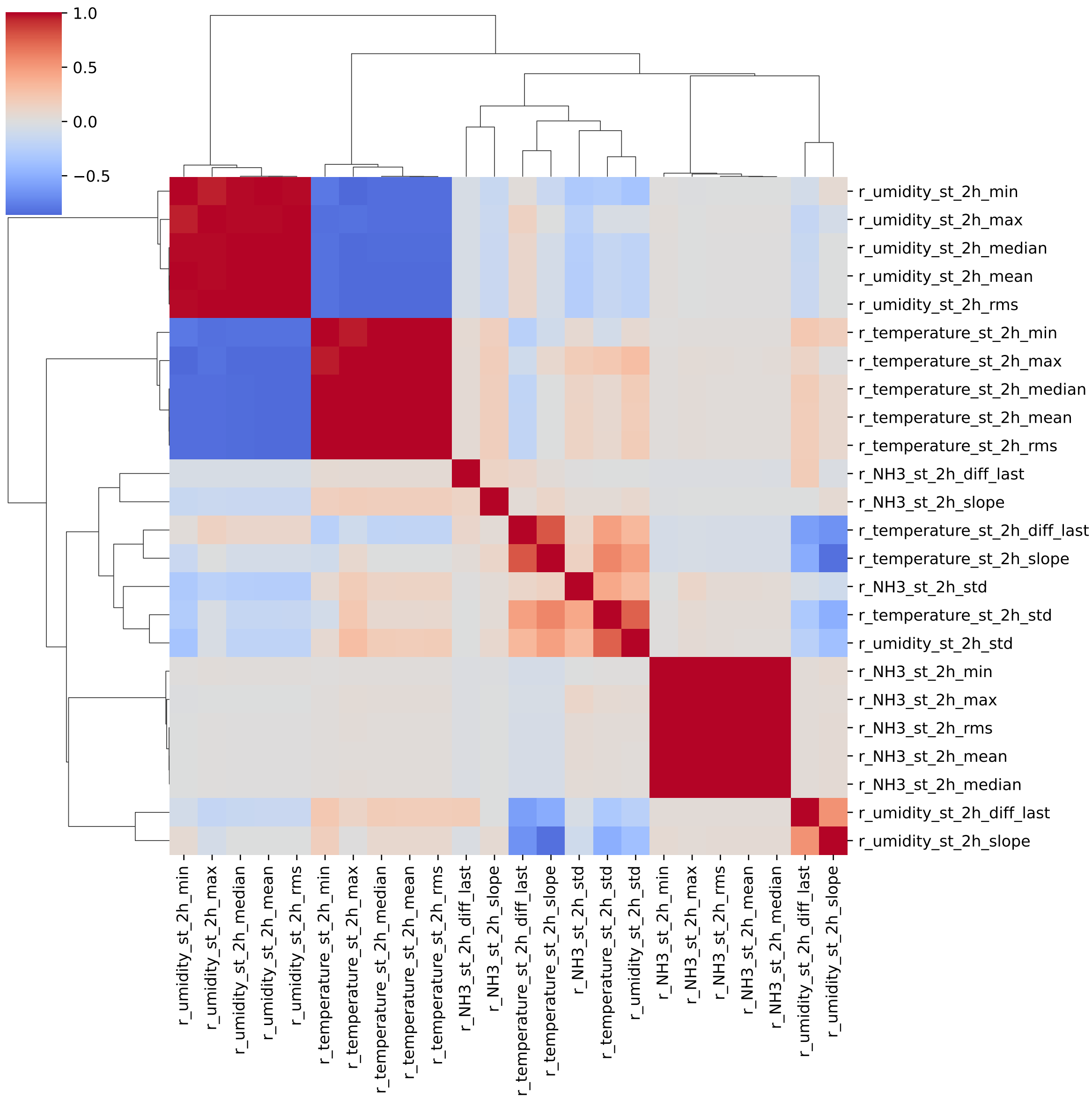
The following section has been done with a threshold choosen of : 0.8

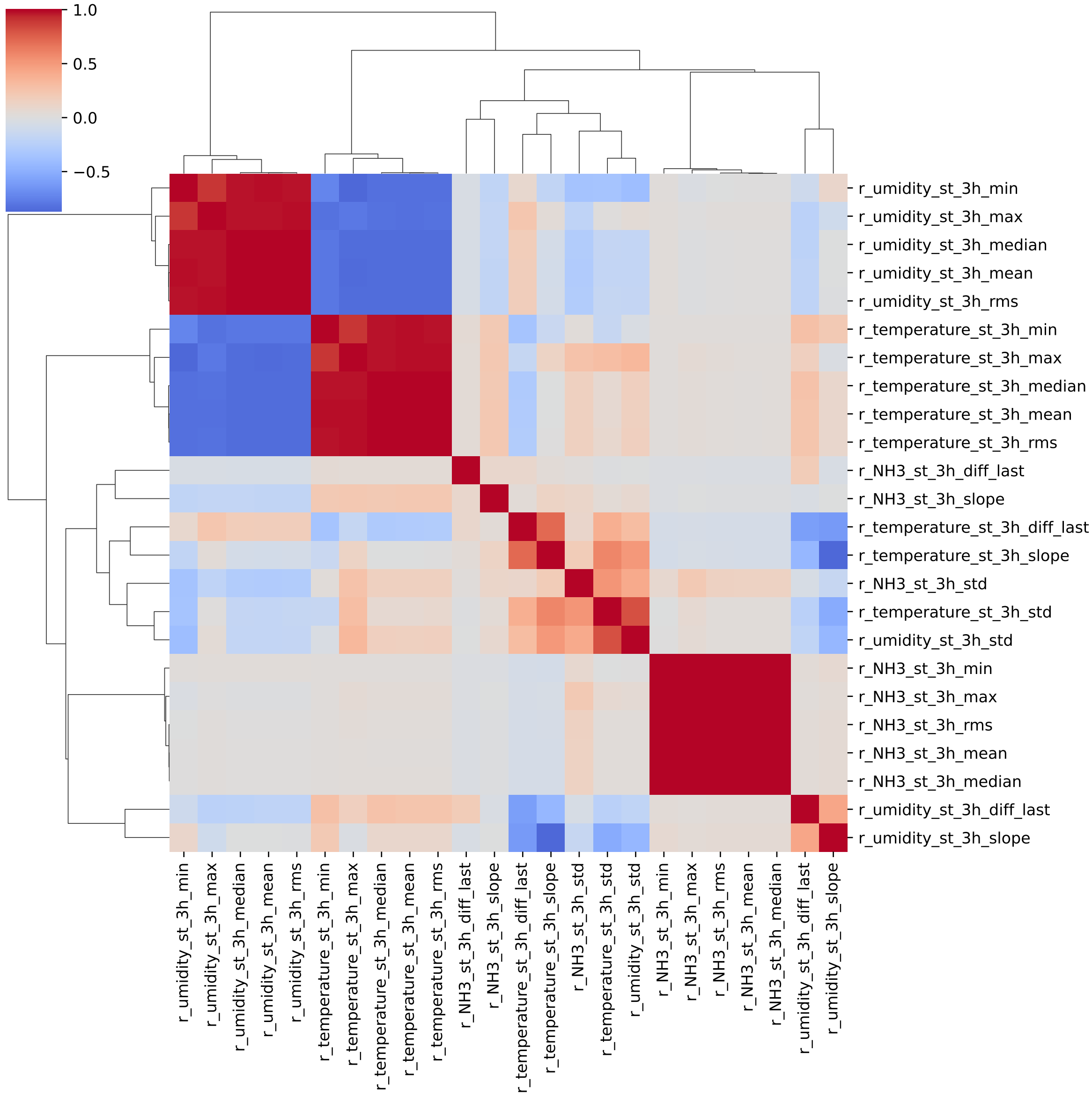
cluster 1: r, *, N, H, 3 ;* ***cluster 2:*** *r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ;* ***cluster 4:*** *r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_st_2h_max, r_NH3_st_2h_min, r_NH3_st_2h_mean, r_NH3_st_2h_rms, r_NH3_st_2h_median ; **cluster 7:** r_NH3_st_2h_std ; **cluster 8:** r_NH3_st_2h_diff_last ; **cluster 9:** r_NH3_st_2h_slope ; **cluster 10:** r_umidity_st_2h_mean, r_temperature_st_2h_rms, r_umidity_st_2h_min, r_umidity_st_2h_median, r_temperature_st_2h_median, r_temperature_st_2h_max, r_temperature_st_2h_mean, r_umidity_st_2h_rms, r_temperature_st_2h_min, r_umidity_st_2h_max ; **cluster 11:** r_temperature_st_2h_std ; **cluster 12:** r_temperature_st_2h_diff_last ; **cluster 13:** r_temperature_st_2h_slope, r_umidity_st_2h_slope ; **cluster 14:** r_umidity_st_2h_std ; **cluster 15:** r_umidity_st_2h_diff_last ; **cluster 16:** r_NH3_st_3h_max, r_NH3_st_3h_min, r_NH3_st_3h_mean, r_NH3_st_3h_median, r_NH3_st_3h_rms ; **cluster 17:** r_NH3_st_3h_std ; **cluster 18:** r_NH3_st_3h_diff_last ; **cluster 19:** r_NH3_st_3h_slope ; **cluster 20:** r_temperature_st_3h_min, r_umidity_st_3h_rms, r_temperature_st_3h_max, r_umidity_st_3h_min, r_temperature_st_3h_median, r_temperature_st_3h_rms, r_umidity_st_3h_mean, r_umidity_st_3h_median, r_temperature_st_3h_mean, r_umidity_st_3h_max ; **cluster 21:** r_temperature_st_3h_std, r_umidity_st_3h_std ; **cluster 22:** r_temperature_st_3h_diff_last ; **cluster 23:** r_umidity_st_3h_slope, r_temperature_st_3h_slope ; **cluster 24:** r_umidity_st_3h_diff_last ; **cluster 25:** r_NH3_st_4h_max, r_NH3_st_4h_rms, r_NH3_st_4h_median, r_NH3_st_4h_min, r_NH3_st_4h_mean ; **cluster 26:** r_NH3_st_4h_std ; **cluster 27:** r_NH3_st_4h_diff_last ; **cluster 28:** r_NH3_st_4h_slope ; **cluster 29:** r_umidity_st_4h_min, r_umidity_st_4h_median, r_temperature_st_4h_max, r_temperature_st_4h_min, r_temperature_st_4h_mean, r_temperature_st_4h_median, r_umidity_st_4h_rms, r_umidity_st_4h_mean, r_umidity_st_4h_max, r_temperature_st_4h_rms ; **cluster 30:** r_umidity_st_4h_std, r_temperature_st_4h_std ; **cluster 31:** r_temperature_st_4h_diff_last ; **cluster 32:** r_temperature_st_4h_slope, r_umidity_st_4h_slope ; **cluster 33:** r_umidity_st_4h_diff_last ;

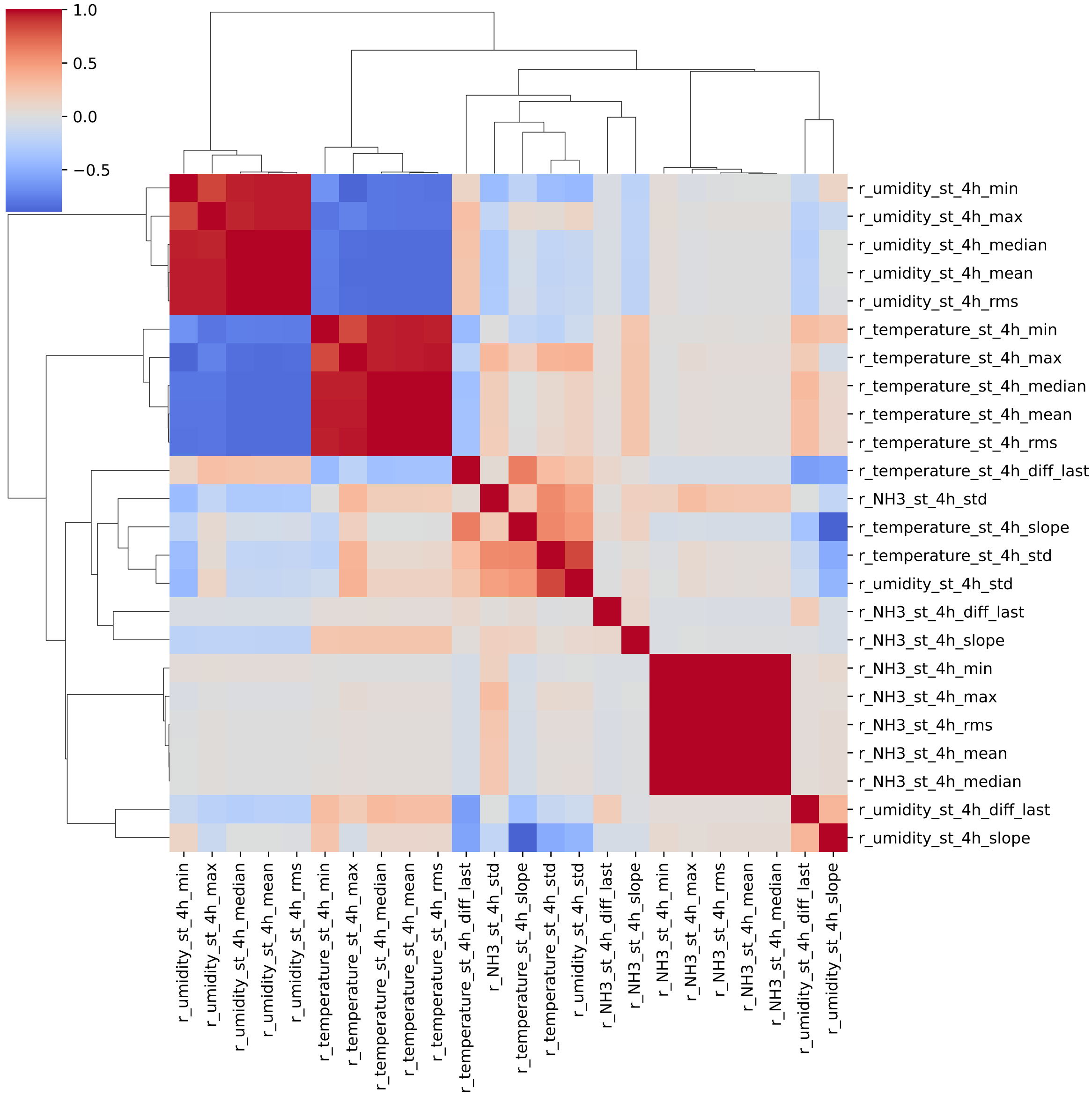
2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_max, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_umidity_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last, r_NH3_st_3h_max, r_NH3_st_3h_std, r_NH3_st_3h_diff_last, r_NH3_st_3h_slope, r_temperature_st_3h_min, r_temperature_st_3h_std, r_temperature_st_3h_diff_last, r_umidity_st_3h_slope, r_umidity_st_3h_diff_last, r_NH3_st_4h_max, r_NH3_st_4h_std, r_NH3_st_4h_diff_last, r_NH3_st_4h_slope, r_umidity_st_4h_min, r_umidity_st_4h_std, r_temperature_st_4h_diff_last, r_temperature_st_4h_slope, r_umidity_st_4h_diff_last


1.0
0.5
0.0
−0.5
r_umidity_st_2h_min
r_umidity_st_2h_max
r_umidity_st_2h_median
r_umidity_st_2h_mean
r_umidity_st_2h_rms
r_temperature_st_2h_min
r_temperature_st_2h_max
r_temperature_st_2h_median
r_temperature_st_2h_mean
r_temperature_st_2h_rms
r_NH3_st_2h_diff_last
r_NH3_st_2h_slope
r_temperature_st_2h_diff_last
r_temperature_st_2h_slope
r_NH3_st_2h_std
r_temperature_st_2h_std
r_umidity_st_2h_std
r_NH3_st_2h_min
r_NH3_st_2h_max
r_NH3_st_2h_rms
r_NH3_st_2h_mean
r_NH3_st_2h_median
r_umidity_st_2h_diff_last
r_umidity_st_2h_slope


1.0
0.5
0.0
−0.5
r_umidity_st_3h_min
r_umidity_st_3h_max
r_umidity_st_3h_median
r_umidity_st_3h_mean
r_umidity_st_3h_rms
r_temperature_st_3h_min
r_temperature_st_3h_max
r_temperature_st_3h_median
r_temperature_st_3h_mean
r_temperature_st_3h_rms
r_NH3_st_3h_diff_last
r_NH3_st_3h_slope
r_temperature_st_3h_diff_last
r_temperature_st_3h_slope
r_NH3_st_3h_std
r_temperature_st_3h_std
r_umidity_st_3h_std
r_NH3_st_3h_min
r_NH3_st_3h_max
r_NH3_st_3h_rms
r_NH3_st_3h_mean
r_NH3_st_3h_median
r_umidity_st_3h_diff_last
r_umidity_st_3h_slope



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_max, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_umidity_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last, r_NH3_st_3h_max, r_NH3_st_3h_std, r_NH3_st_3h_diff_last, r_NH3_st_3h_slope, r_temperature_st_3h_min, r_temperature_st_3h_std, r_temperature_st_3h_diff_last, r_umidity_st_3h_slope, r_umidity_st_3h_diff_last, r_NH3_st_4h_max, r_NH3_st_4h_std, r_NH3_st_4h_diff_last, r_NH3_st_4h_slope, r_umidity_st_4h_min, r_umidity_st_4h_std, r_temperature_st_4h_diff_last, r_temperature_st_4h_slope, r_umidity_st_4h_diff_last

Len(Features_X): 33

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

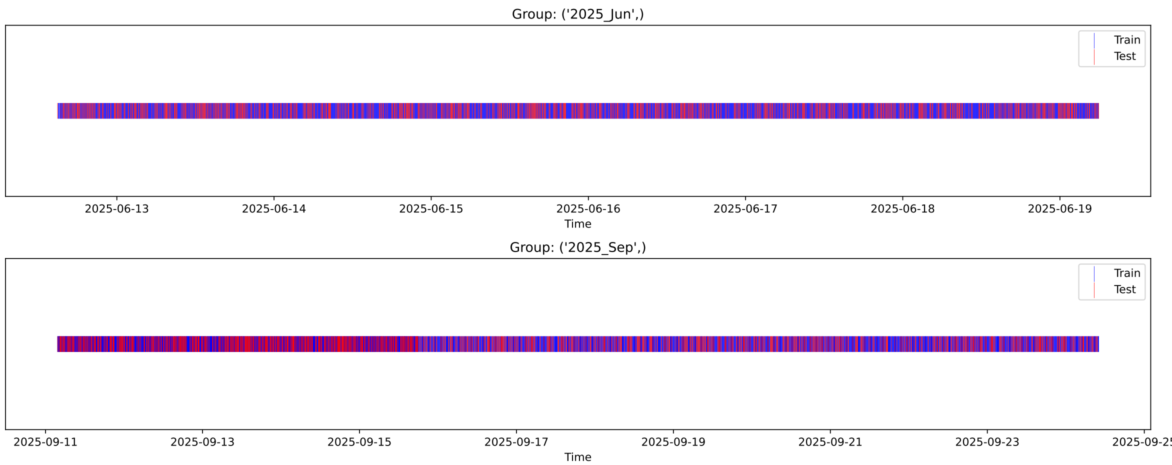
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

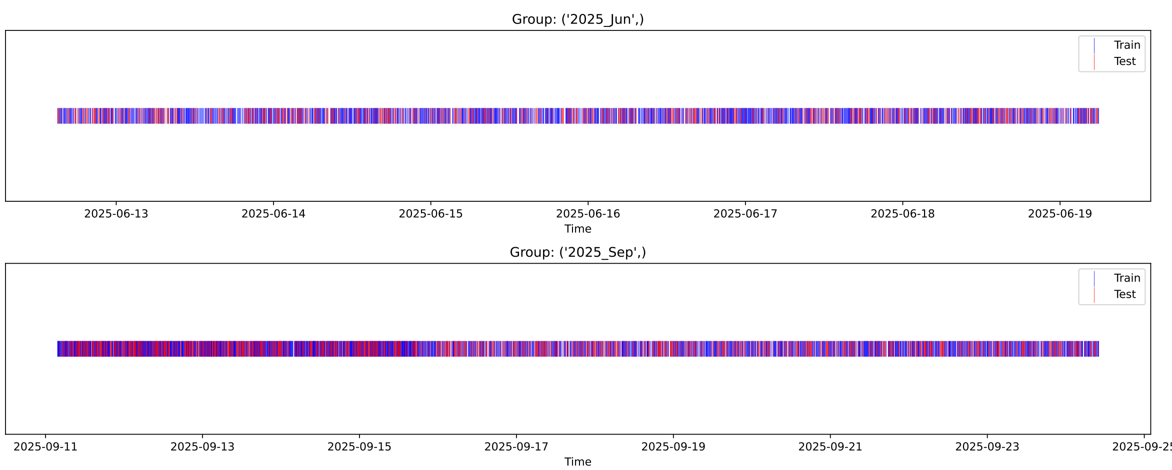
External Split: Train: 5619 samples Test: 2409 samples



3.3.2 Internal CV Folds

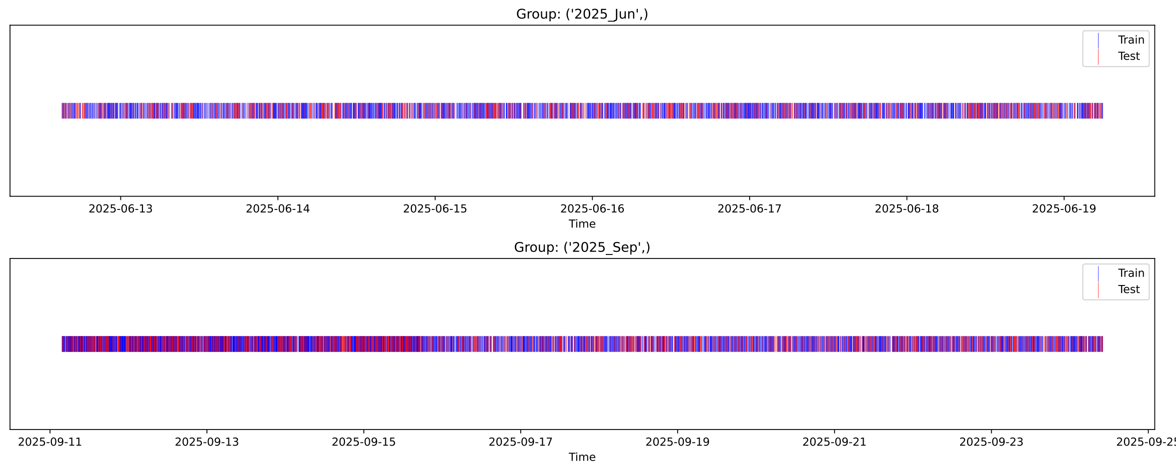
Fold 1/5

Train: 3933 samples Test: 1686 samples Excluded: 0 samples Window exclusion: 0h



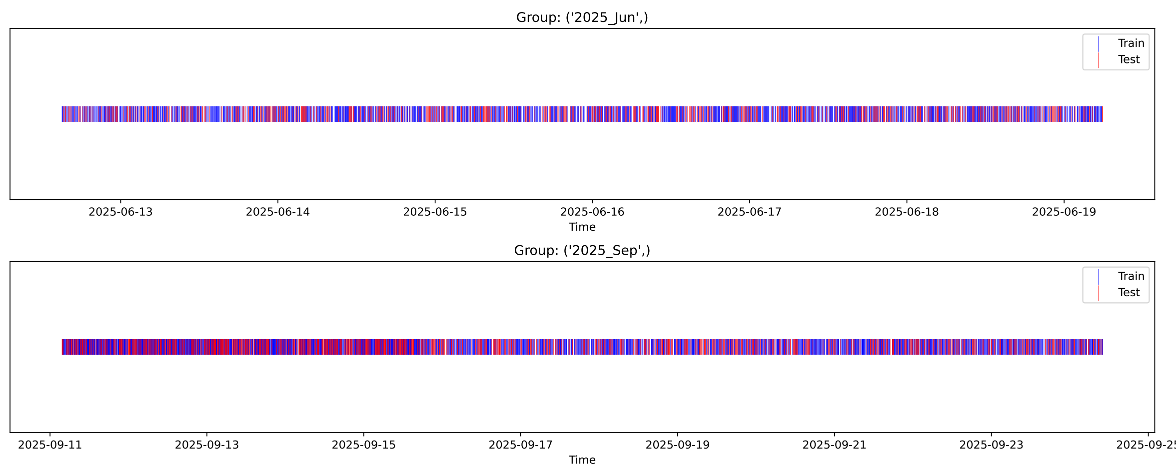
Fold 2/5

Train: 3933 samples Test: 1686 samples Excluded: 0 samples Window exclusion: 0h



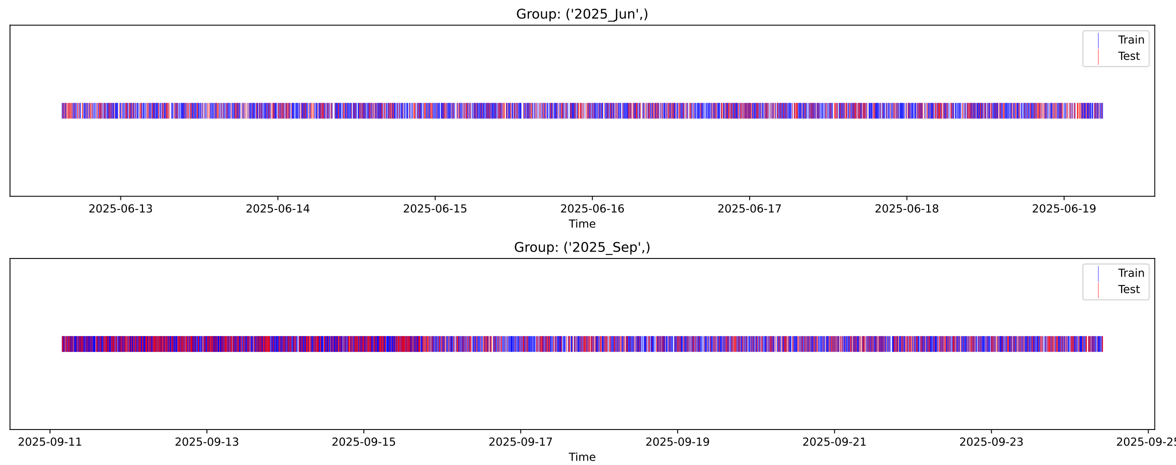
Fold 3/5

Train: 3933 samples Test: 1686 samples Excluded: 0 samples Window exclusion: 0h



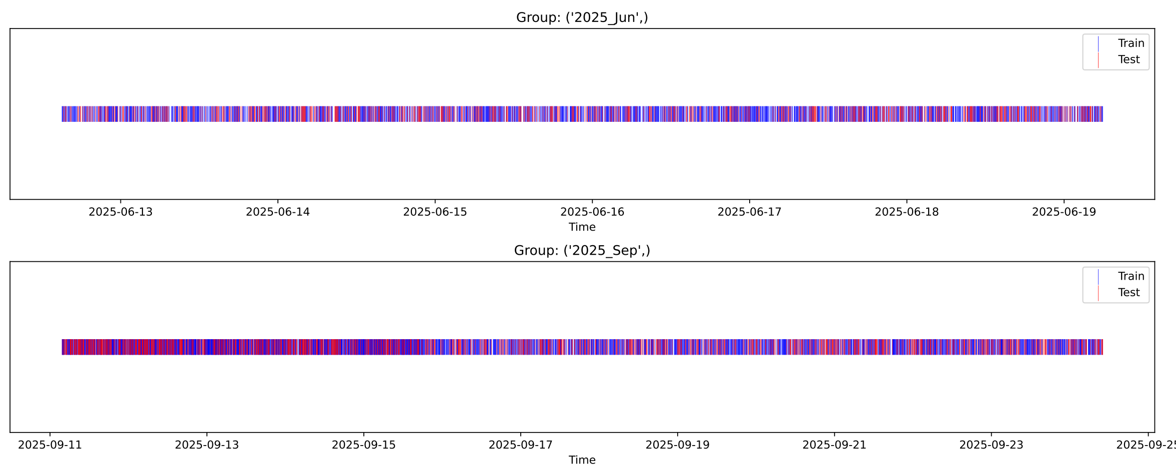
Fold 4/5

Train: 3933 samples Test: 1686 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3933 samples Test: 1686 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.350, std=0.806, min=0.959, max=13.845 y_test: mean=2.355, std=0.778, min=1.011, max=7.822

Fold 0 — y_test: mean=2.34, std=0.79, y_train: mean=2.36, std=0.81

Fold 1 — y_test: mean=2.39, std=0.84, y_train: mean=2.33, std=0.79

Fold 2 — y_test: mean=2.35, std=0.79, y_train: mean=2.35, std=0.81

Fold 3 — y_test: mean=2.35, std=0.86, y_train: mean=2.35, std=0.78

Fold 4 — y_test: mean=2.35, std=0.81, y_train: mean=2.35, std=0.81

train scores CV: [0.56709432 0.57493514 0.55710113 0.58898758 0.55796594]

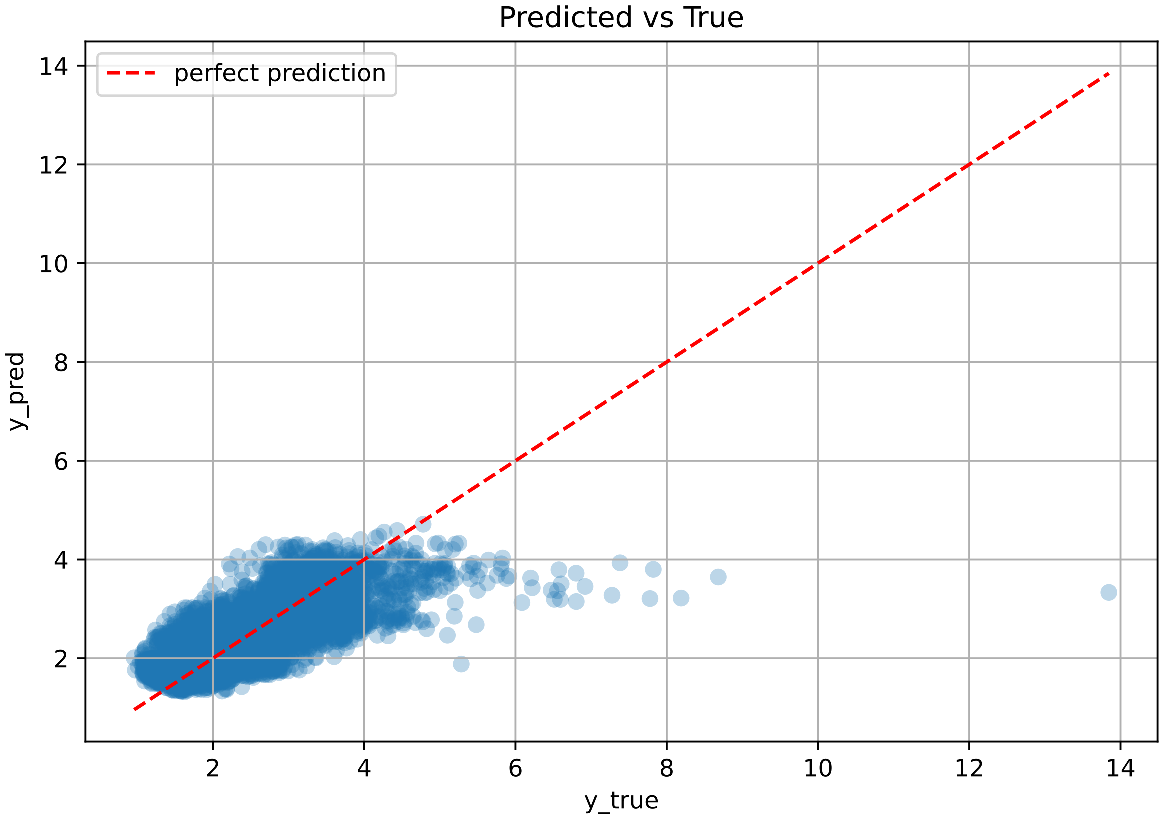
test scores CV: [0.5453468 0.52828213 0.56960852 0.50616786 0.56727233]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5692	0.5433	0.024	0.5774	0.5777	0.0344	1.0633	1.0476

Coefficients:

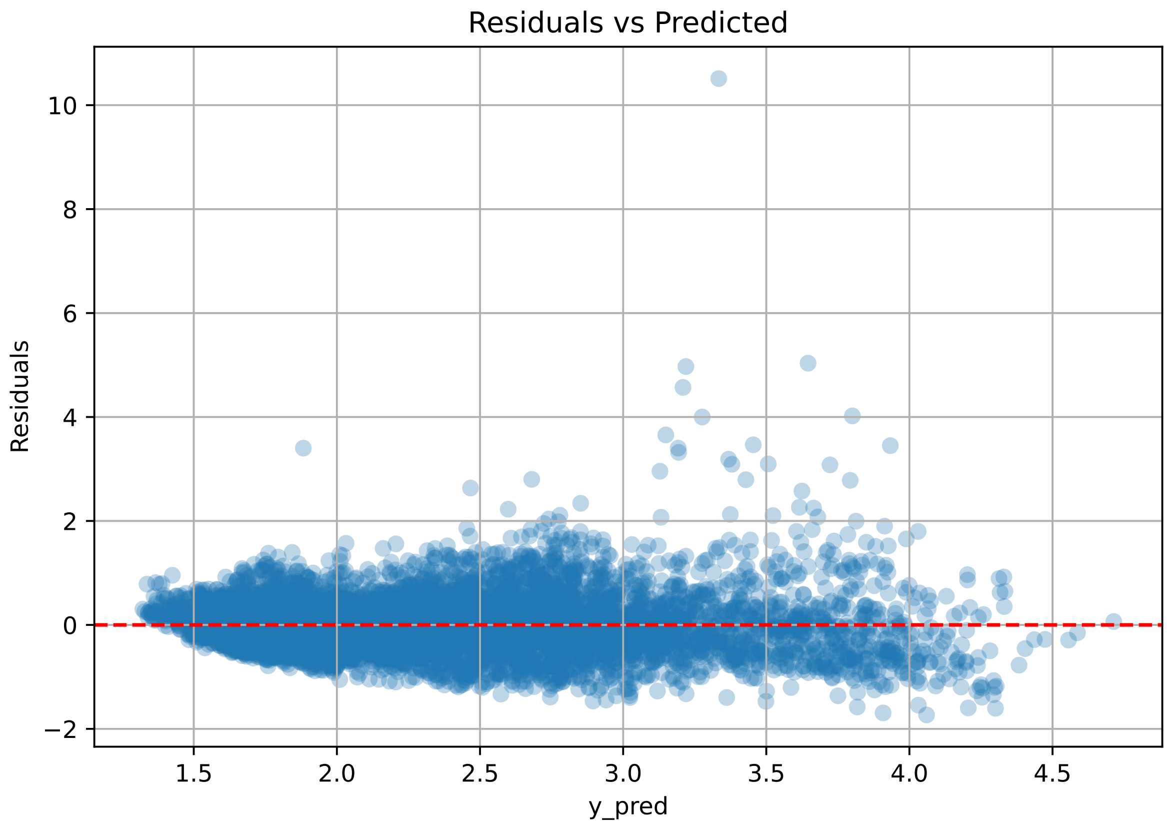
coef_r_CO2 = -0.0386, coef_r_NH3 = -0.0138, coef_r_NH3_st_2h_diff_last = 0.0051, coef_r_NH3_st_2h_max = -0.3927, coef_r_NH3_st_2h_slope = 0.0061, coef_r_NH3_st_2h_std = 0.0688, coef_r_NH3_st_3h_diff_last = 0.0051, coef_r_NH3_st_3h_max = -0.0898, coef_r_NH3_st_3h_slope = -0.0228, coef_r_NH3_st_3h_std = 0.0470, coef_r_NH3_st_4h_diff_last = 0.0051, coef_r_NH3_st_4h_max = 0.4517, coef_r_NH3_st_4h_slope = 0.0641, coef_r_NH3_st_4h_std = -0.0510, coef_r_THI = 0.4765, coef_r_temperature = 0.7827, coef_r_temperature_st_2h_diff_last = -0.0104, coef_r_temperature_st_2h_slope = -0.1929, coef_r_temperature_st_2h_std = -0.0217, coef_r_temperature_st_3h_diff_last = -0.0104, coef_r_temperature_st_3h_min = -1.2071, coef_r_temperature_st_3h_std = -0.0988, coef_r_temperature_st_4h_diff_last = -0.0104, coef_r_temperature_st_4h_slope = -0.1602, coef_r_umidity = -0.7464, coef_r_umidity_st_2h_diff_last = -0.0028, coef_r_umidity_st_2h_mean = 0.4671, coef_r_umidity_st_2h_std = 0.0647, coef_r_umidity_st_3h_diff_last = -0.0028, coef_r_umidity_st_3h_slope = -0.0314, coef_r_umidity_st_4h_diff_last = -0.0028, coef_r_umidity_st_4h_min = -0.1370, coef_r_umidity_st_4h_std = -0.1687, intercept = 2.3503,

3.4.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

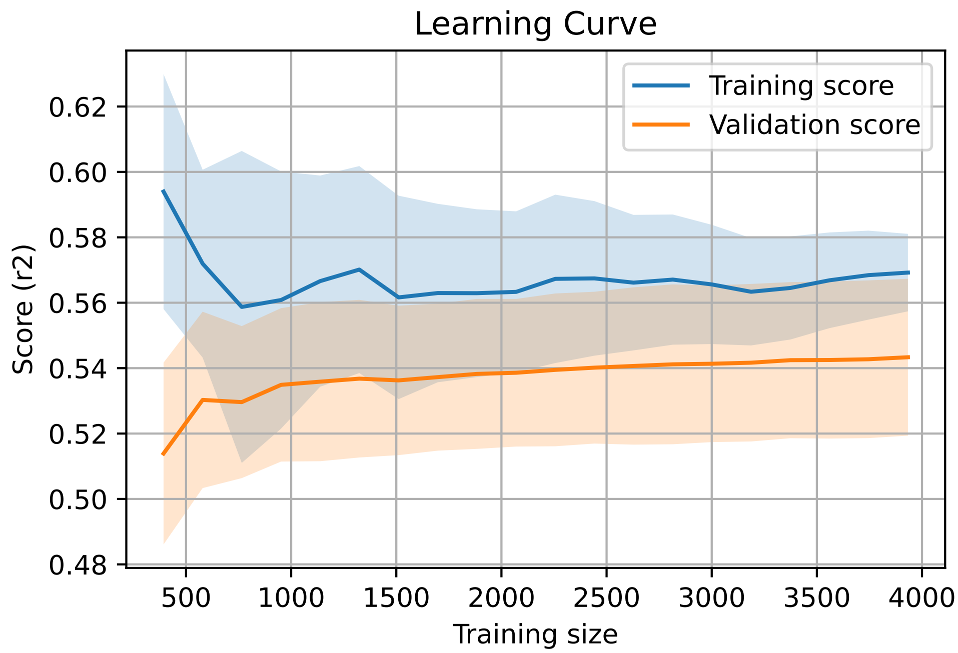
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 8028, **Len(training)** : 5620,

Len(training_real) : 3934, **Len(test_of_training)** : 1686, **Len(test)** : 2408

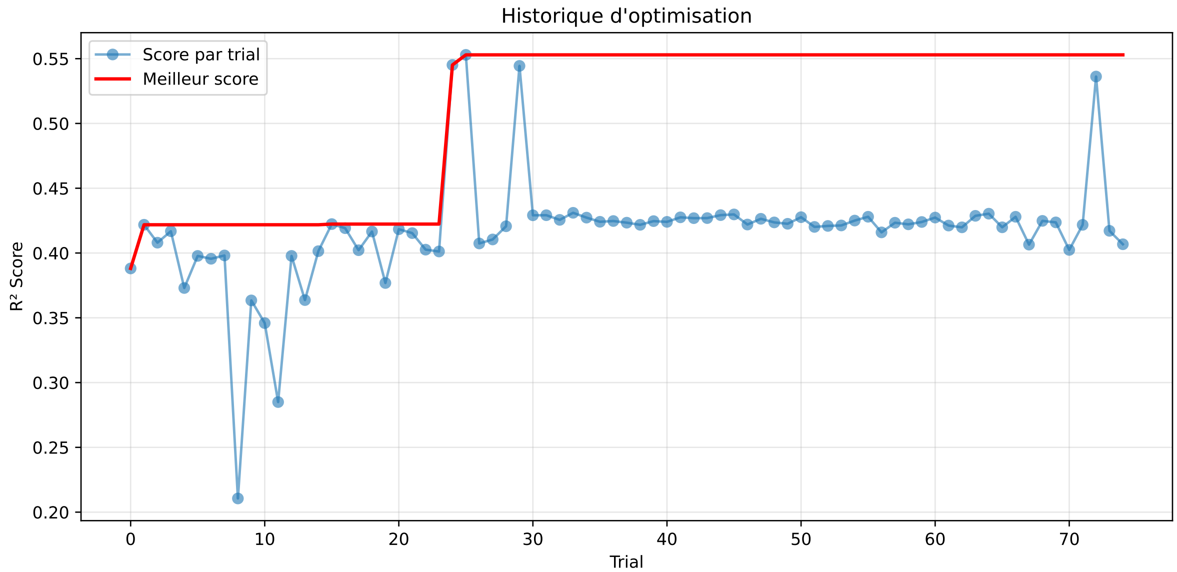


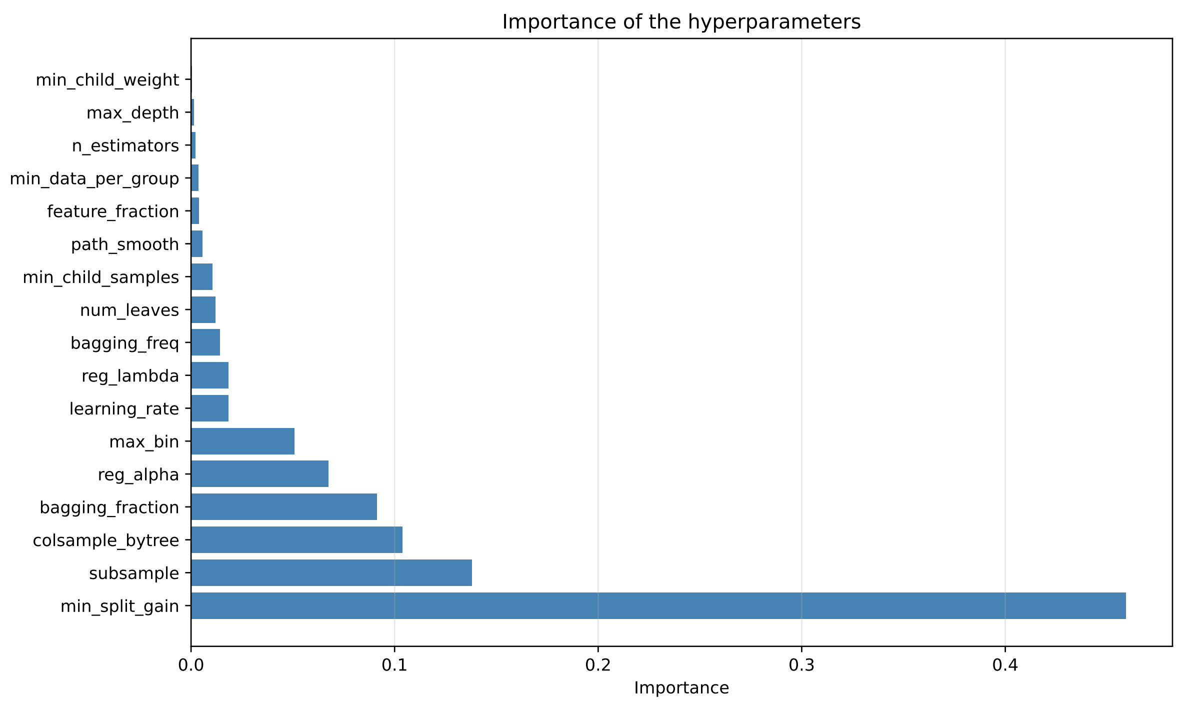
3.5 Model : LGBM

3.5.1 Gridsearch

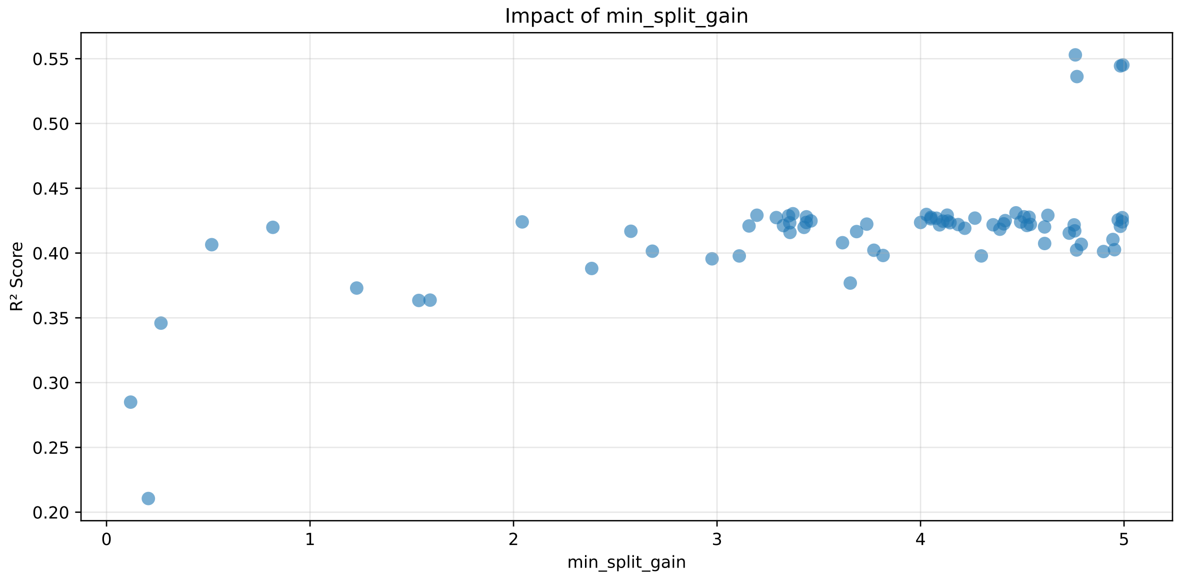
Distribution y_train vs y_test: y_train: mean=2.350, std=0.806, min=0.959, max=13.845 y_test: mean=2.355, std=0.778, min=1.011, max=7.822

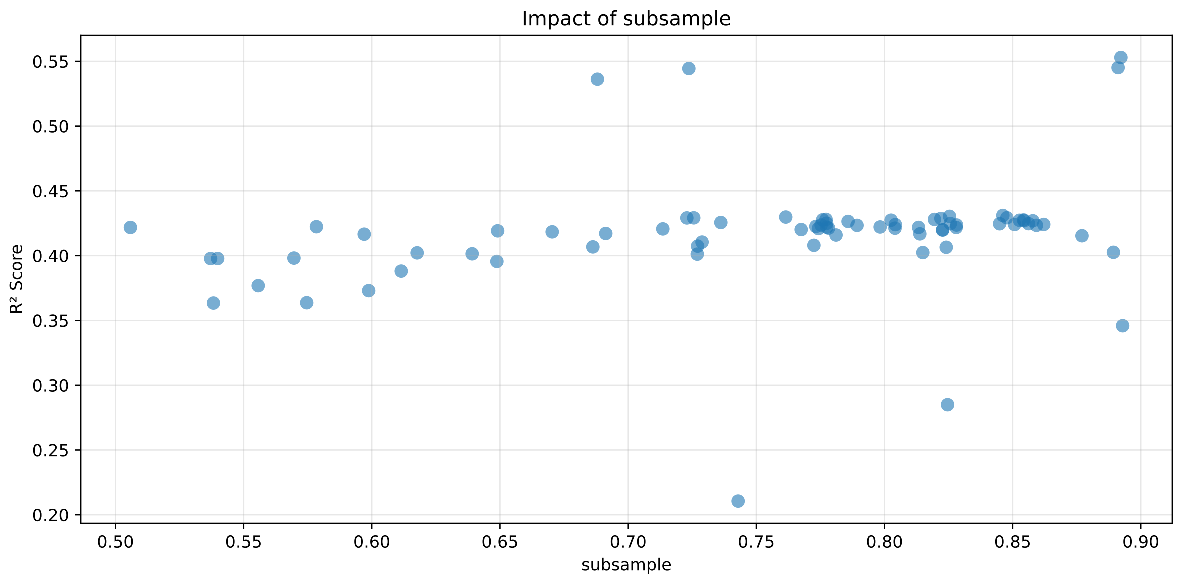
==================== TOP Importance FEATURES ==================== feature importance r_umidity 149 r_temperature_st_4h_slope 121 r_CO2 80 r_NH3_st_2h_std 73 r_temperature_st_2h_slope 67 r_temperature_st_2h_diff_last 56 r_umidity_st_2h_mean 52 r_temperature_st_2h_std 52 r_NH3_st_3h_std 51 r_umidity_st_4h_min 42 r_NH3_st_4h_max 35 r_temperature_st_3h_std 32 r_NH3_st_4h_std 30 r_NH3_st_3h_max 24 r_NH3_st_2h_slope 23 r_NH3_st_2h_diff_last 22 r_umidity_st_2h_diff_last 22 r_temperature_st_3h_min 20 r_umidity_st_3h_slope 20 r_NH3 16 r_NH3_st_3h_slope 14 r_temperature 12 r_NH3_st_4h_slope 12 r_temperature_st_3h_diff_last 11 r_NH3_st_2h_max 11 r_umidity_st_3h_diff_last 8 r_umidity_st_4h_std 8 r_umidity_st_2h_std 5 r_NH3_st_3h_diff_last 4 r_NH3_st_4h_diff_last 3 r_temperature_st_4h_diff_last 3 r_THI 2 r_umidity_st_4h_diff_last 1

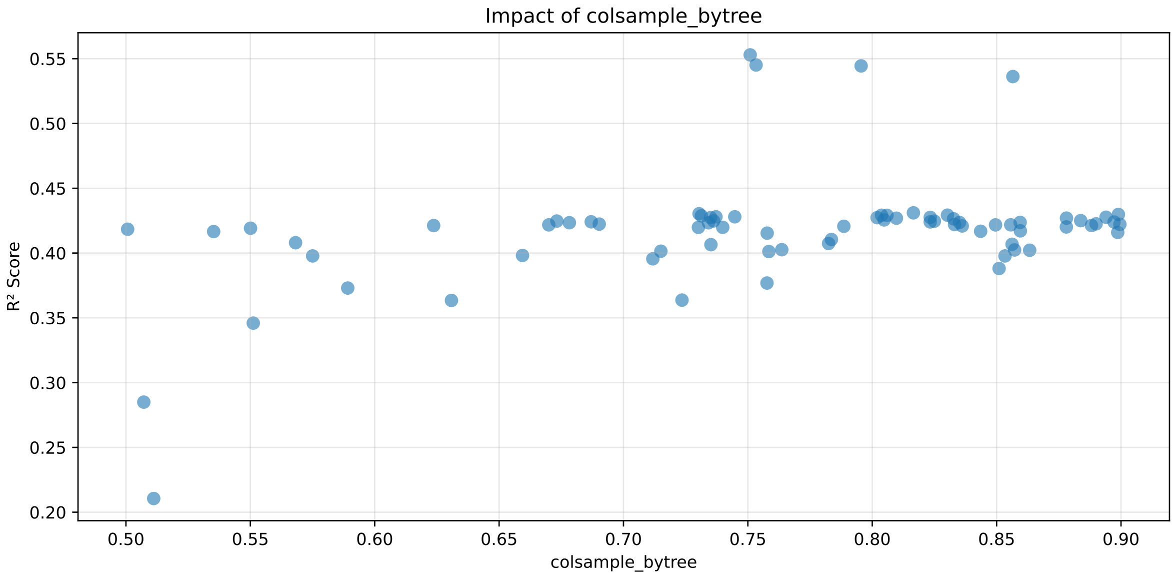


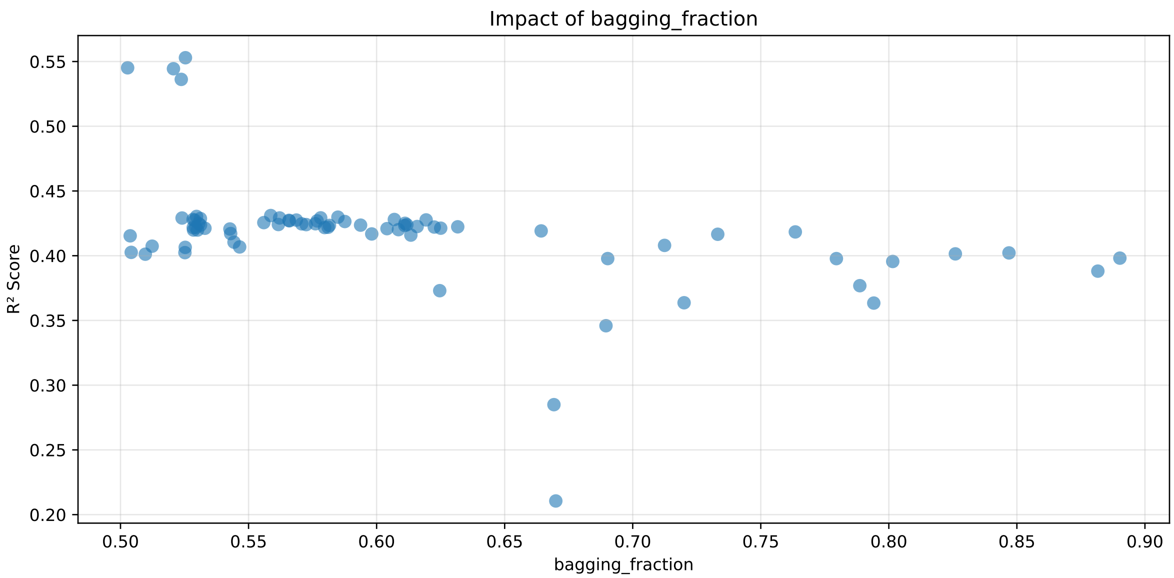


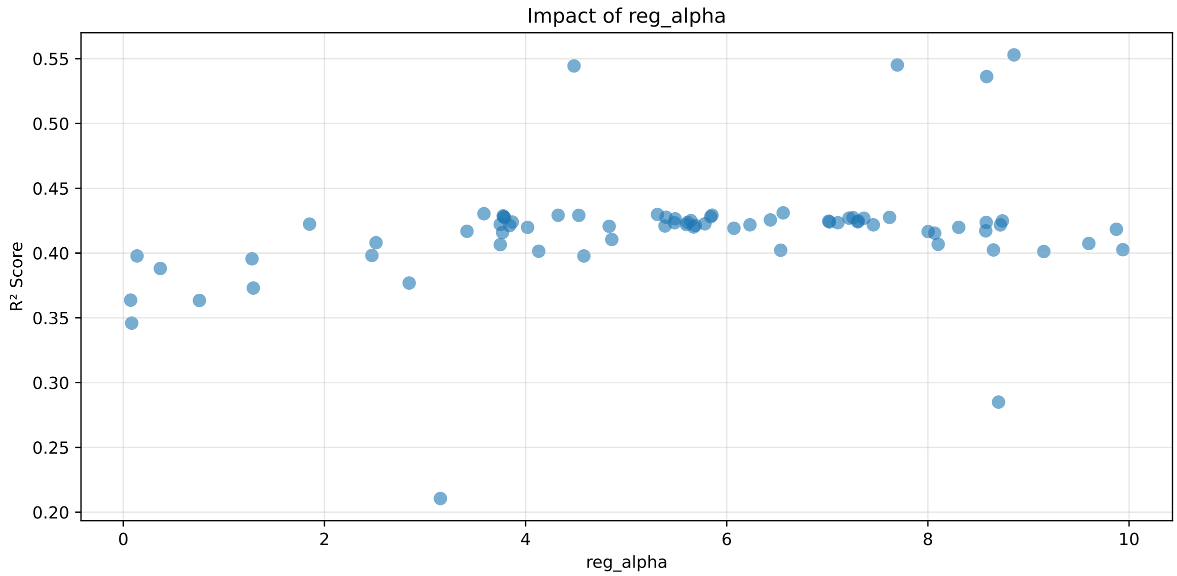
Paramètres avec >5% d'importance : ['min_split_gain', 'subsample', 'colsample_bytree', 'bagging_fraction', 'reg_alpha', 'max_bin']

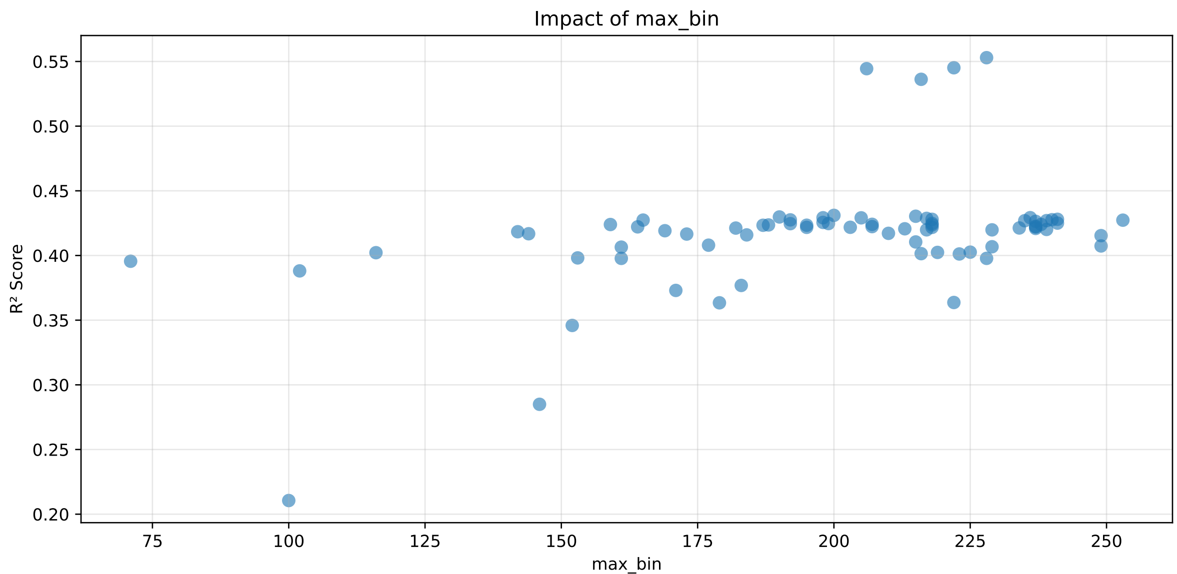

Impact of min_split_gain
R² Score
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0
1
2
3
4
5
min_split_gain


Impact of subsample
R² Score
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
subsample


Impact of colsample_bytree
R² Score
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
colsample_bytree

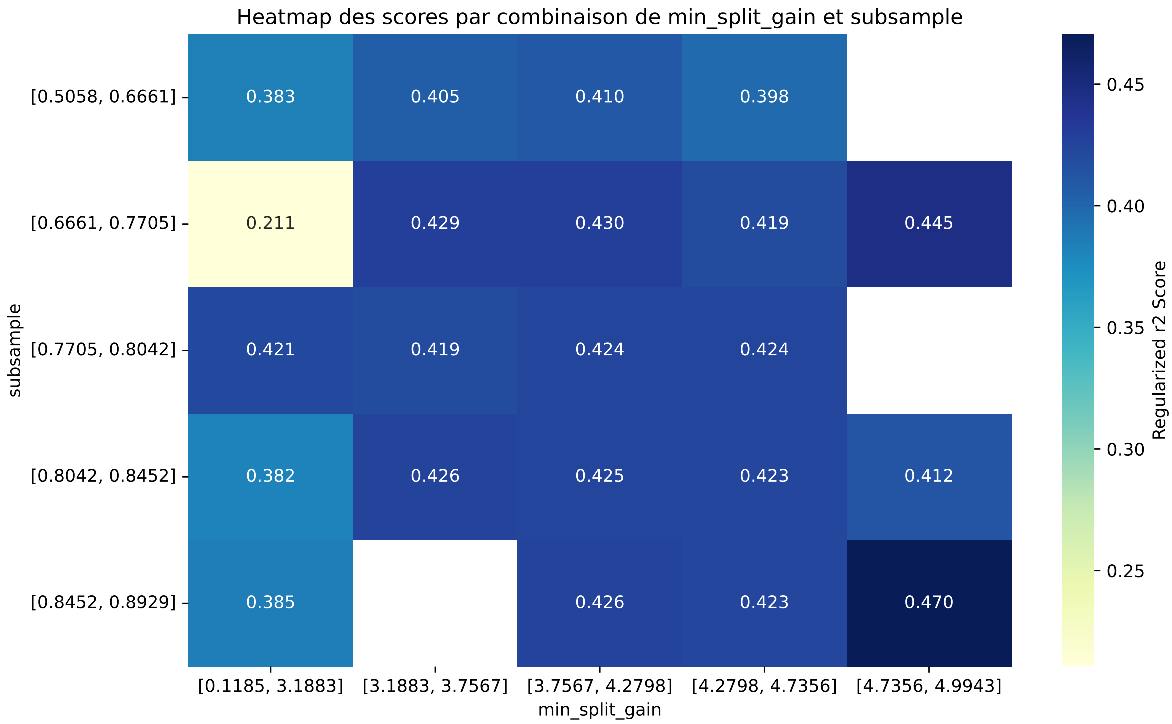

Impact of bagging_fraction
R² Score
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
bagging_fraction






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5803	0.5529	0.5785	0.0291	0.5529	0.6031	0.0501	1.0907	1.0495
2	0.5718	0.5452	0.5701	0.0293	0.5452	0.5963	0.0511	1.0937	1.049
3	0.5716	0.5444	0.5701	0.0293	0.5444	0.5935	0.049	1.0901	1.0498
4	0.5627	0.5362	0.5615	0.0287	0.5362	0.5815	0.0453	1.0844	1.0493
5	0.6195	0.5881	0.616	0.0294	0.431	0.6328	0.0447	1.076	1.0534
6	0.6356	0.6014	0.6298	0.0273	0.4303	0.6482	0.0468	1.0779	1.0569
7	0.6255	0.5928	0.6202	0.0251	0.4298	0.6404	0.0475	1.0802	1.055
8	0.6281	0.5949	0.6227	0.0284	0.4292	0.64	0.0451	1.0758	1.0557
9	0.6392	0.6042	0.6328	0.0284	0.4292	0.6516	0.0474	1.0784	1.0579
10	0.6119	0.5814	0.6079	0.0305	0.4291	0.6314	0.0499	1.0859	1.0524
---	---	---	---	---	---	---	---	---	---
75	0.8212	0.7194	0.7554	0.0346	0.2106	0.7725	0.0531	1.0738	1.1415

Hyperparameters:

Rank 1: {'reg___n_estimators': 200, 'reg___max_depth': 8, 'reg___learning_rate': 0.012368100985015324, 'reg___num_leaves': 40, 'reg___subsample': 0.8922321506784028, 'reg___colsample_bytree': 0.750931605712374, 'reg___min_child_samples': 49, 'reg___reg_alpha': 8.856264921544685, 'reg___reg_lambda': 48.54831816093221, 'reg___min_split_gain': 4.760814481199997, 'reg___path_smooth': 0.10789652323444221, 'reg___min_child_weight':

0.26998915379077, ’reg__feature_fraction’: 0.6577871243003972, ’reg__bagging_fraction’: 0.5253591055985517, ’reg__bagging_freq’: 7, ’reg__max_bin’: 228, ’reg__min_data_per_group’: 105}

Rank 2: {’reg__n_estimators’: 200, ’reg__max_depth’: 8, ’reg__learning_rate’: 0.011489653199465289, ’reg__num_leaves’: 39, ’reg__subsample’: 0.8911256680036673, ’reg__colsample_bytree’: 0.7533008178046687, ’reg__min_child_samples’: 50, ’reg__reg_alpha’: 7.696436887795401, ’reg__reg_lambda’: 49.610457519914355, ’reg__min_split_gain’: 4.99428694384397, ’reg__path_smooth’: 0.24000647201799552, ’reg__min_child_weight’: 0.6180091775199283, ’reg__feature_fraction’: 0.6728335127508291, ’reg__bagging_fraction’: 0.5027879373957379, ’reg__bagging_freq’: 7, ’reg__max_bin’: 222, ’reg__min_data_per_group’: 113}

Rank 3: {’reg__n_estimators’: 200, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.01028795289643142, ’reg__num_leaves’: 42, ’reg__subsample’: 0.7236960904754843, ’reg__colsample_bytree’: 0.7955323367371425, ’reg__min_child_samples’: 49, ’reg__reg_alpha’: 4.481639977016341, ’reg__reg_lambda’: 49.86236415607604, ’reg__min_split_gain’: 4.982622132332178, ’reg__path_smooth’: 0.006310983432350037, ’reg__min_child_weight’: 0.46823342157022757, ’reg__feature_fraction’: 0.8025857177168073, ’reg__bagging_fraction’: 0.5206780979688755, ’reg__bagging_freq’: 6, ’reg__max_bin’: 206, ’reg__min_data_per_group’: 112}

Rank 4: {’reg__n_estimators’: 200, ’reg__max_depth’: 8, ’reg__learning_rate’: 0.01004573969367508, ’reg__num_leaves’: 35, ’reg__subsample’: 0.6880203778189696, ’reg__colsample_bytree’: 0.8565328573753722, ’reg__min_child_samples’: 43, ’reg__reg_alpha’: 8.584806708975663, ’reg__reg_lambda’: 47.330782521851084, ’reg__min_split_gain’: 4.768611190232208, ’reg__path_smooth’: 0.6238209517864197, ’reg__min_child_weight’: 0.28805847938900614, ’reg__feature_fraction’: 0.6476754075043613, ’reg__bagging_fraction’: 0.5237295320131822, ’reg__bagging_freq’: 5, ’reg__max_bin’: 216, ’reg__min_data_per_group’: 110}

Rank 5: {’reg__n_estimators’: 300, ’reg__max_depth’: 7, ’reg__learning_rate’: 0.016214695812009537, ’reg__num_leaves’: 42, ’reg__subsample’: 0.8461717467017182, ’reg__colsample_bytree’: 0.8165466075797199, ’reg__min_child_samples’: 50, ’reg__reg_alpha’: 6.561122325260685, ’reg__reg_lambda’: 37.95620666272198, ’reg__min_split_gain’: 4.468999690655012, ’reg__path_smooth’: 2.2162604536332653, ’reg__min_child_weight’: 0.38117901419665545, ’reg__feature_fraction’: 0.615746689185062, ’reg__bagging_fraction’: 0.5586710852653107, ’reg__bagging_freq’: 6, ’reg__max_bin’: 200, ’reg__min_data_per_group’: 102}

Rank 6: {’reg__n_estimators’: 600, ’reg__max_depth’: 6, ’reg__learning_rate’: 0.019365791736273046, ’reg__num_leaves’: 51, ’reg__subsample’: 0.8253810993791022, ’reg__colsample_bytree’: 0.7303808657331876, ’reg__min_child_samples’: 42, ’reg__reg_alpha’: 3.58584278209981, ’reg__reg_lambda’: 47.11036047724813, ’reg__min_split_gain’: 3.373026784996114, ’reg__path_smooth’: 1.2755112100633315, ’reg__min_child_weight’:

0.8292171591002461, ’reg__feature_fraction’: 0.6438648656634689, ’reg__bagging_fraction’: 0.5296897720605226, ’reg__bagging_freq’: 5, ’reg__max_bin’: 215, ’reg__min_data_per_group’: 116}

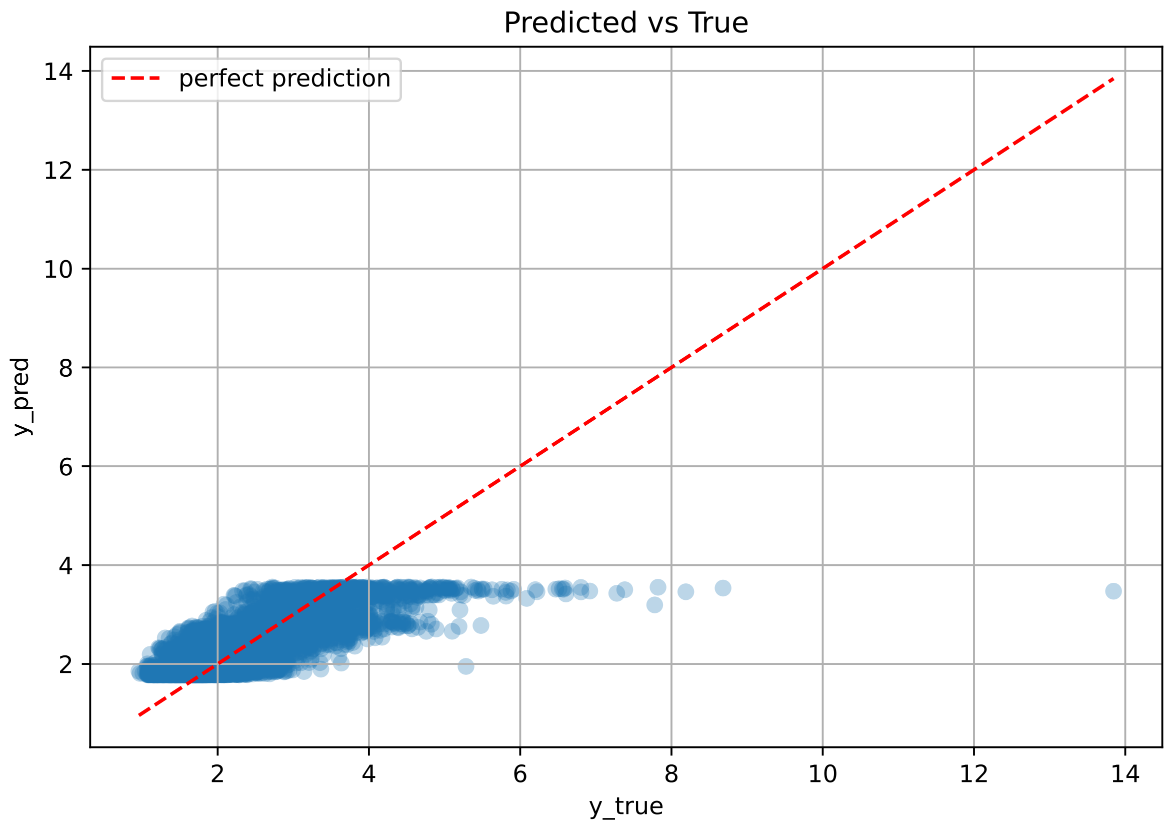
Rank 7: {’reg__n_estimators’: 300, ’reg__max_depth’: 7, ’reg__learning_rate’: 0.025024569356643142, ’reg__num_leaves’: 45, ’reg__subsample’: 0.7614843122386636, ’reg__colsample_bytree’: 0.8989466392746849, ’reg__min_child_samples’: 45, ’reg__reg_alpha’: 5.311379454957386, ’reg__reg_lambda’: 45.270171807438274, ’reg__min_split_gain’: 4.02880140800138, ’reg__path_smooth’: 0.2518936614195765, ’reg__min_child_weight’: 1.0284579109159586, ’reg__feature_fraction’: 0.7096891261924968, ’reg__bagging_fraction’: 0.58489748099983, ’reg__bagging_freq’: 7, ’reg__max_bin’: 190, ’reg__min_data_per_group’: 129}

Rank 8: {’reg__n_estimators’: 300, ’reg__max_depth’: 7, ’reg__learning_rate’: 0.014627160784028105, ’reg__num_leaves’: 45, ’reg__subsample’: 0.8477468468769519, ’reg__colsample_bytree’: 0.8302354781246394, ’reg__min_child_samples’: 45, ’reg__reg_alpha’: 5.853713244133902, ’reg__reg_lambda’: 41.497708144659605, ’reg__min_split_gain’: 4.131067647163776, ’reg__path_smooth’: 0.4346804353531366, ’reg__min_child_weight’: 0.11447068671770452, ’reg__feature_fraction’: 0.7205374769179798, ’reg__bagging_fraction’: 0.5781653271464738, ’reg__bagging_freq’: 6, ’reg__max_bin’: 236, ’reg__min_data_per_group’: 127}

Rank 9: {’reg__n_estimators’: 300, ’reg__max_depth’: 7, ’reg__learning_rate’: 0.016075000332299128, ’reg__num_leaves’: 42, ’reg__subsample’: 0.7256023807388654, ’reg__colsample_bytree’: 0.8037381119244775, ’reg__min_child_samples’: 49, ’reg__reg_alpha’: 4.3237257826449955, ’reg__reg_lambda’: 49.467848806891354, ’reg__min_split_gain’: 3.196007926317961, ’reg__path_smooth’: 2.0716623930514095, ’reg__min_child_weight’: 0.5417559881700028, ’reg__feature_fraction’: 0.7914929170065695, ’reg__bagging_fraction’: 0.5621171516466091, ’reg__bagging_freq’: 6, ’reg__max_bin’: 198, ’reg__min_data_per_group’: 177}

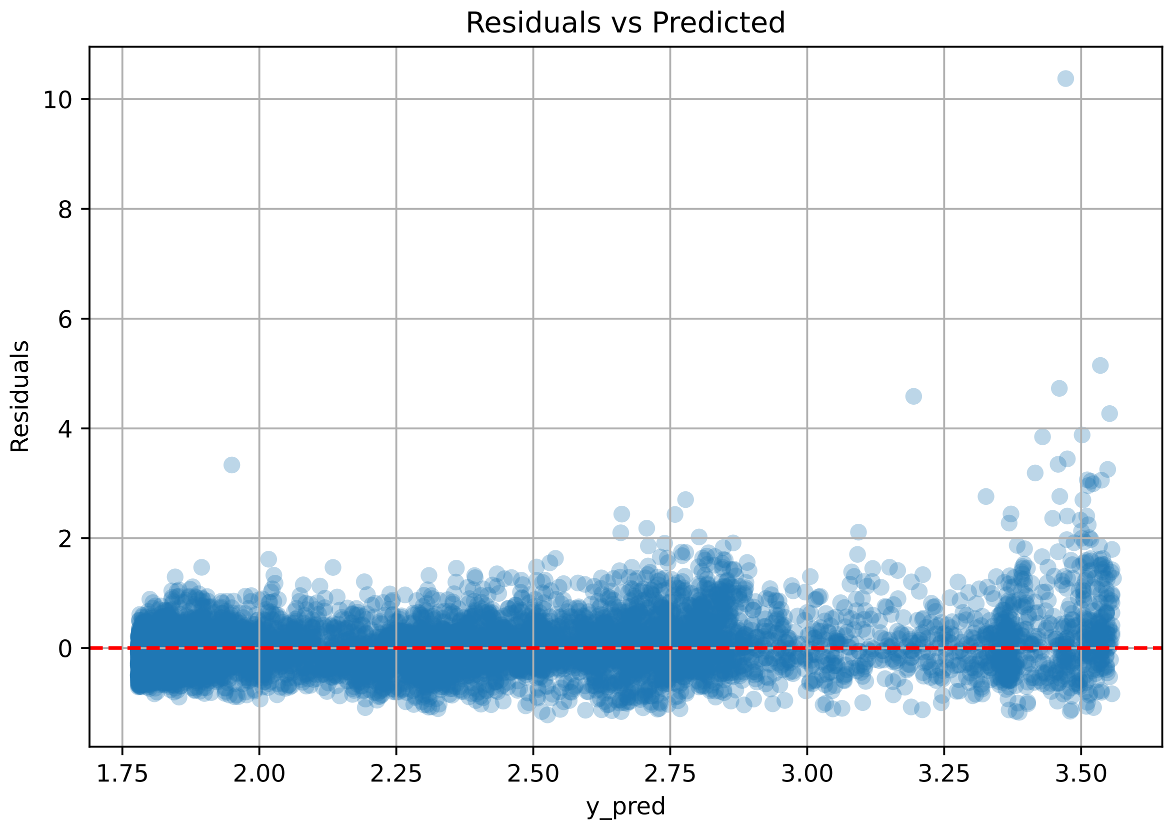
Rank 10: {’reg__n_estimators’: 200, ’reg__max_depth’: 7, ’reg__learning_rate’: 0.016488013305908066, ’reg__num_leaves’: 41, ’reg__subsample’: 0.7229165173645702, ’reg__colsample_bytree’: 0.8059322152168652, ’reg__min_child_samples’: 49, ’reg__reg_alpha’: 4.529179631374969, ’reg__reg_lambda’: 37.35291687052662, ’reg__min_split_gain’: 4.625676405934896, ’reg__path_smooth’: 1.9056772846141325, ’reg__min_child_weight’: 0.5100333899755409, ’reg__feature_fraction’: 0.7993152673628747, ’reg__bagging_fraction’: 0.5240420625902732, ’reg__bagging_freq’: 6, ’reg__max_bin’: 205, ’reg__min_data_per_group’: 177}

3.5.2 Scatter



Predicted vs True
perfect prediction
14
12
10
8
6
4
2
y_pred
2
4
6
8
10
12
14
y_true

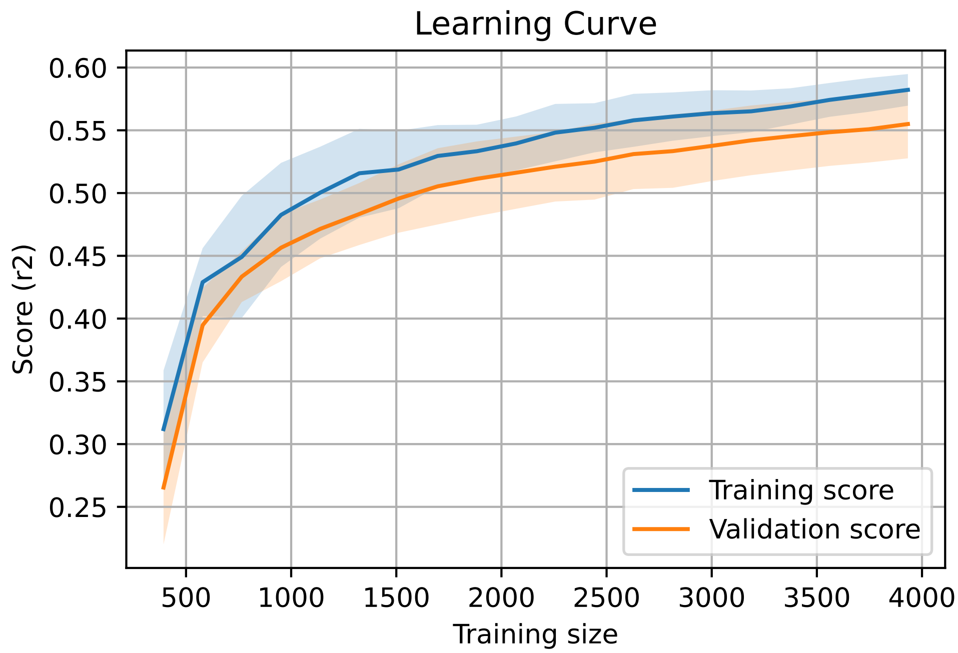
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 8028, **Len(training)** : 5620,

Len(training_real) : 3934, **Len(test_of_training)** : 1686, **Len(test)** : 2408

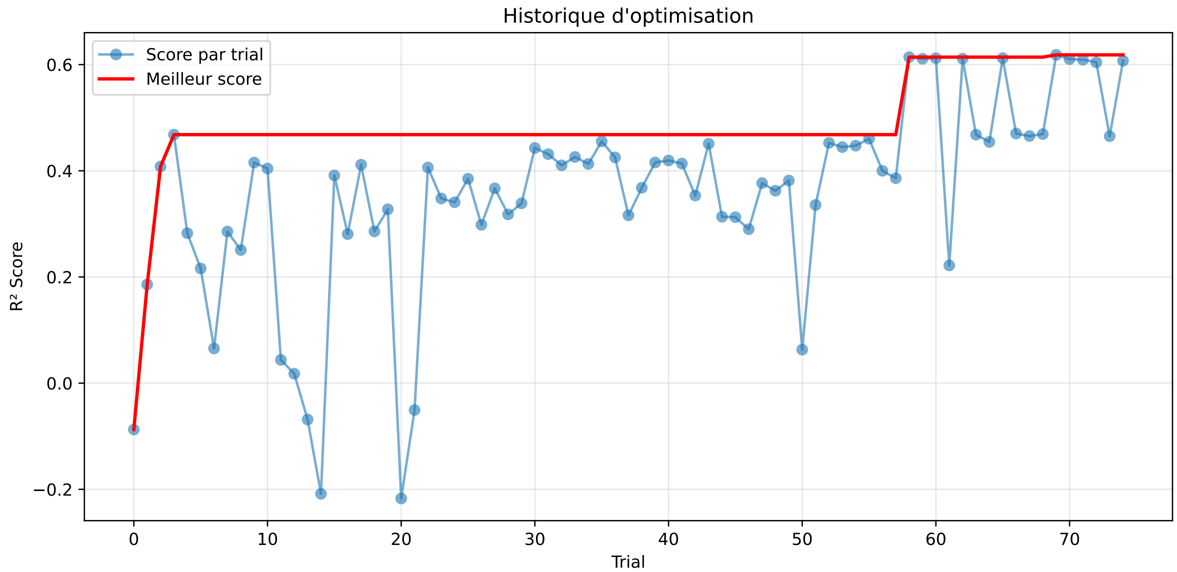


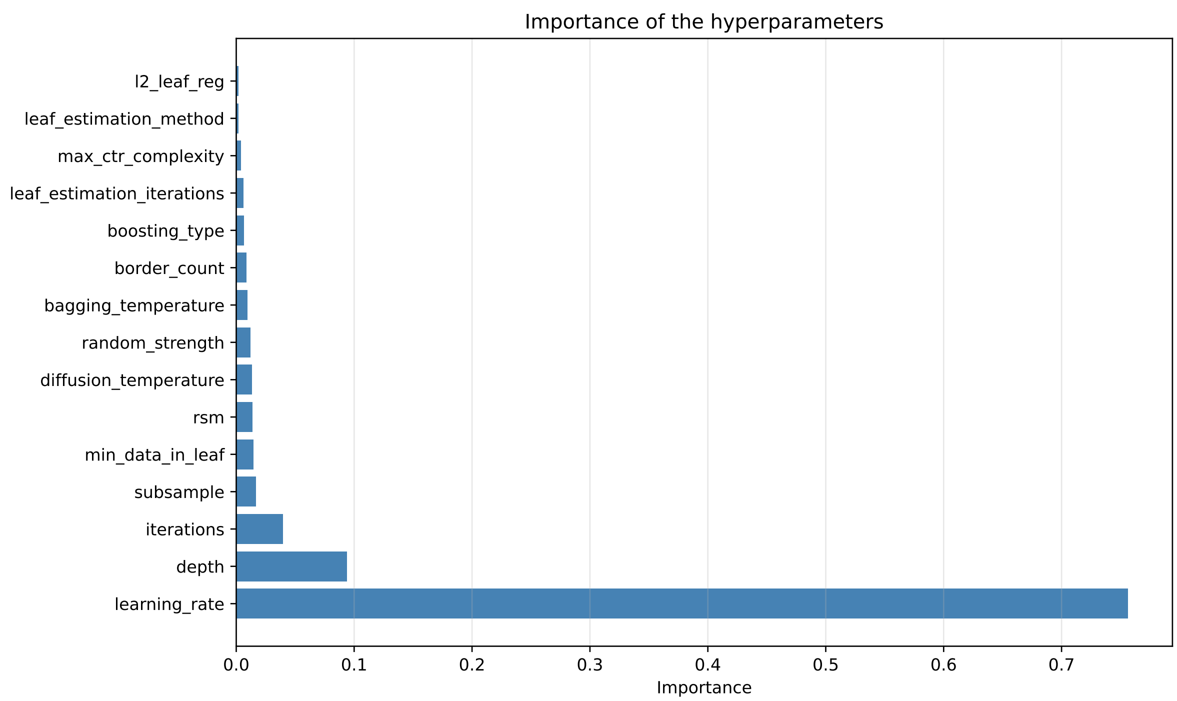
3.6 Model : CatBoost

3.6.1 Gridsearch

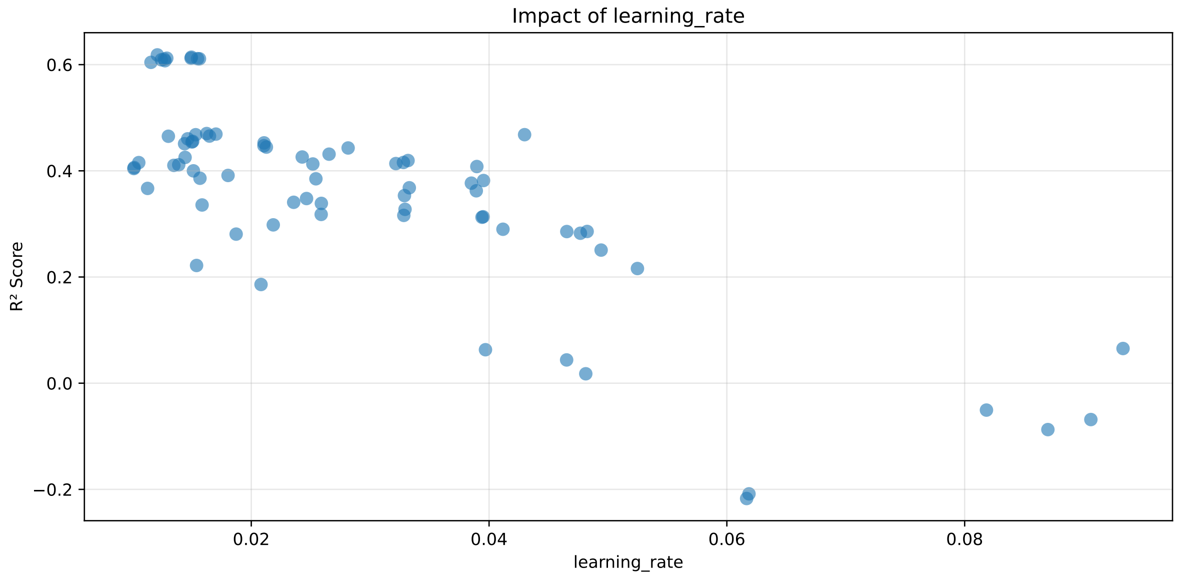
Distribution y_train vs y_test: y_train: mean=2.350, std=0.806, min=0.959, max=13.845 y_test: mean=2.355, std=0.778, min=1.011, max=7.822

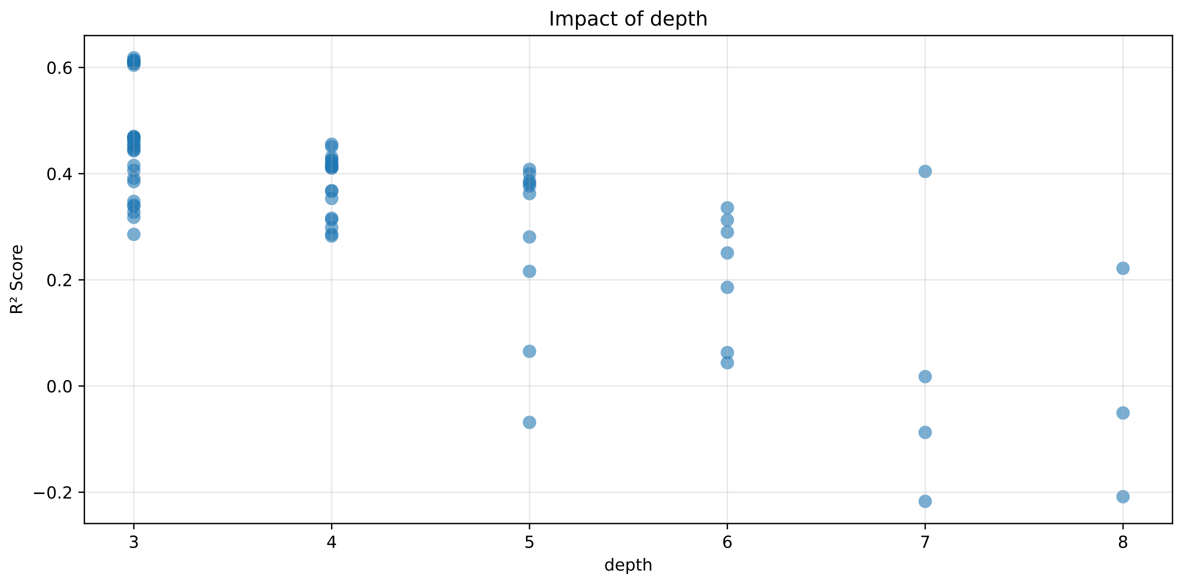
==================== TOP Importance FEATURES ==================== feature importance r_umidity 15.102298 r_temperature_st_4h_slope 10.974614 r_temperature_st_2h_slope 9.338951 r_umidity_st_3h_slope 7.756741 r_umidity_st_2h_mean 6.014468 r_temperature_st_2h_std 5.968355 r_CO2 4.180909 r_temperature_st_3h_std 3.953515 r_temperature 3.803388 r_temperature_st_2h_diff_last 3.601035 r_umidity_st_4h_min 2.879458 r_NH3_st_2h_std 2.249920 r_NH3_st_3h_std 2.247498 r_temperature_st_3h_min 1.873428 r_temperature_st_4h_diff_last 1.847582 r_NH3_st_4h_max 1.739303 r_temperature_st_3h_diff_last 1.715550 r_NH3 1.668221 r_NH3_st_3h_max 1.552051 r_NH3_st_4h_std 1.467047 r_NH3_st_2h_max 1.442645 r_THI 1.115989 r_umidity_st_3h_diff_last 1.000664 r_NH3_st_2h_slope 0.980467 r_NH3_st_3h_slope 0.877170 r_umidity_st_2h_diff_last 0.824394 r_NH3_st_4h_slope 0.776369 r_NH3_st_3h_diff_last 0.660368 r_umidity_st_2h_std 0.575332 r_NH3_st_4h_diff_last 0.549955 r_umidity_st_4h_diff_last 0.470470 r_umidity_st_4h_std 0.453707 r_NH3_st_2h_diff_last 0.338138





Paramètres avec >5% d'importance : ['learning_rate', 'depth']





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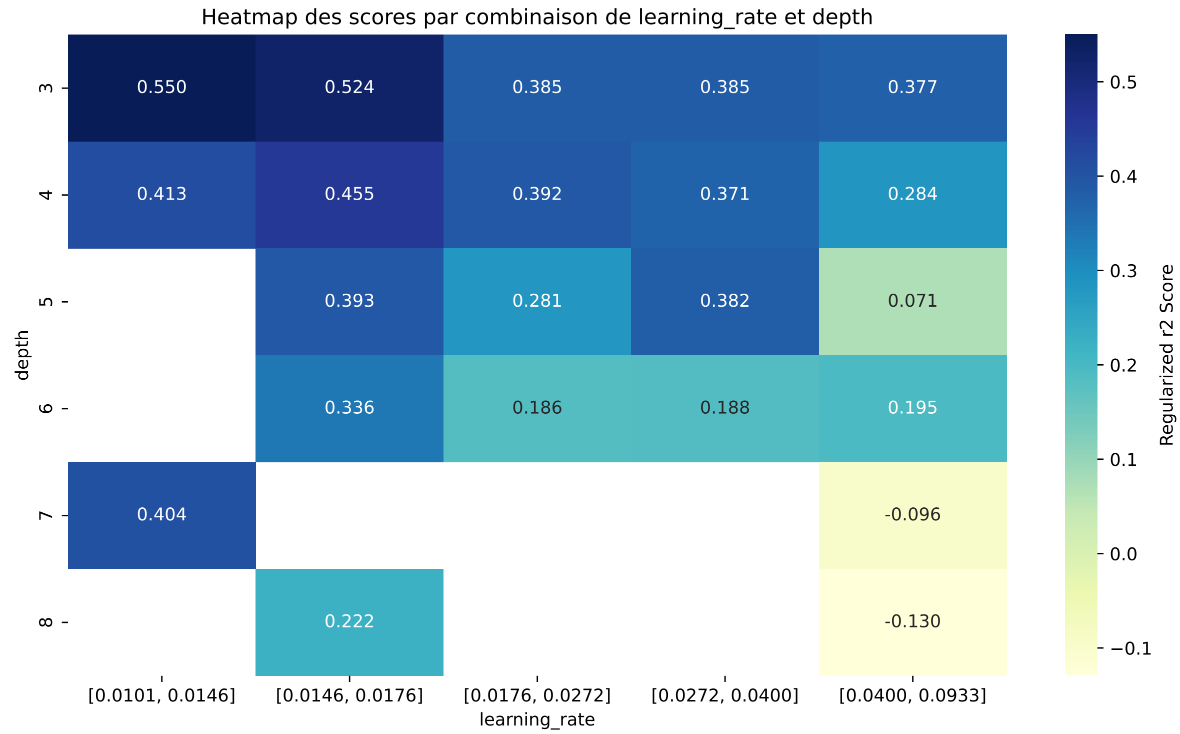
The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.648	0.6184	0.6433	0.0264	0.6184	0.6483	0.03	1.0484	1.0479
2	0.6447	0.6141	0.6394	0.0271	0.6141	0.6435	0.0294	1.0479	1.0498
3	0.6403	0.6121	0.6371	0.0271	0.6121	0.6432	0.0311	1.0507	1.0459
4	0.6397	0.6121	0.6362	0.027	0.6121	0.6387	0.0265	1.0434	1.045
5	0.6402	0.6112	0.6348	0.0273	0.6112	0.6416	0.0304	1.0497	1.0474
6	0.6402	0.611	0.6363	0.0266	0.611	0.6374	0.0264	1.0432	1.0478
7	0.639	0.6103	0.6364	0.0273	0.6103	0.6418	0.0315	1.0516	1.0471
8	0.6374	0.6092	0.6343	0.027	0.6092	0.6411	0.0319	1.0524	1.0464
9	0.6356	0.6073	0.6318	0.0265	0.6073	0.6351	0.0278	1.0458	1.0465
10	0.6319	0.6043	0.6289	0.0267	0.6043	0.6347	0.0304	1.0503	1.0457
---	---	---	---	---	---	---	---	---	---
75	0.9755	0.7767	0.8117	0.037	-0.2172	0.8244	0.0477	1.0614	1.2559

Hyperparameters:

Rank 1: {’reg___iterations’: 700, ’reg___depth’: 3, ’reg___learning_rate’: 0.01210068980081506, ’reg___l2_leaf_reg’: 14.436936987267678, ’reg___bagging_temperature’: 2.1655399701517295, ’reg___random_strength’: 3.9257573053058614, ’reg___subsample’: 0.5008916668775042, ’reg___min_data_in_leaf’: 45, ’reg___leaf_estimation_iterations’: 8, ’reg___rsm’: 0.7856816966556072, ’reg___border_count’: 128, ’reg___leaf_estimation_method’: ‘Gradient’,

’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 2751.1470704770713}

Rank 2: {’reg__iterations’: 500, ’reg__depth’: 3, ’reg__learning_rate’: 0.014964222442232236, ’reg__l2_leaf_reg’: 15.838268056449596, ’reg__bagging_temperature’: 3.341366294597587, ’reg__random_strength’: 3.6940392171136143, ’reg__subsample’: 0.6846996226264596, ’reg__min_data_in_leaf’: 28, ’reg__leaf_estimation_iterations’: 8, ’reg__rsm’: 0.797420185062919, ’reg__border_count’: 103, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 1707.452282177747}

Rank 3: {’reg__iterations’: 600, ’reg__depth’: 3, ’reg__learning_rate’: 0.012894612171532599, ’reg__l2_leaf_reg’: 11.312634403245088, ’reg__bagging_temperature’: 2.3711422707467404, ’reg__random_strength’: 4.077352808348798, ’reg__subsample’: 0.6854291428446536, ’reg__min_data_in_leaf’: 48, ’reg__leaf_estimation_iterations’: 8, ’reg__rsm’: 0.7812818773251621, ’reg__border_count’: 123, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 3220.8668909692105}

Rank 4: {’reg__iterations’: 500, ’reg__depth’: 3, ’reg__learning_rate’: 0.01493517720402781, ’reg__l2_leaf_reg’: 17.12911165813437, ’reg__bagging_temperature’: 3.34114569974794, ’reg__random_strength’: 3.65539792031575, ’reg__subsample’: 0.5467184913928005, ’reg__min_data_in_leaf’: 47, ’reg__leaf_estimation_iterations’: 8, ’reg__rsm’: 0.8056594670782543, ’reg__border_count’: 102, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 3, ’reg__diffusion_temperature’: 1765.9382061800636}

Rank 5: {’reg__iterations’: 500, ’reg__depth’: 3, ’reg__learning_rate’: 0.01548135230673944, ’reg__l2_leaf_reg’: 12.848707754802852, ’reg__bagging_temperature’: 3.4103454667276547, ’reg__random_strength’: 4.341451529328445, ’reg__subsample’: 0.5200706104888373, ’reg__min_data_in_leaf’: 48, ’reg__leaf_estimation_iterations’: 8, ’reg__rsm’: 0.804922638441654, ’reg__border_count’: 127, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 3, ’reg__diffusion_temperature’: 3106.3144515001613}

Rank 6: {’reg__iterations’: 500, ’reg__depth’: 3, ’reg__learning_rate’: 0.01564375169340389, ’reg__l2_leaf_reg’: 14.470873111647332, ’reg__bagging_temperature’: 3.4090241139903212, ’reg__random_strength’: 4.391478621584909, ’reg__subsample’: 0.851691914558832, ’reg__min_data_in_leaf’: 28, ’reg__leaf_estimation_iterations’: 8, ’reg__rsm’: 0.8167042333509243, ’reg__border_count’: 128, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 3, ’reg__diffusion_temperature’: 2852.384696009573}

Rank 7: {’reg__iterations’: 600, ’reg__depth’: 3, ’reg__learning_rate’: 0.012672227989496701, ’reg__l2_leaf_reg’: 12.176068745707255, ’reg__bagging_temperature’: 2.4081408195381933, ’reg__random_strength’: 3.935482197030665, ’reg__subsample’: 0.5033635965882592,

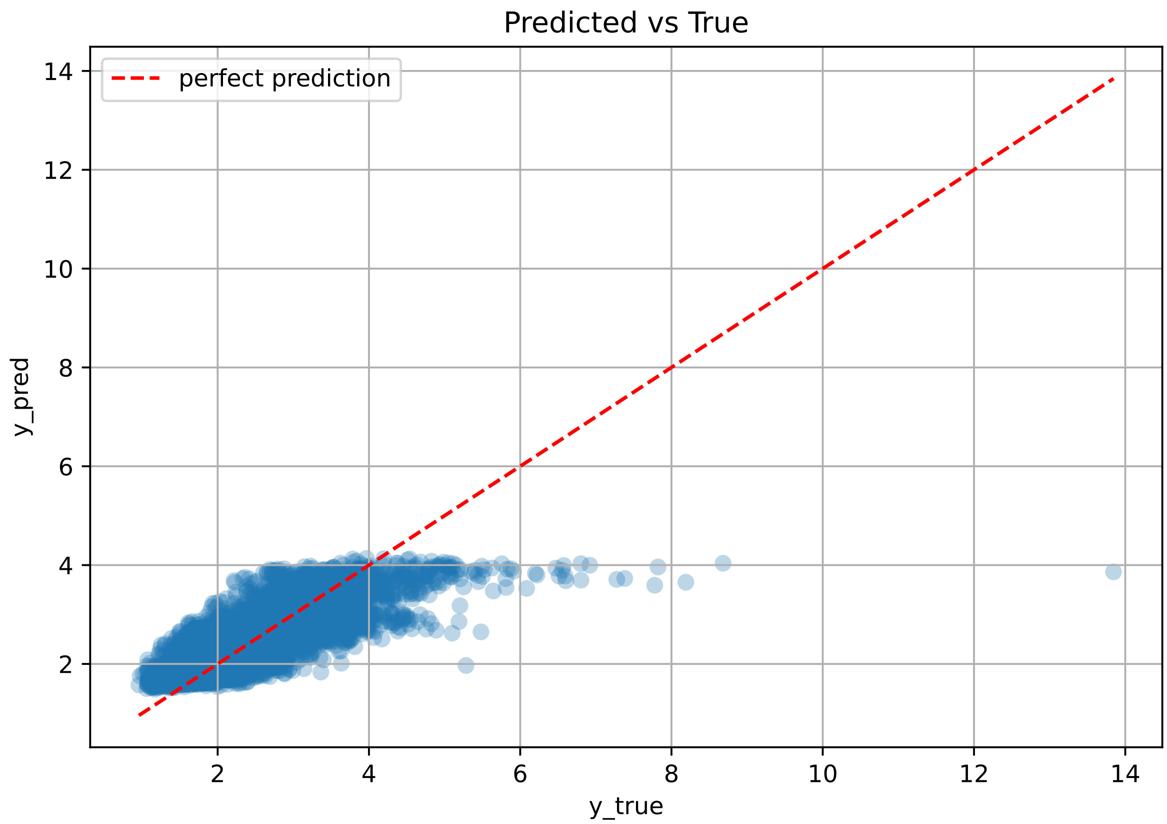
’reg___min_data_in_leaf’: 45, ’reg___leaf_estimation_iterations’: 5, ’reg___rsm’: 0.8747610905741742, ’reg___border_count’: 128, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Ordered’, ’reg___max_ctr_complexity’: 2, ’reg___diffusion_temperature’: 2875.4485390894843}

Rank 8: {’reg___iterations’: 600, ’reg___depth’: 3, ’reg___learning_rate’: 0.012437770884842188, ’reg___l2_leaf_reg’: 12.159428167743812, ’reg___bagging_temperature’: 2.2667098654908293, ’reg___random_strength’: 4.054537546763915, ’reg___subsample’: 0.5028085940777256, ’reg___min_data_in_leaf’: 45, ’reg___leaf_estimation_iterations’: 5, ’reg___rsm’: 0.8794254102680419, ’reg___border_count’: 126, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Ordered’, ’reg___max_ctr_complexity’: 1, ’reg___diffusion_temperature’: 3206.3043586918698}

Rank 9: {’reg___iterations’: 600, ’reg___depth’: 3, ’reg___learning_rate’: 0.012743363507933666, ’reg___l2_leaf_reg’: 10.294619429178997, ’reg___bagging_temperature’: 2.1157130061518545, ’reg___random_strength’: 4.660476130007891, ’reg___subsample’: 0.5022897588782248, ’reg___min_data_in_leaf’: 44, ’reg___leaf_estimation_iterations’: 8, ’reg___rsm’: 0.8205522913217664, ’reg___border_count’: 119, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Ordered’, ’reg___max_ctr_complexity’: 1, ’reg___diffusion_temperature’: 2961.2450615300863}

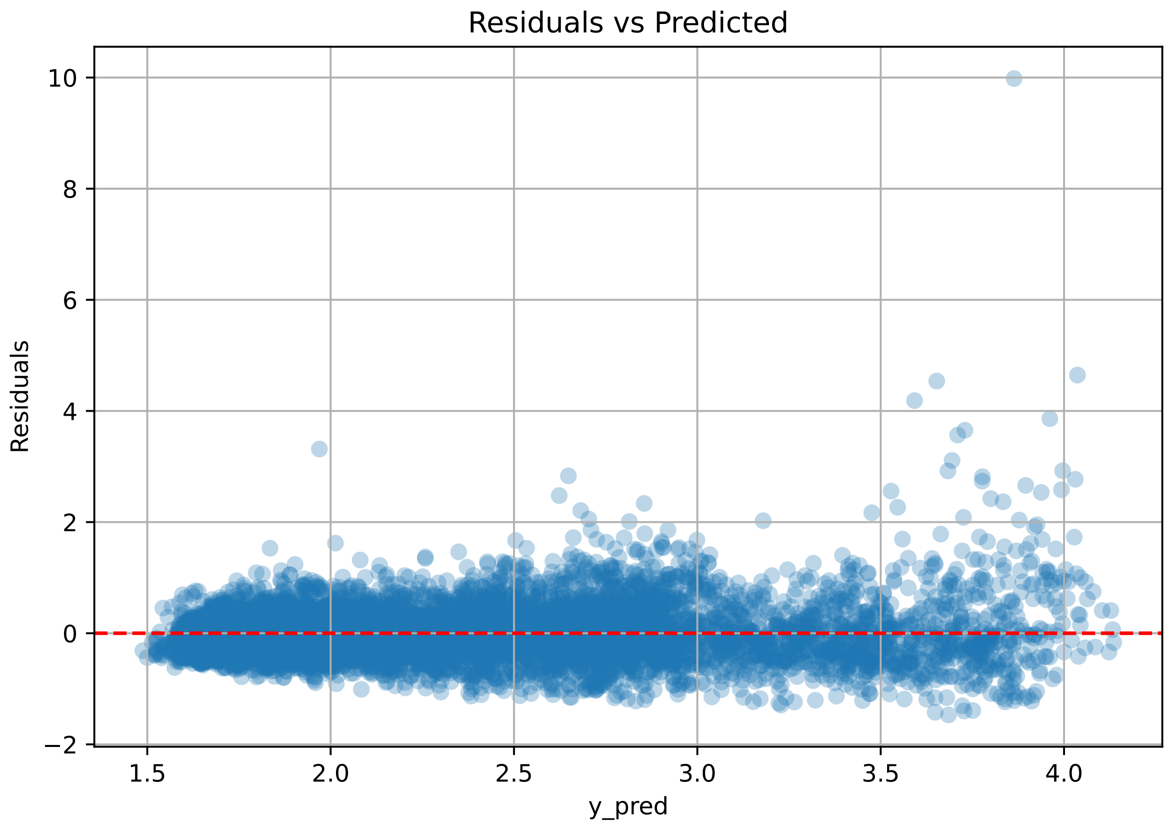
Rank 10: {’reg___iterations’: 600, ’reg___depth’: 3, ’reg___learning_rate’: 0.011563403990433398, ’reg___l2_leaf_reg’: 11.331537668221445, ’reg___bagging_temperature’: 2.2103805872945954, ’reg___random_strength’: 3.970845711488929, ’reg___subsample’: 0.506541224248736, ’reg___min_data_in_leaf’: 45, ’reg___leaf_estimation_iterations’: 8, ’reg___rsm’: 0.8214710222772676, ’reg___border_count’: 125, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Ordered’, ’reg___max_ctr_complexity’: 1, ’reg___diffusion_temperature’: 3242.302846004878}

3.6.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

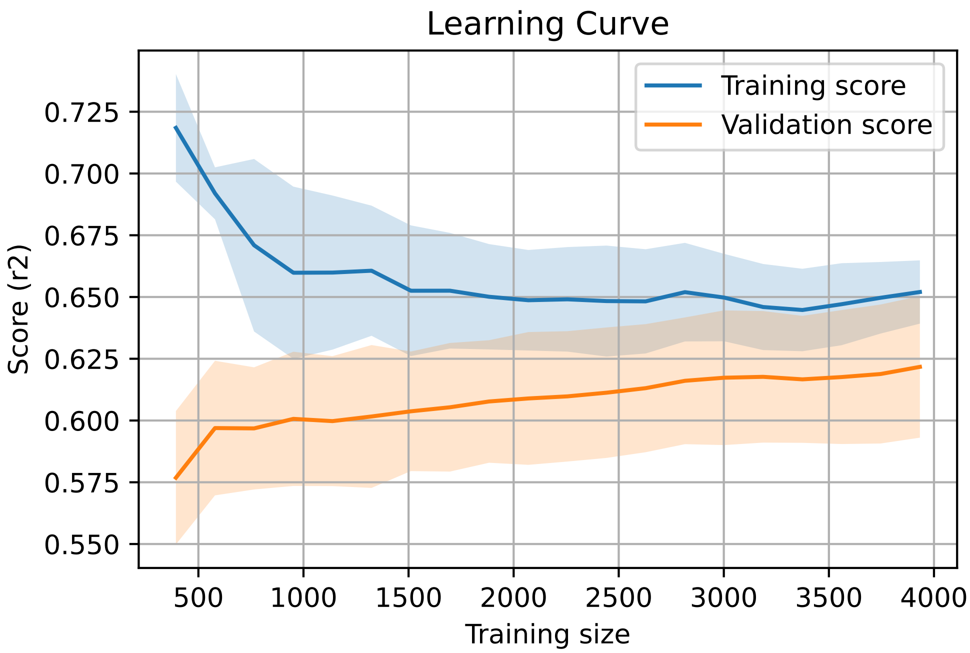
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 8028, **Len(training)** : 5620,

Len(training_real) : 3934, **Len(test_of_training)** : 1686, **Len(test)** : 2408



3.7 Sumup-table

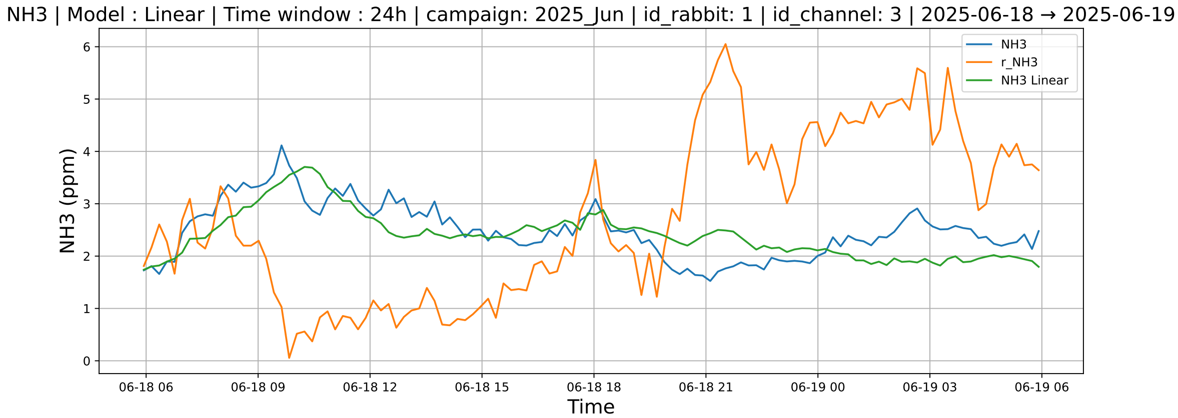
model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.569	0.543	0.024	0.577	0.578	0.034	1.063	1.048	nan
LGBM	0.58	0.553	0.029	0.578	0.603	0.05	1.091	1.05	0.5529
CatBoost	0.648	0.618	0.026	0.643	0.648	0.03	1.048	1.048	0.6184

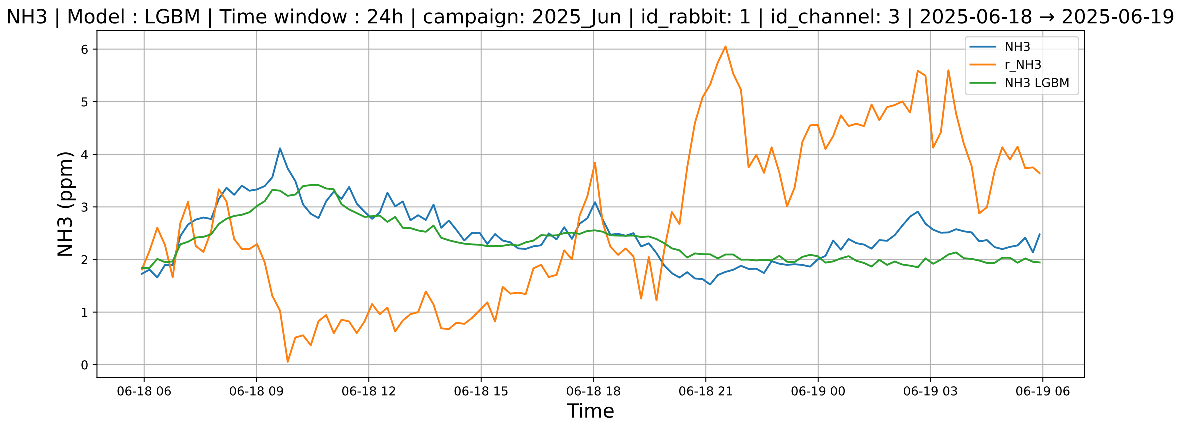
4 Plot

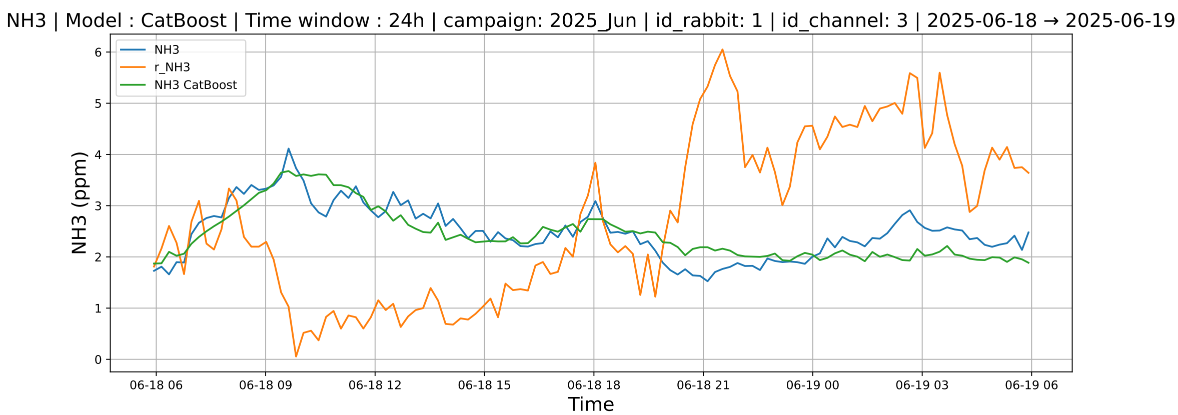
4.1 Standalone plots

4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

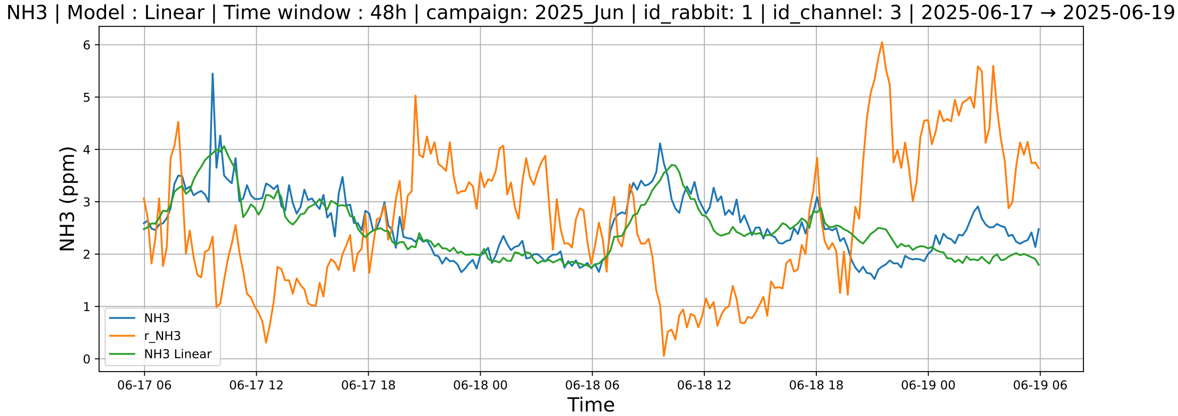
4.1.1.1 Time window : 24

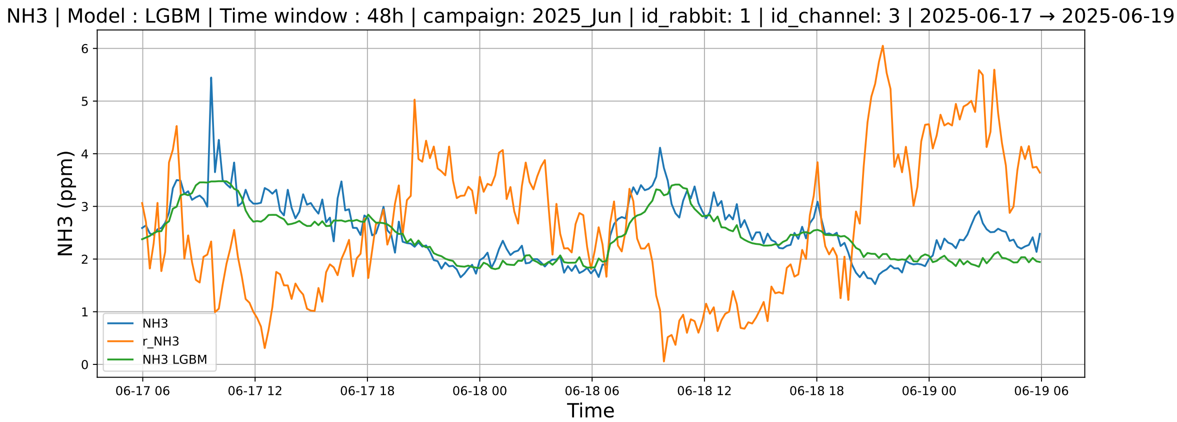


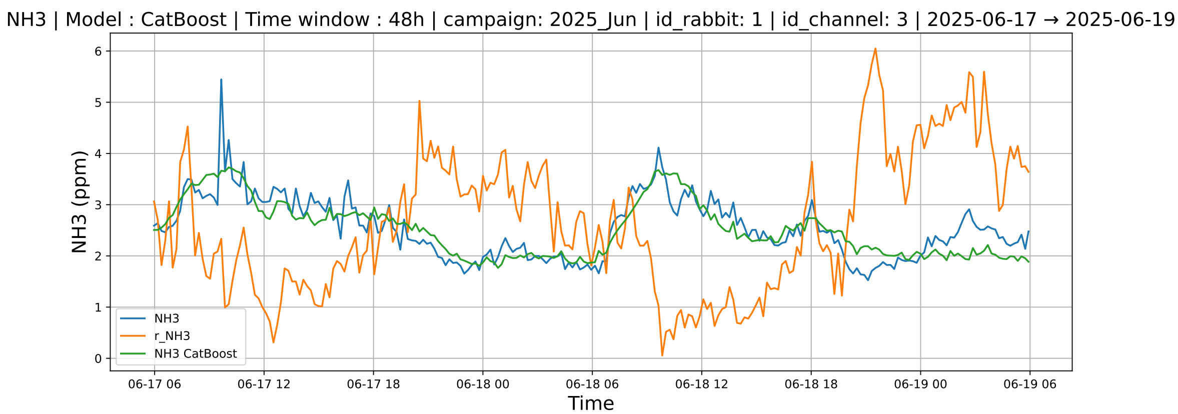




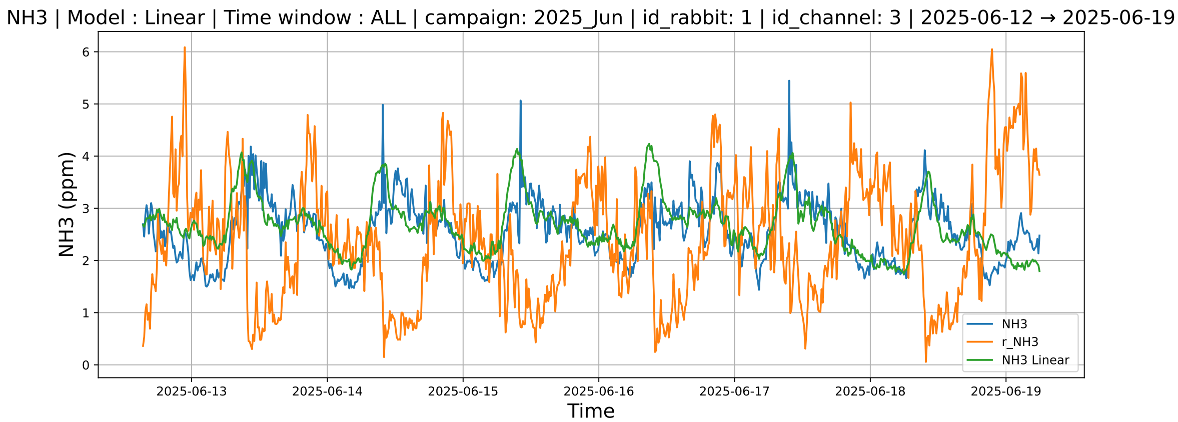
4.1.1.2 Time window : 48

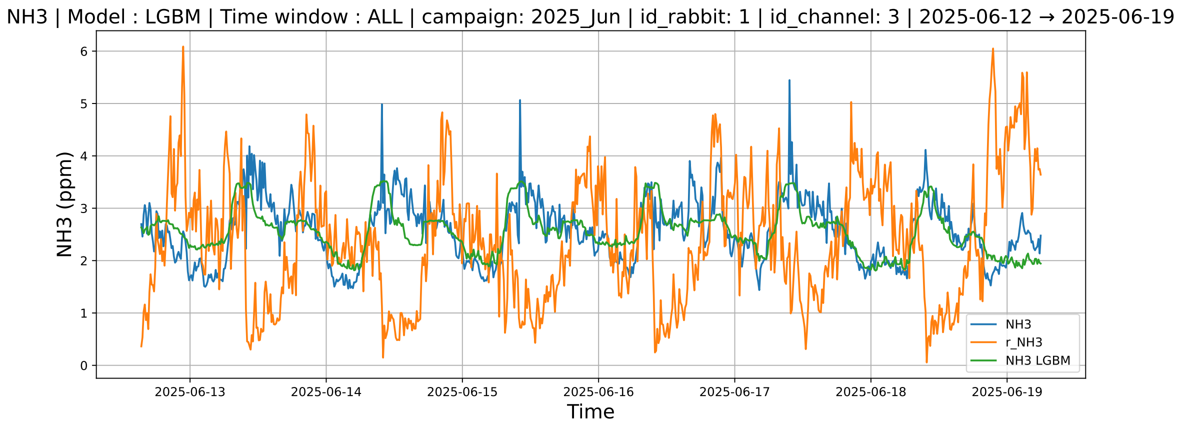


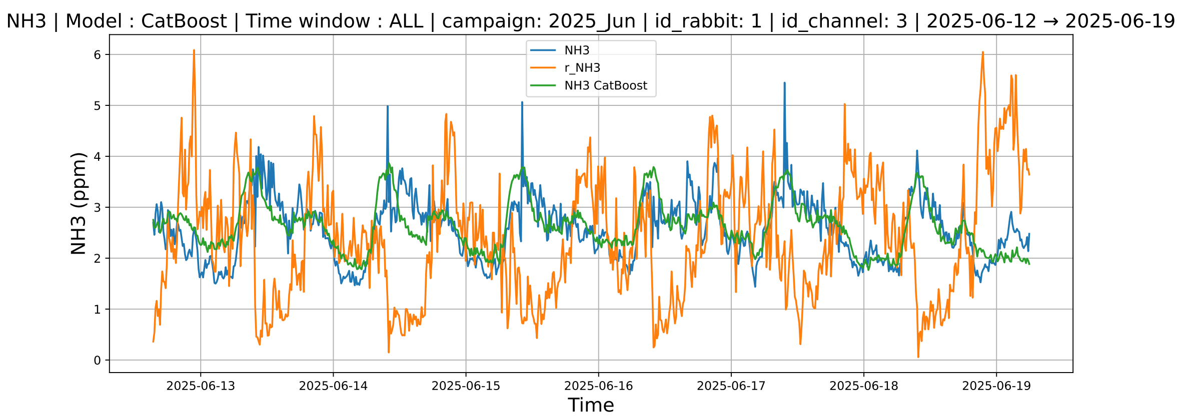




4.1.1.3 Time window : ALL



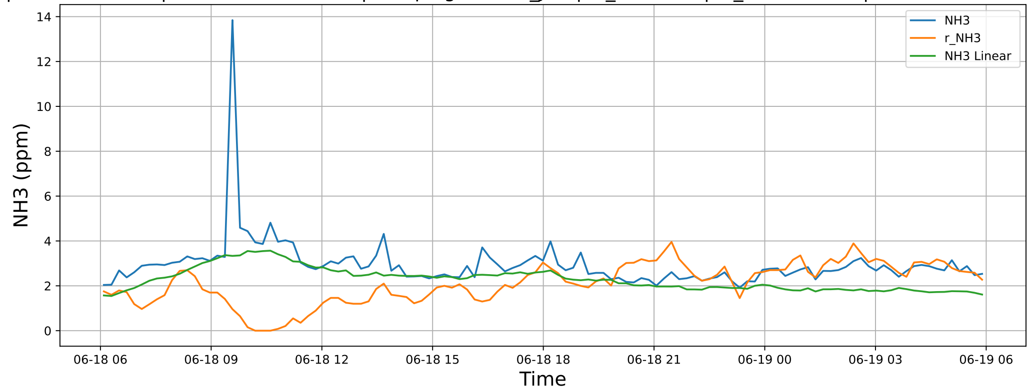




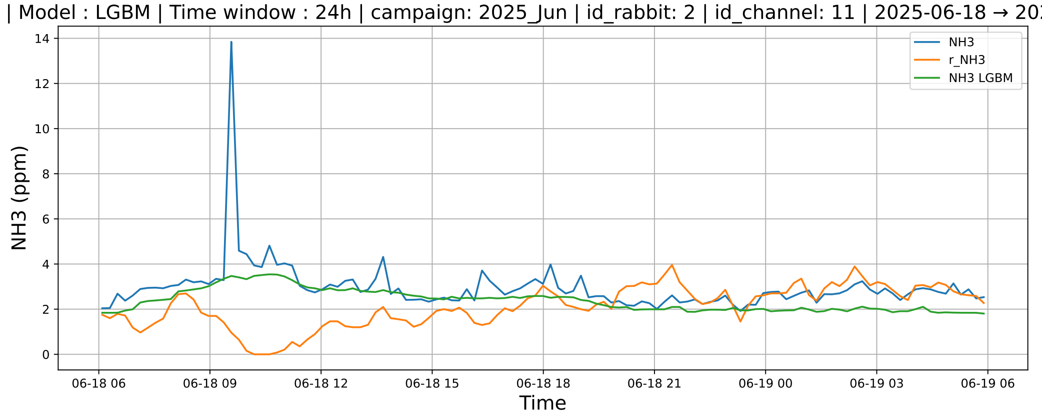
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

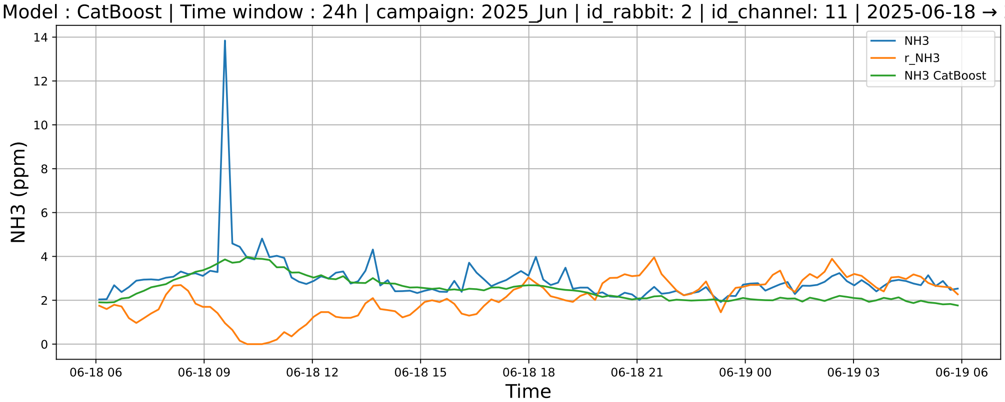
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



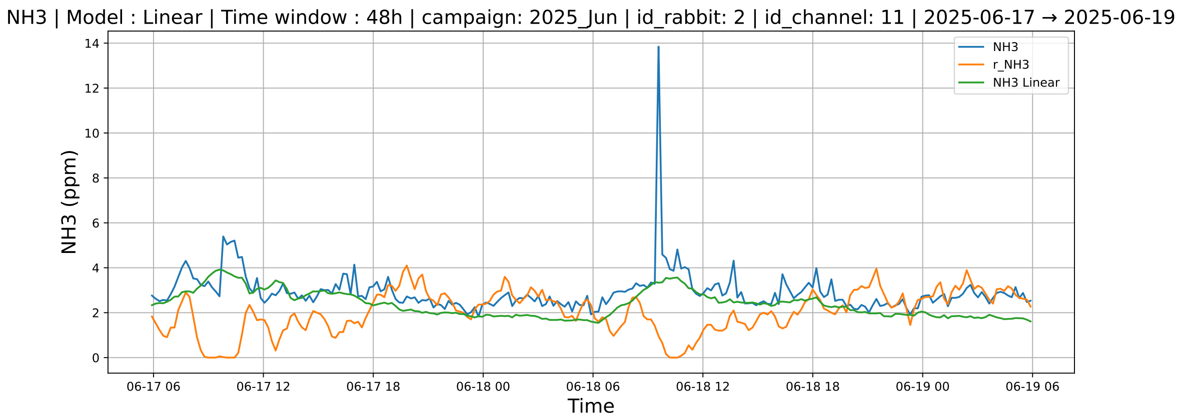
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

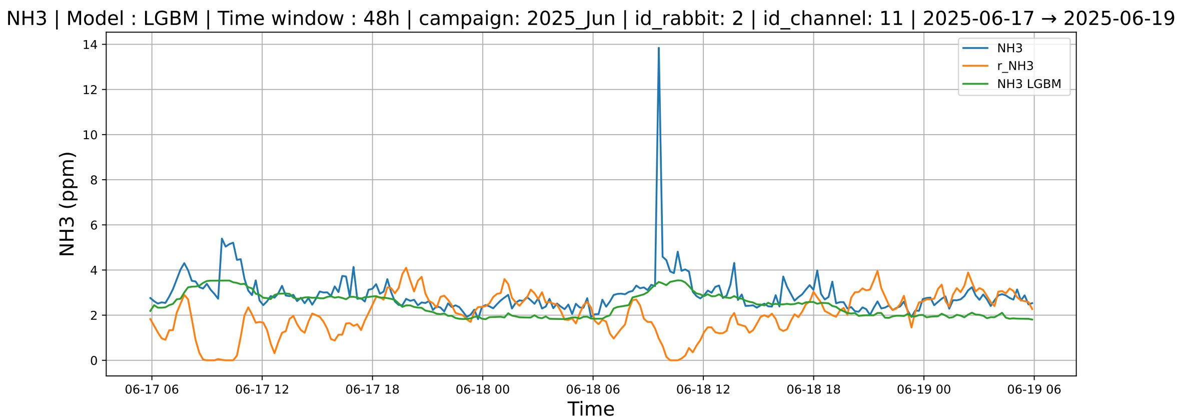


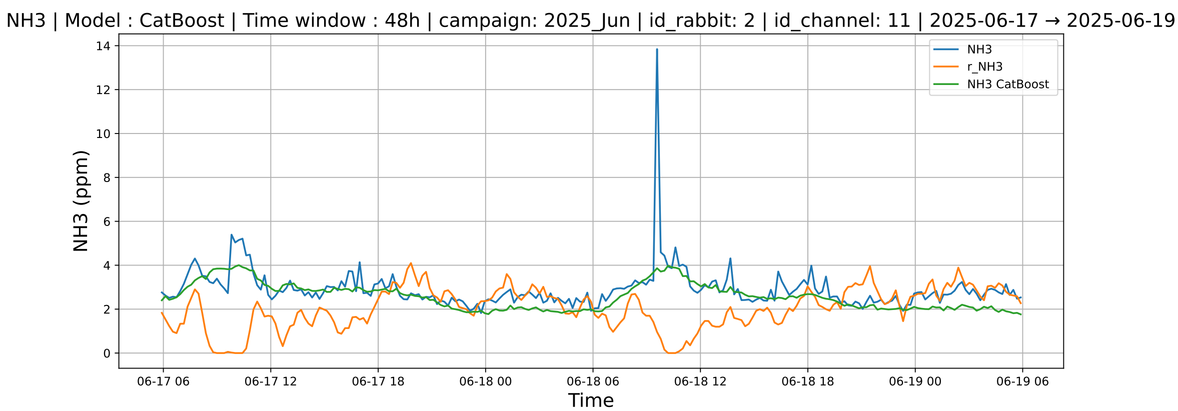
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



4.1.2.2 Time window : 48

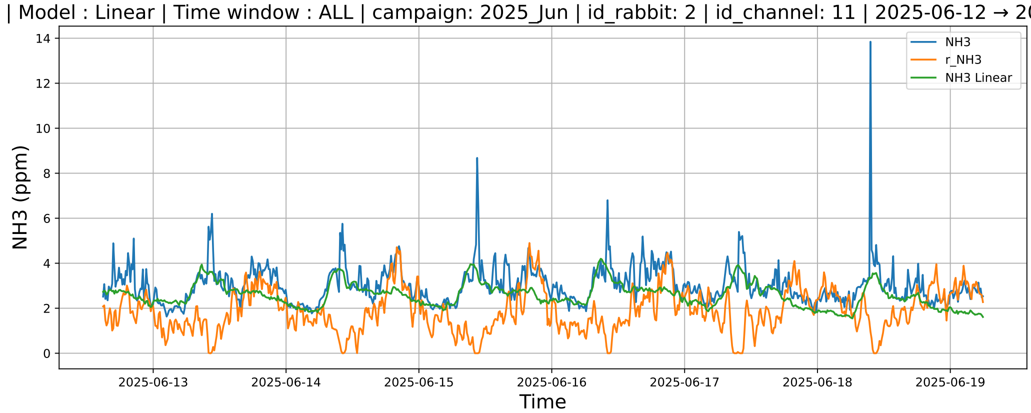




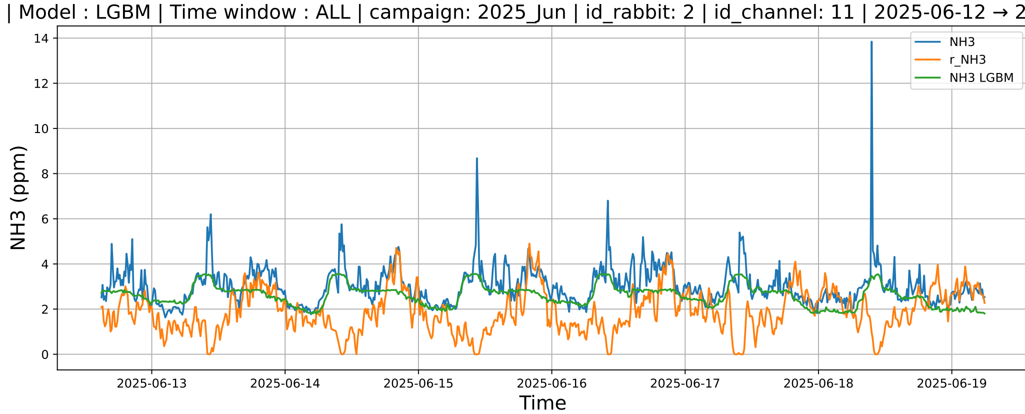


4.1.2.3 Time window : ALL

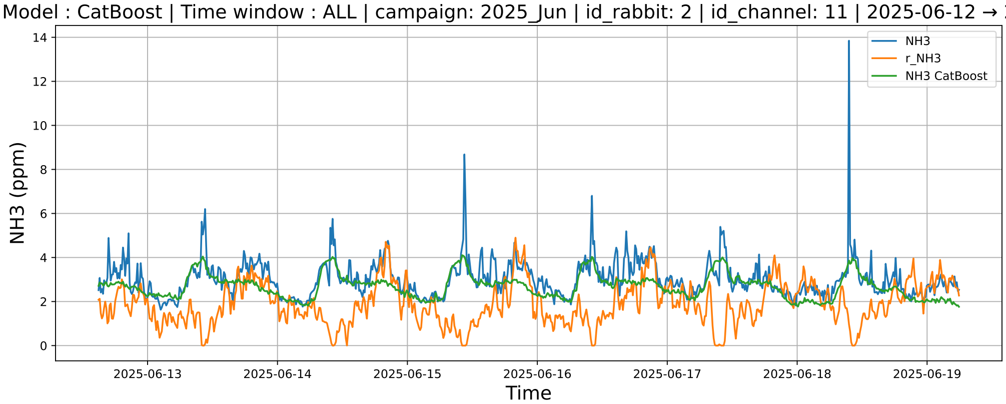
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

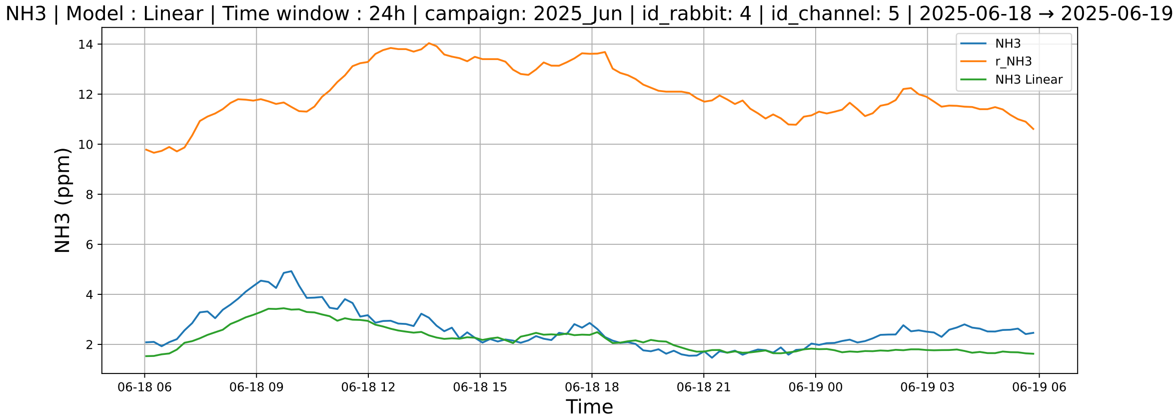


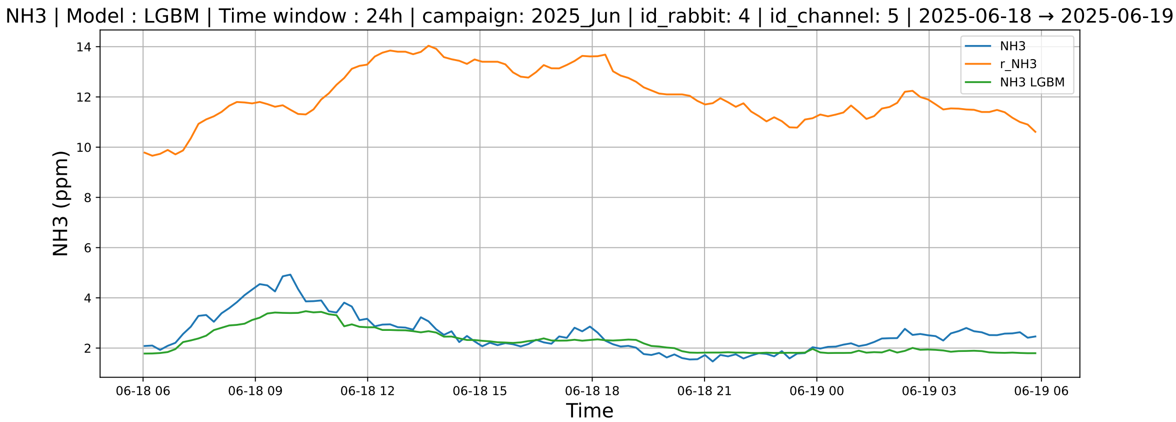
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

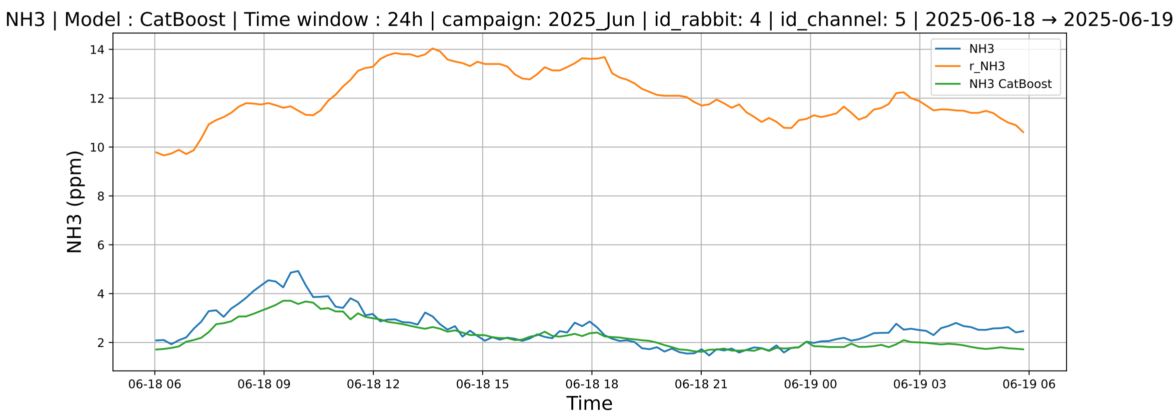


4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

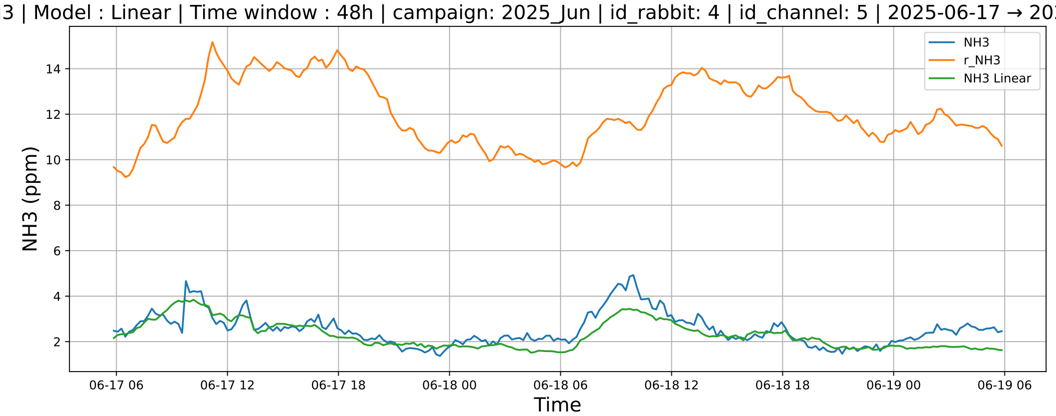




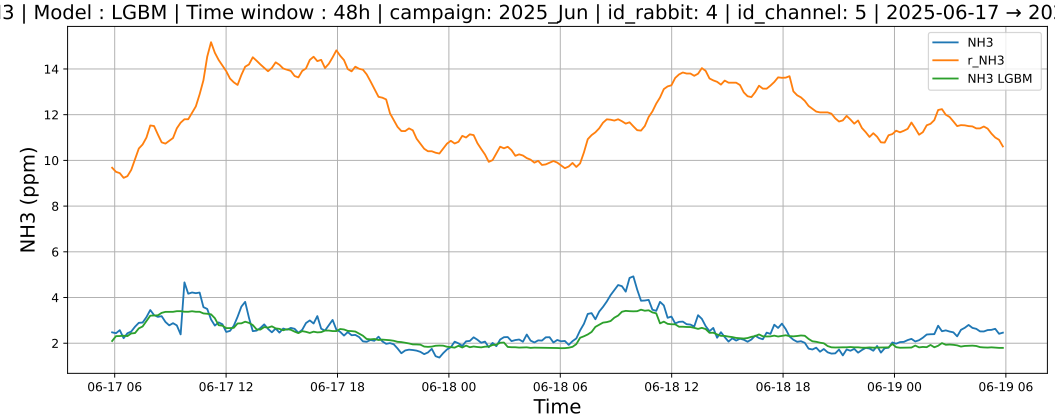


4.1.3.2 Time window : 48

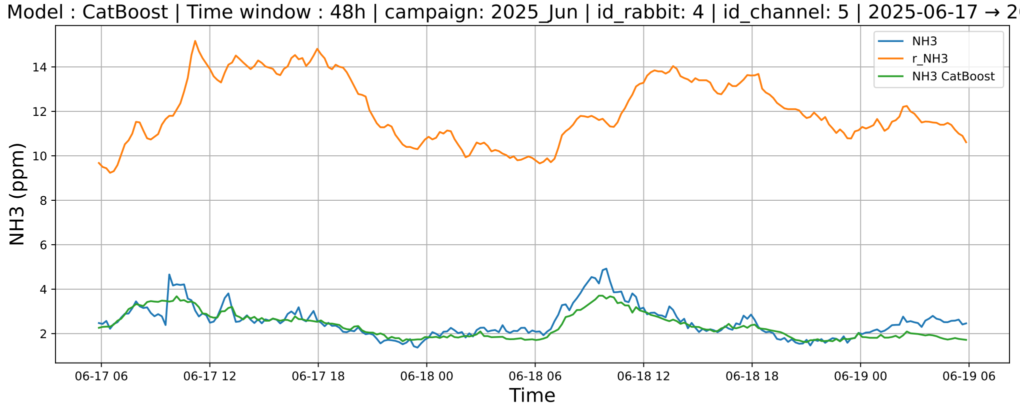
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



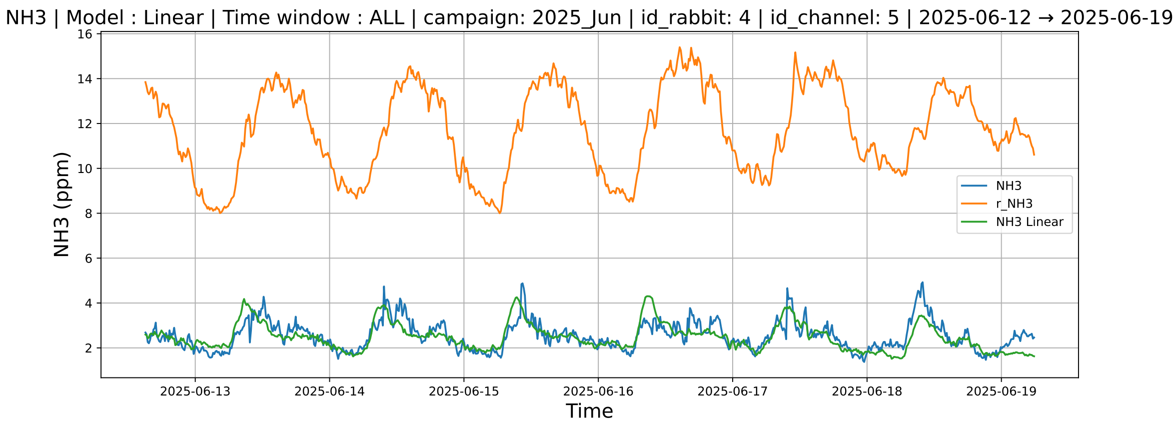
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

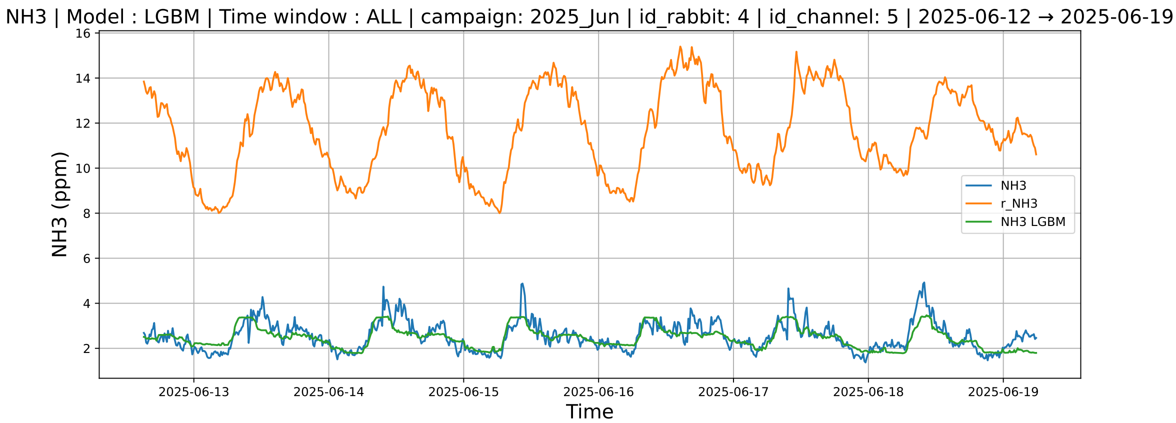


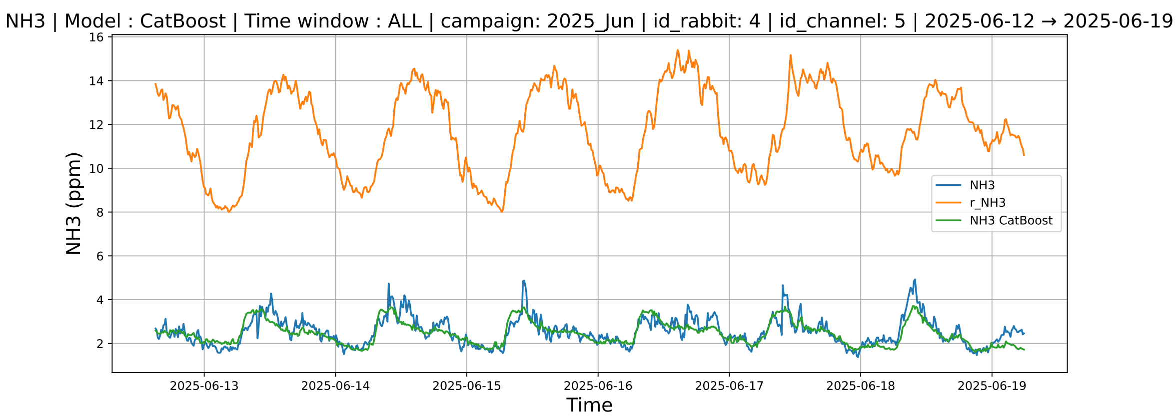
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.1.3.3 Time window : ALL

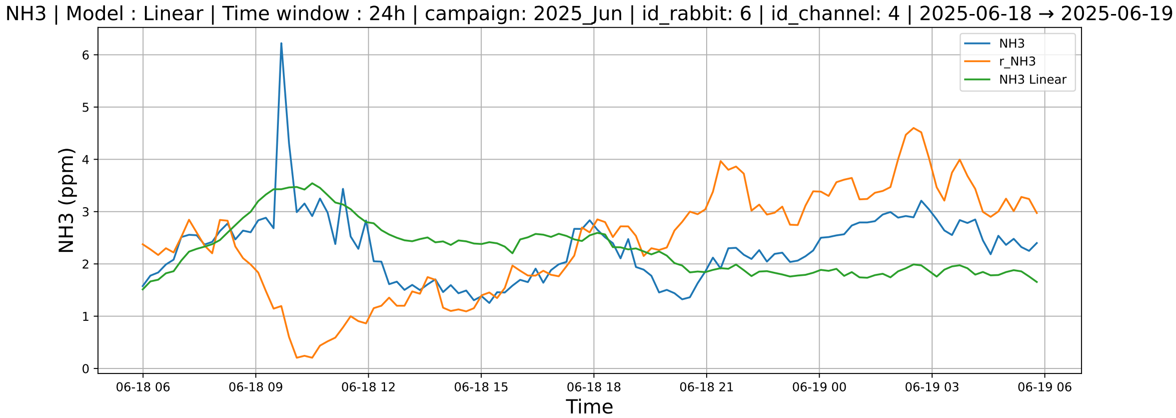


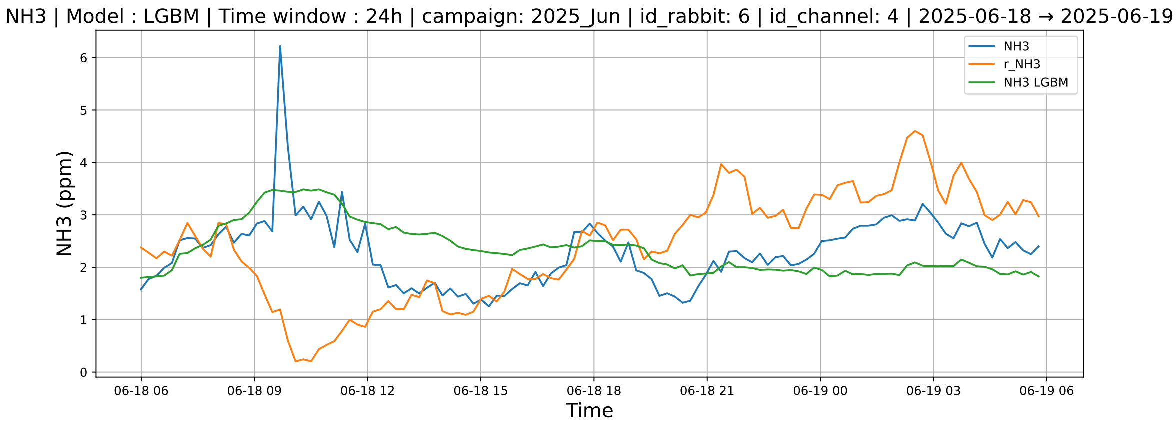


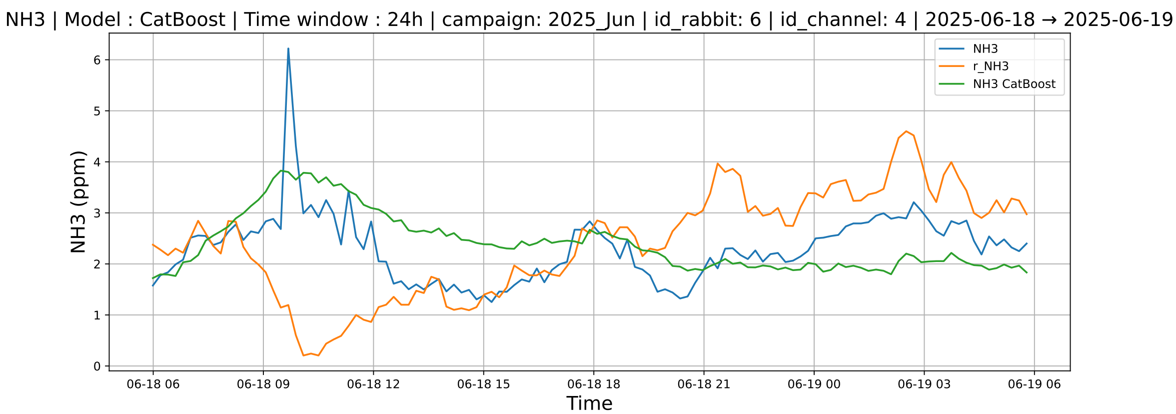


4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

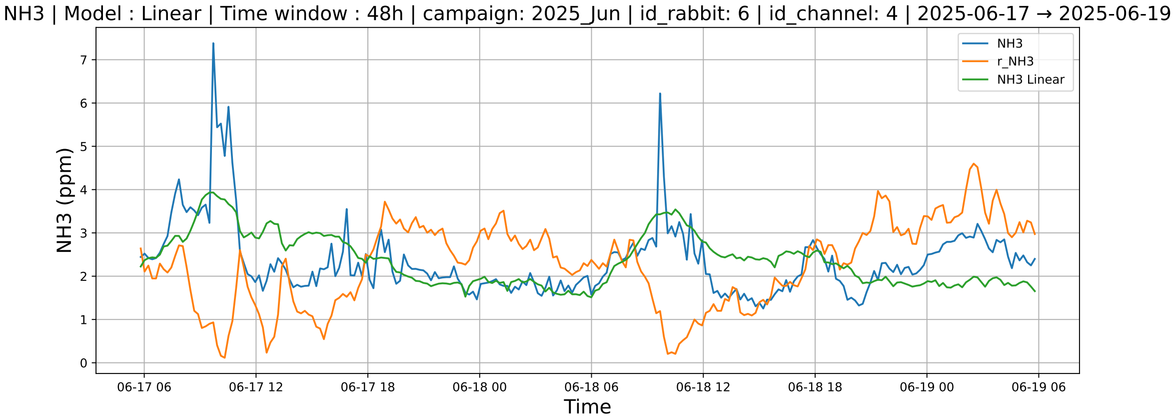
4.1.4.1 Time window : 24

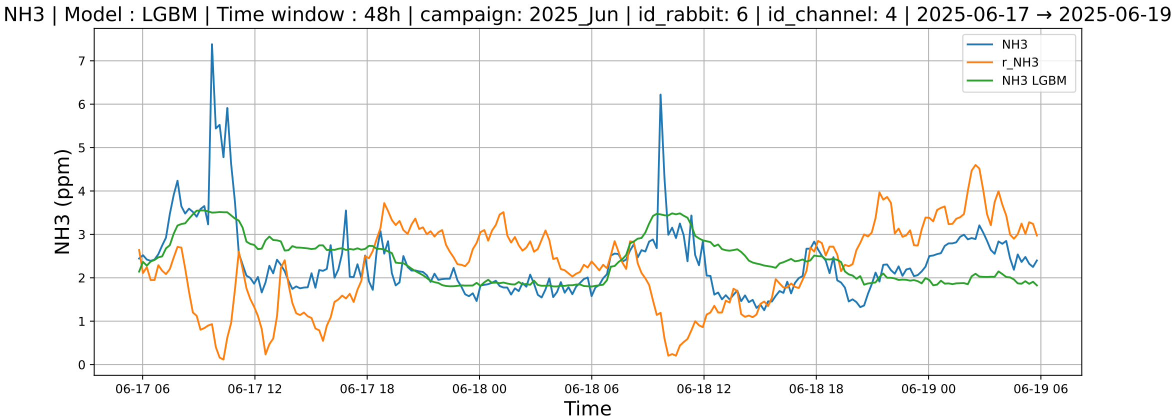


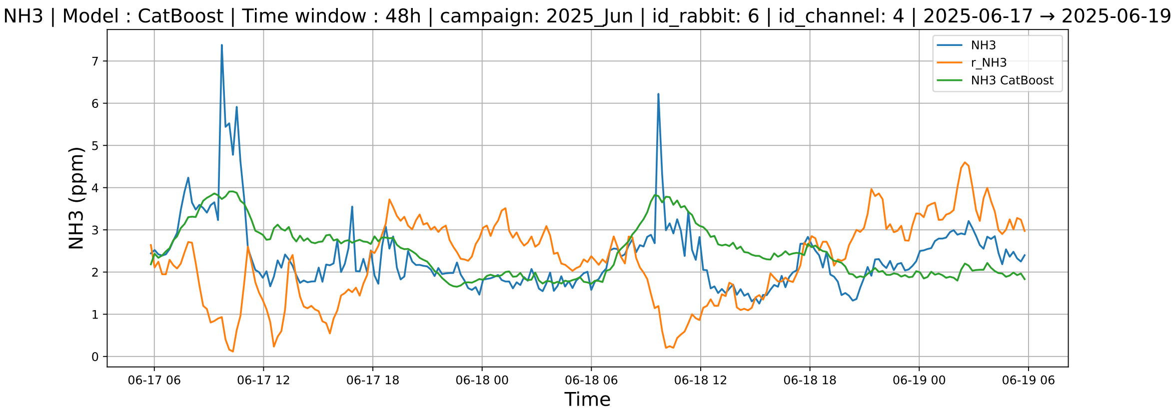




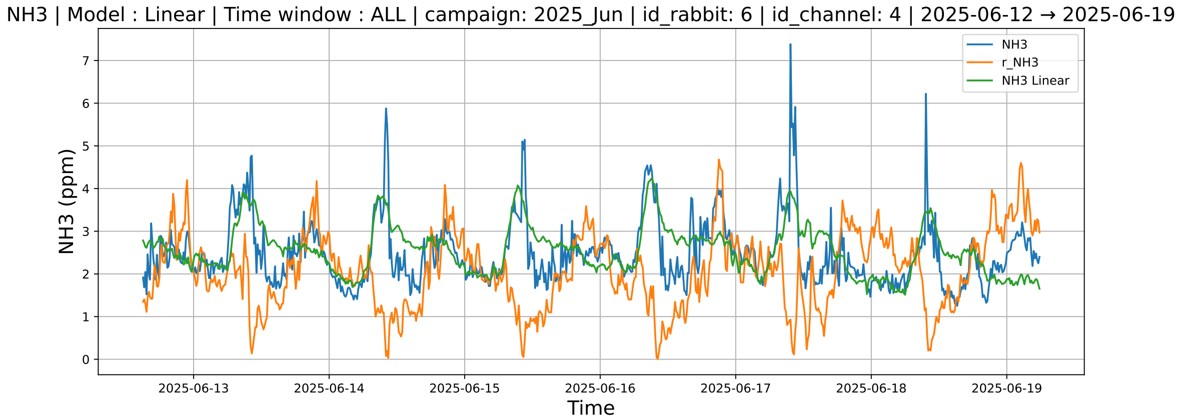
4.1.4.2 Time window : 48

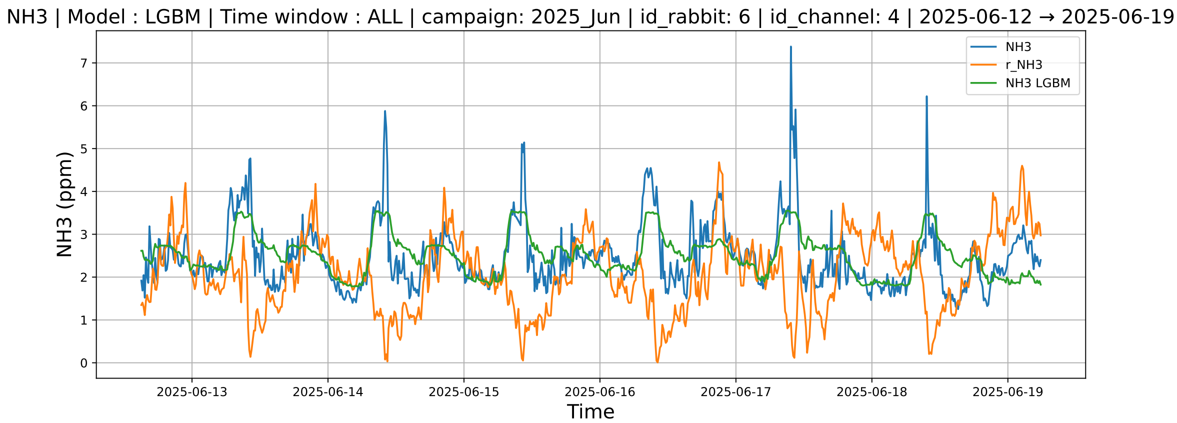


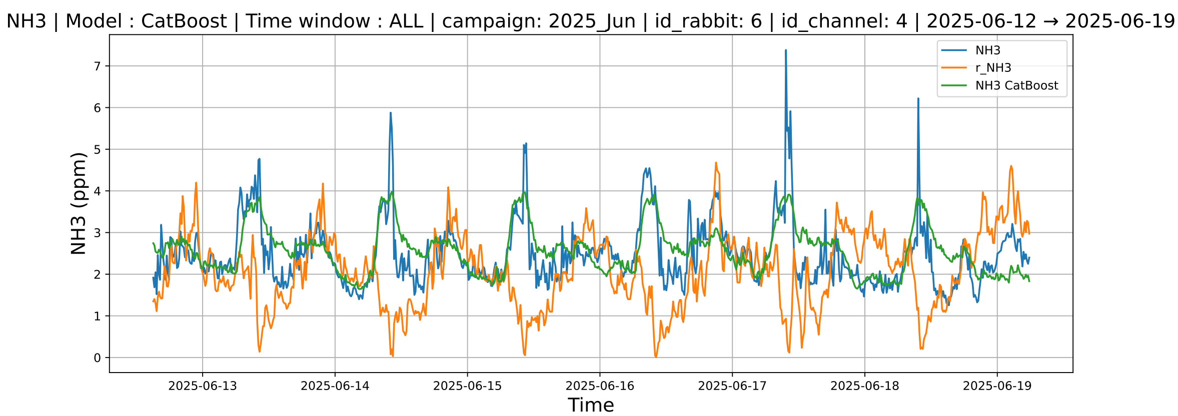




4.1.4.3 Time window : ALL

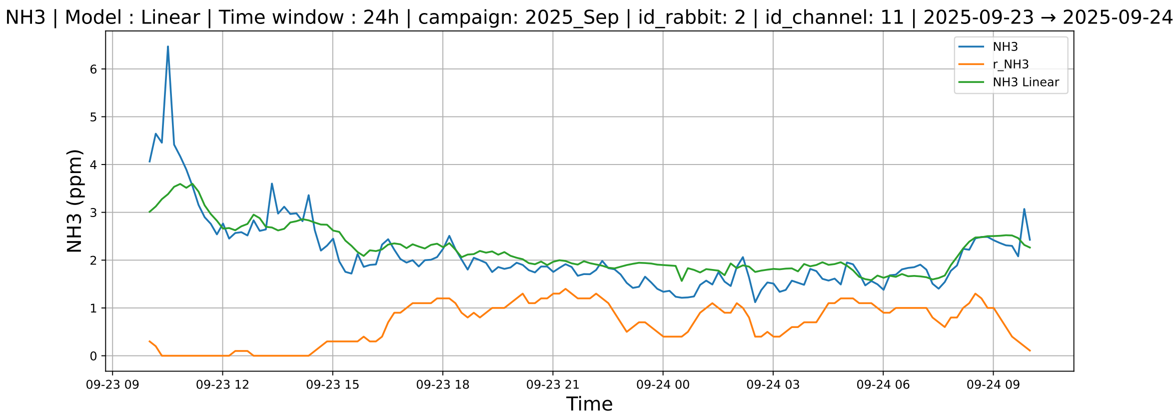


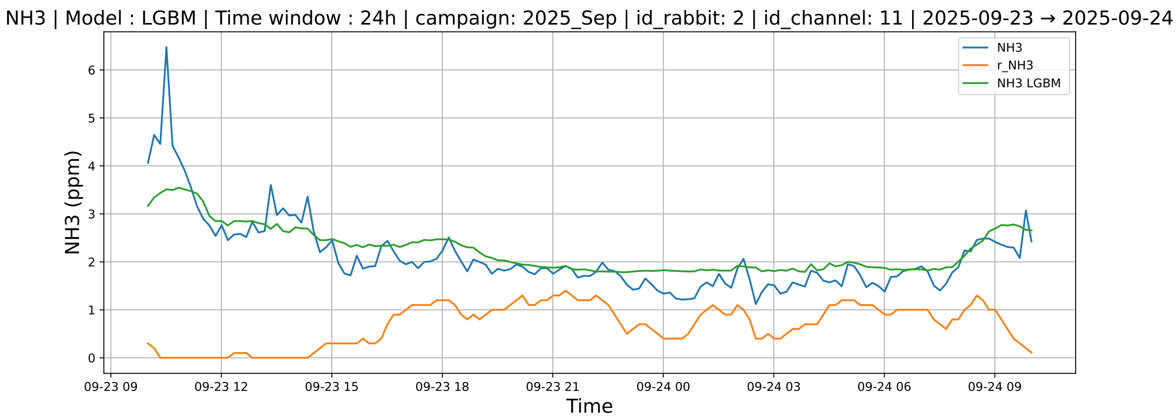


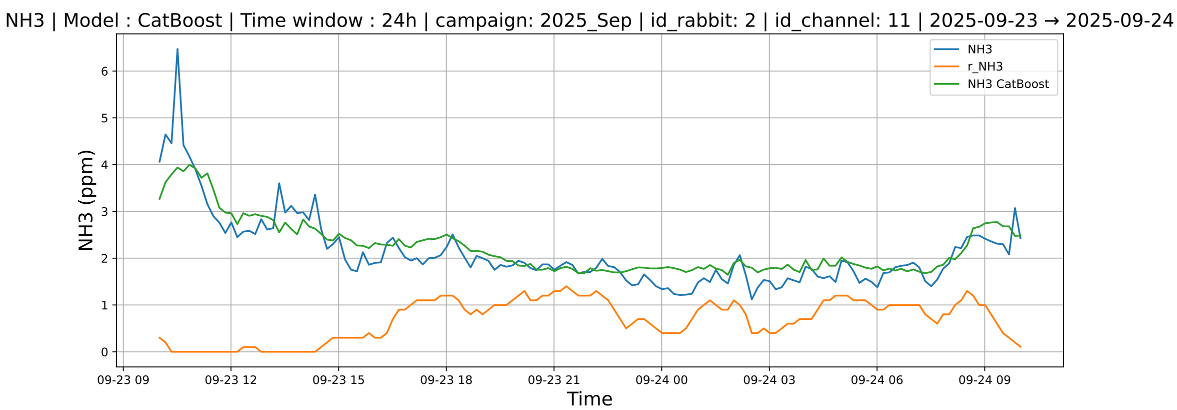


4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

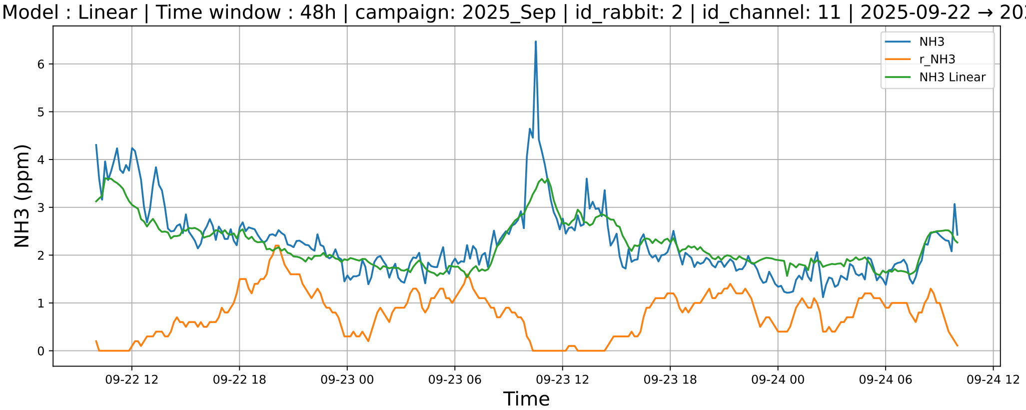




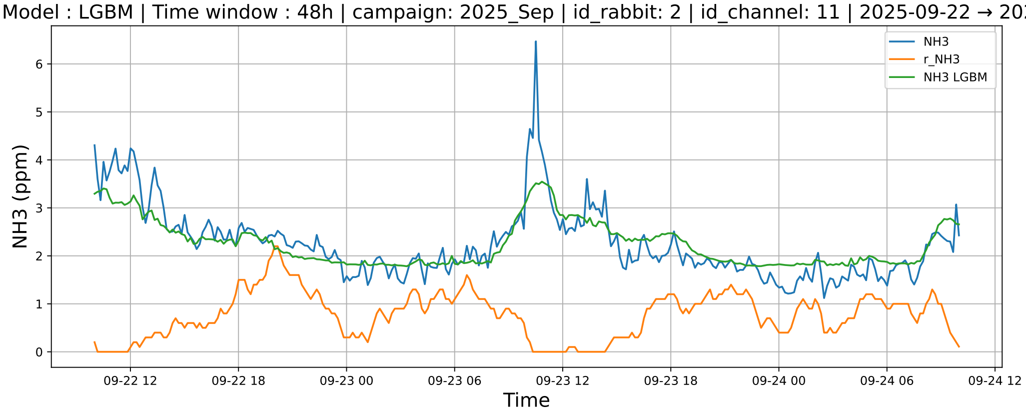


4.1.5.2 Time window : 48

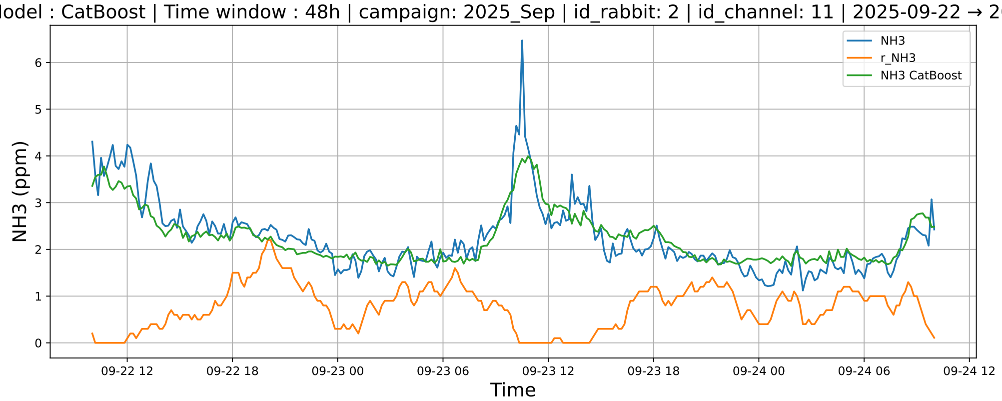
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

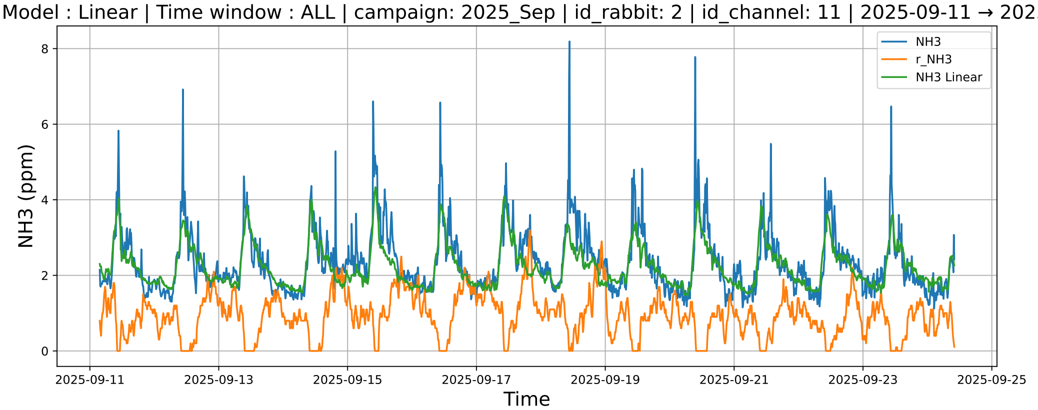


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

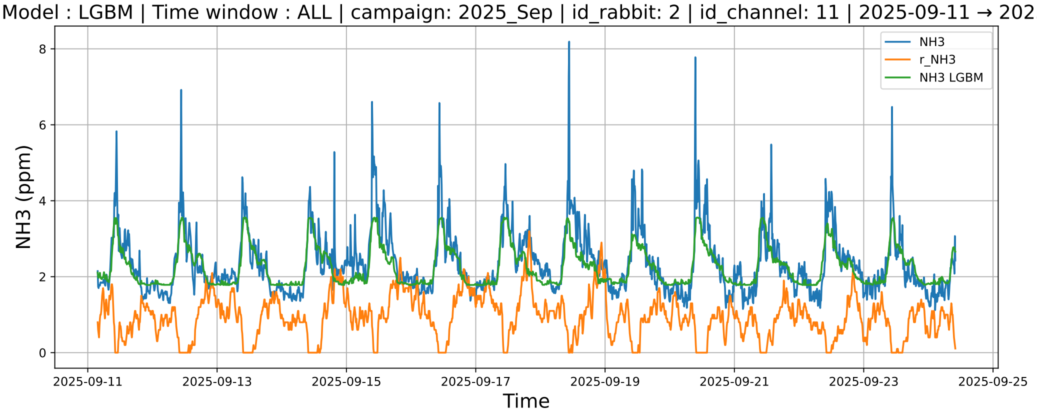


4.1.5.3 Time window : ALL

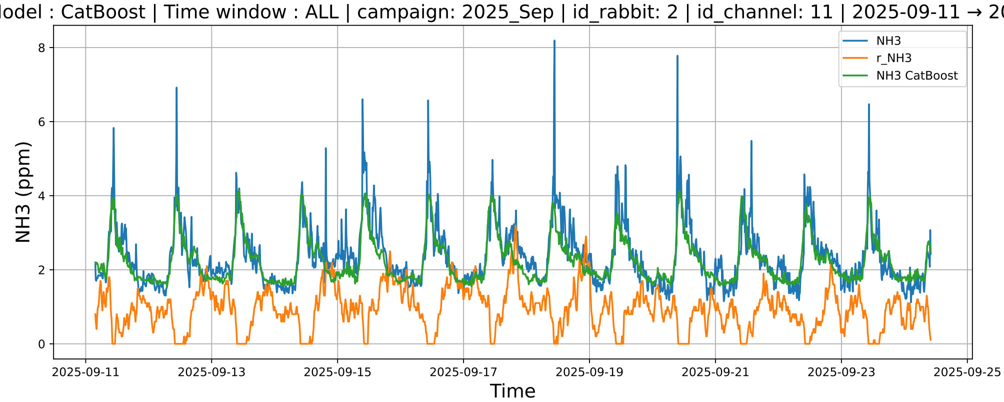
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24

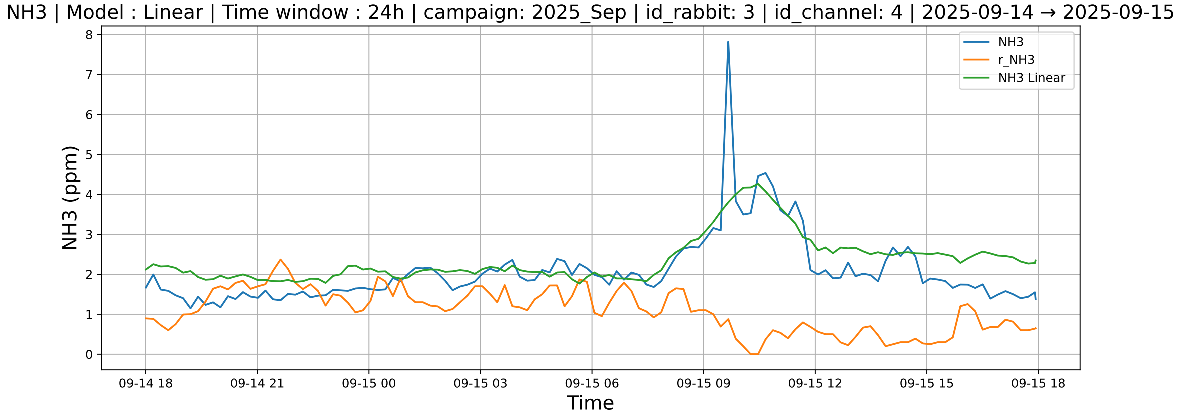


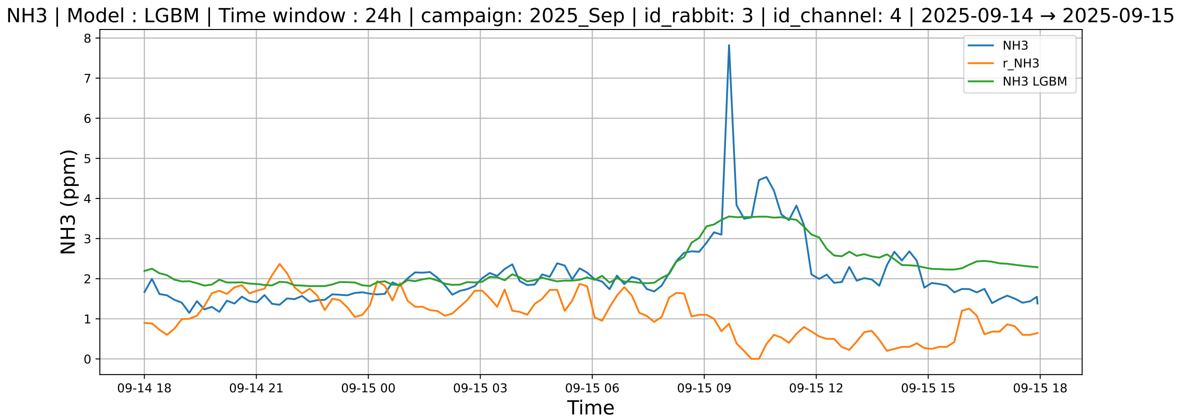
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24

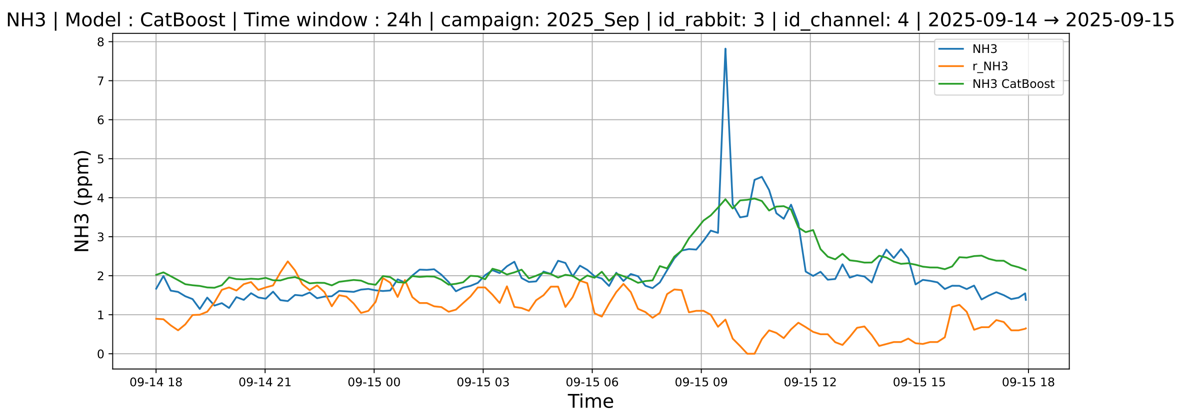


4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

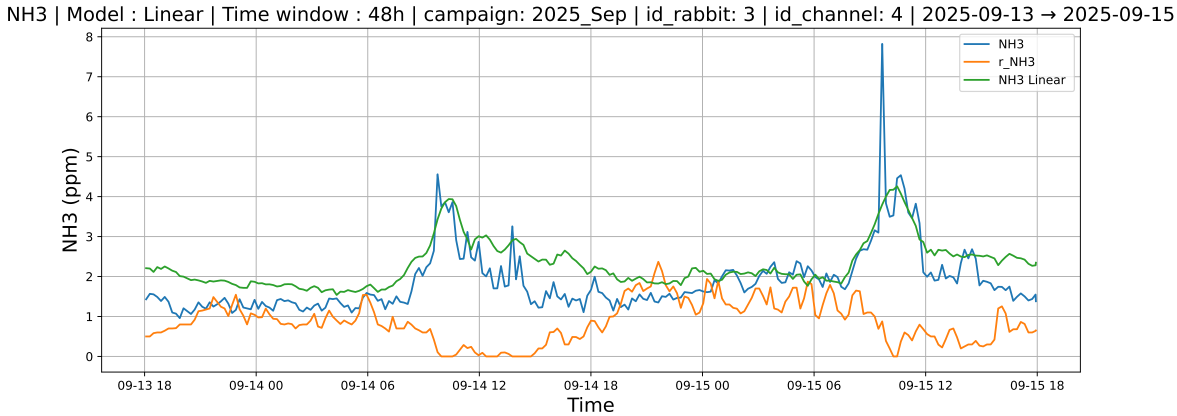
4.1.6.1 Time window : 24

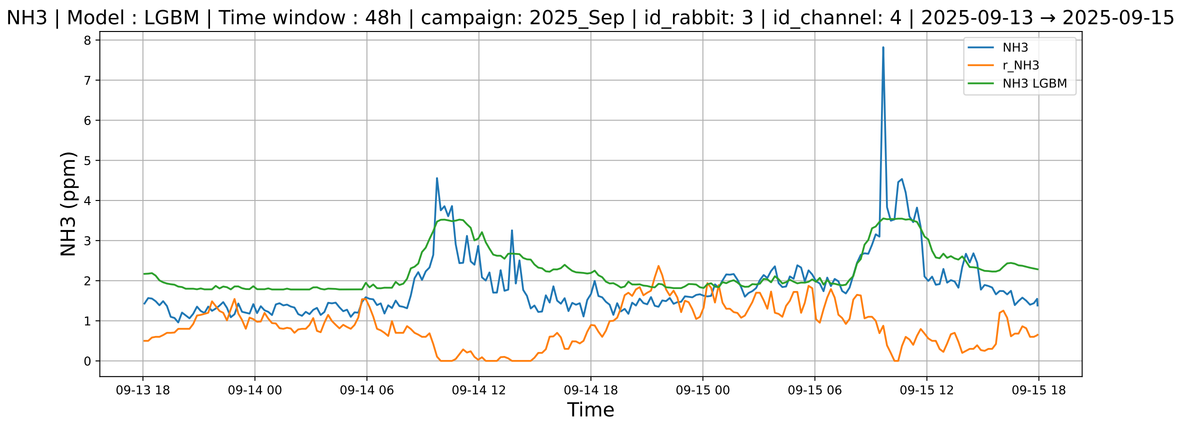


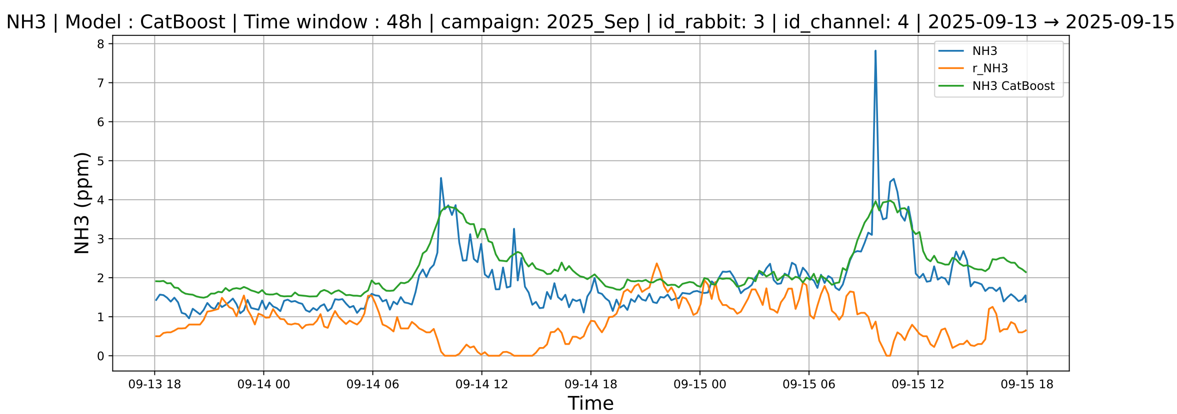




4.1.6.2 Time window : 48

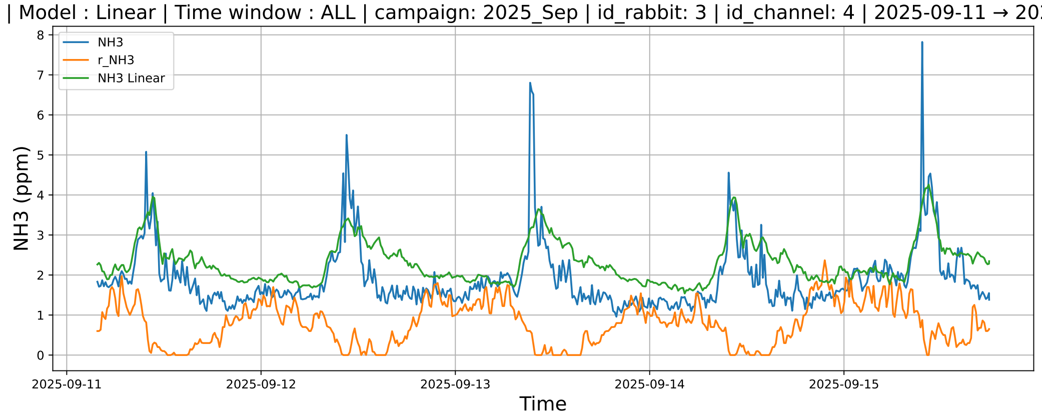




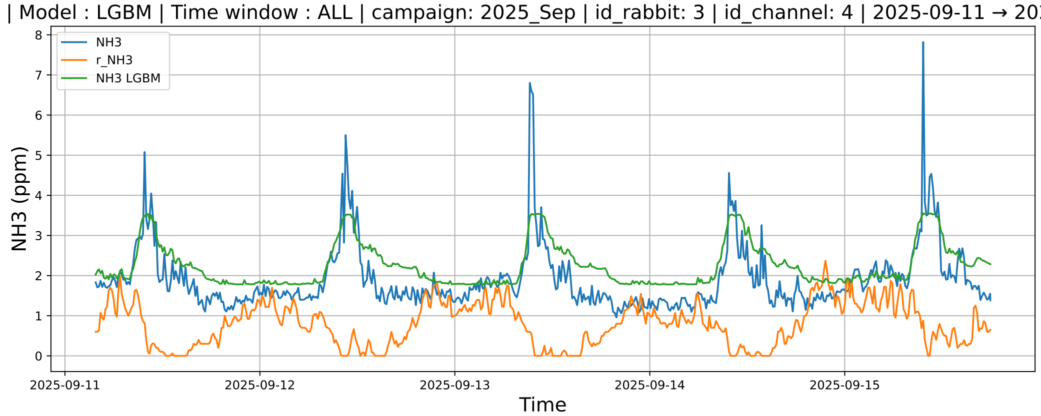


4.1.6.3 Time window : ALL

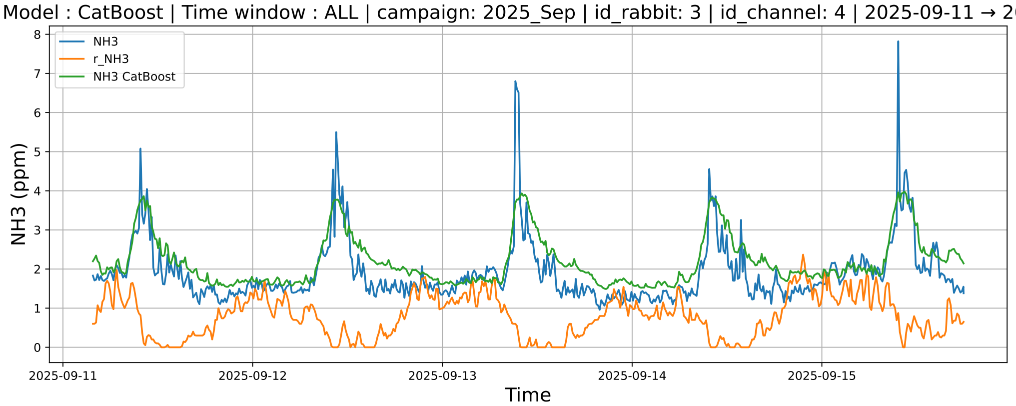
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

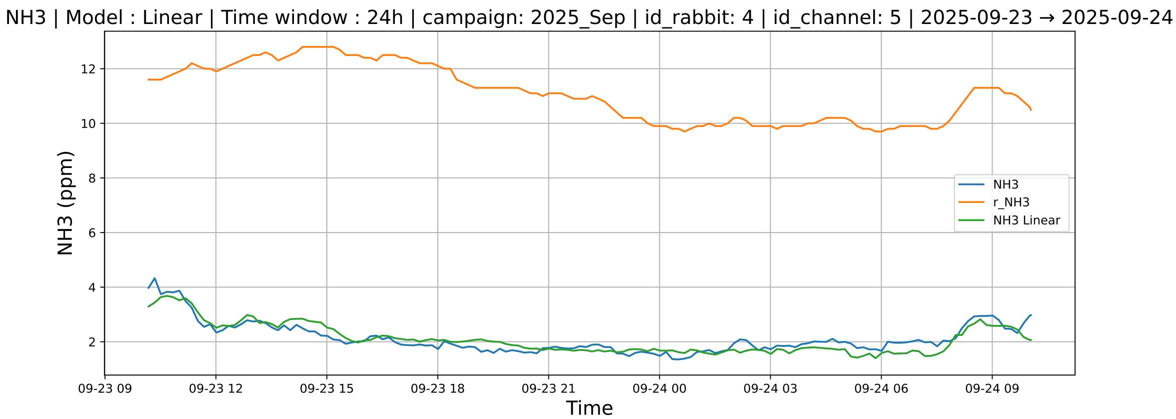


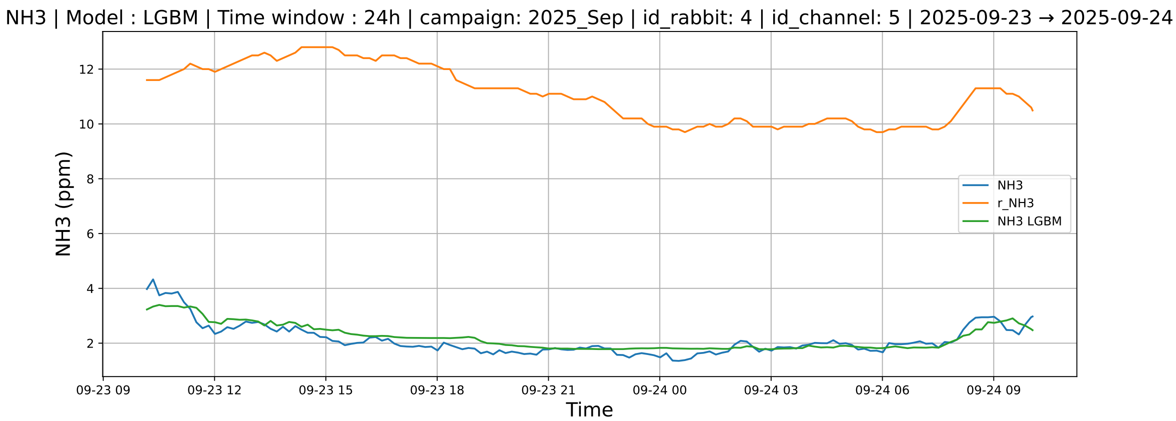
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

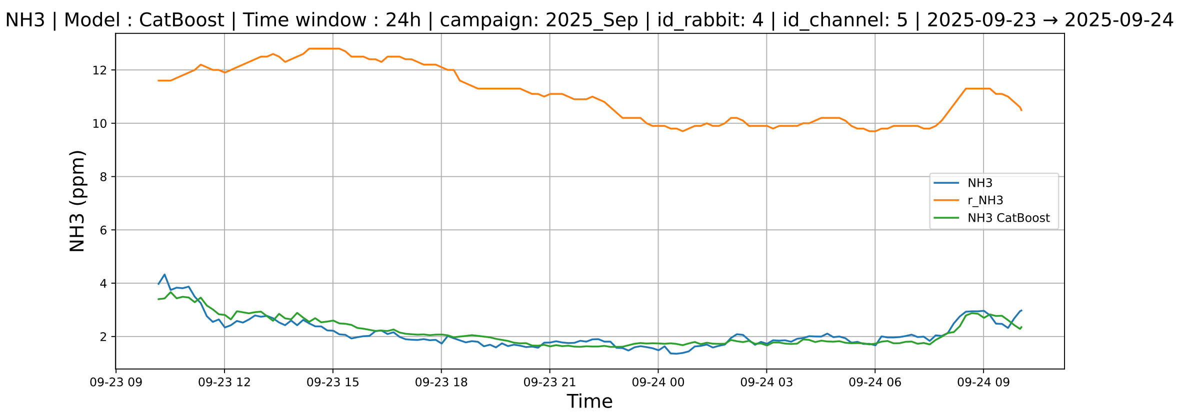


4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

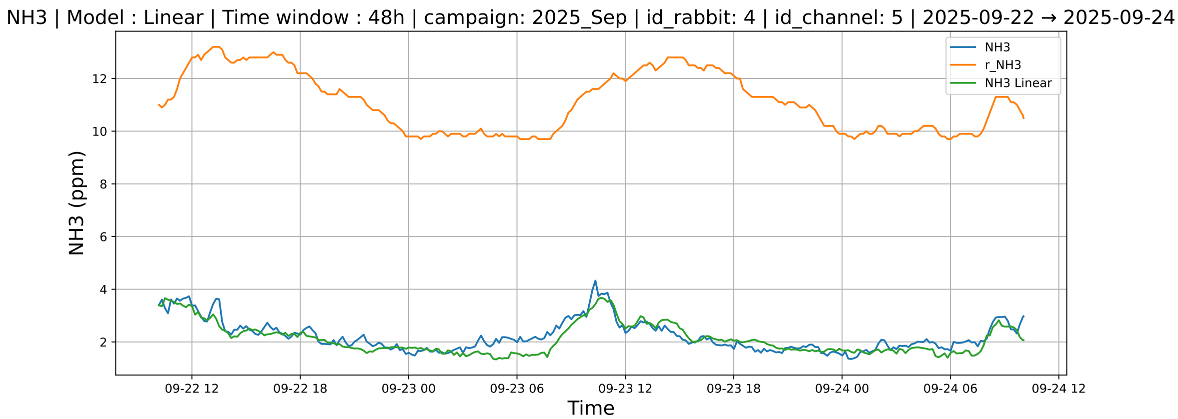
4.1.7.1 Time window : 24

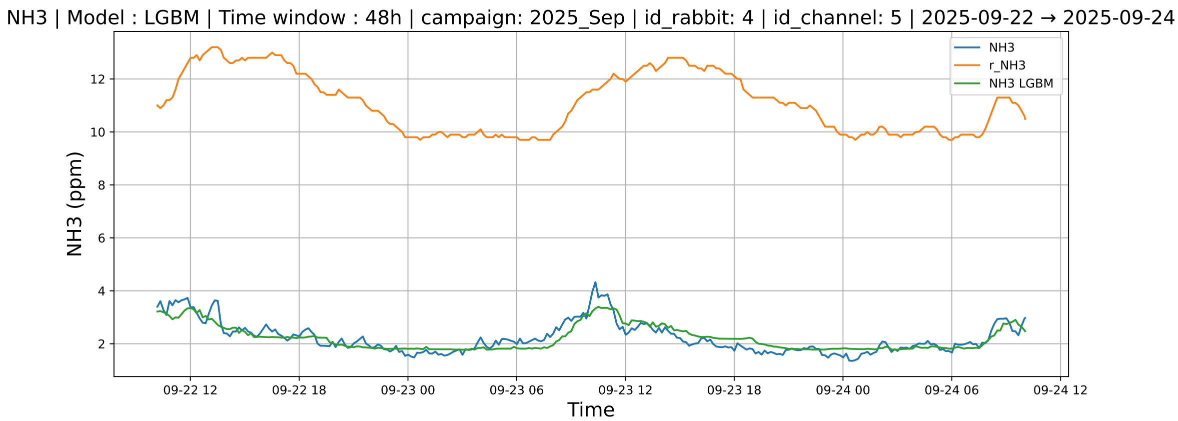


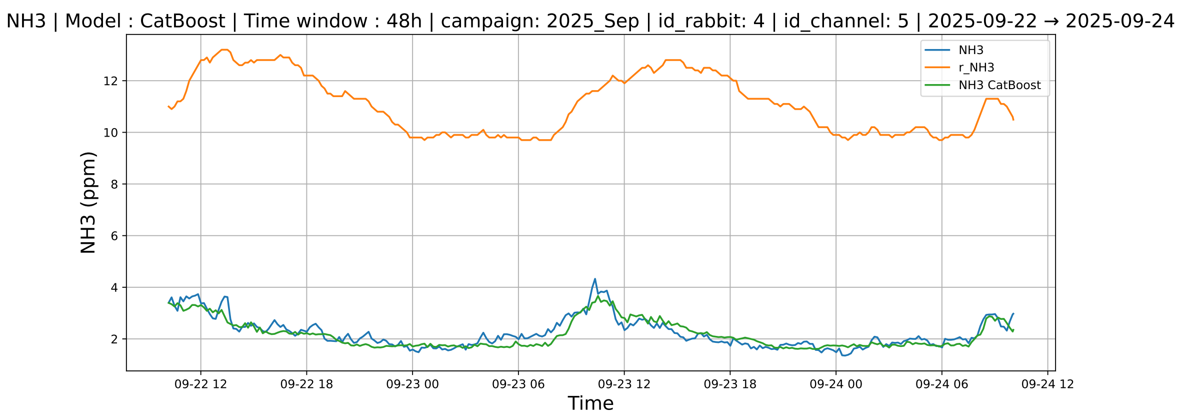




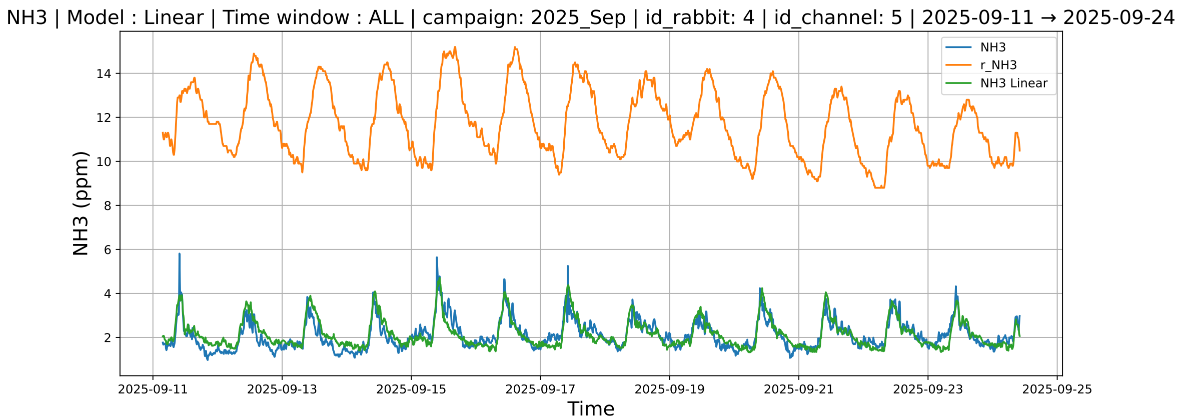
4.1.7.2 Time window : 48

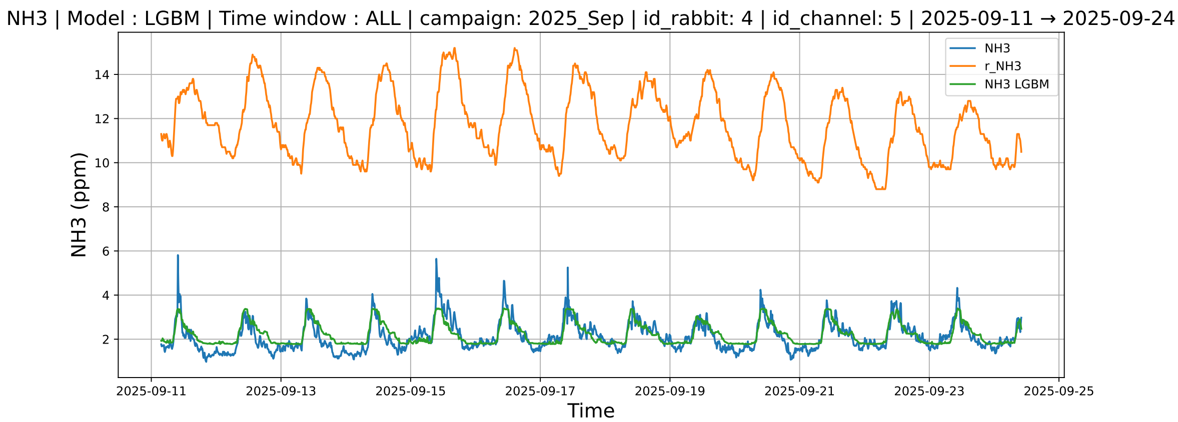


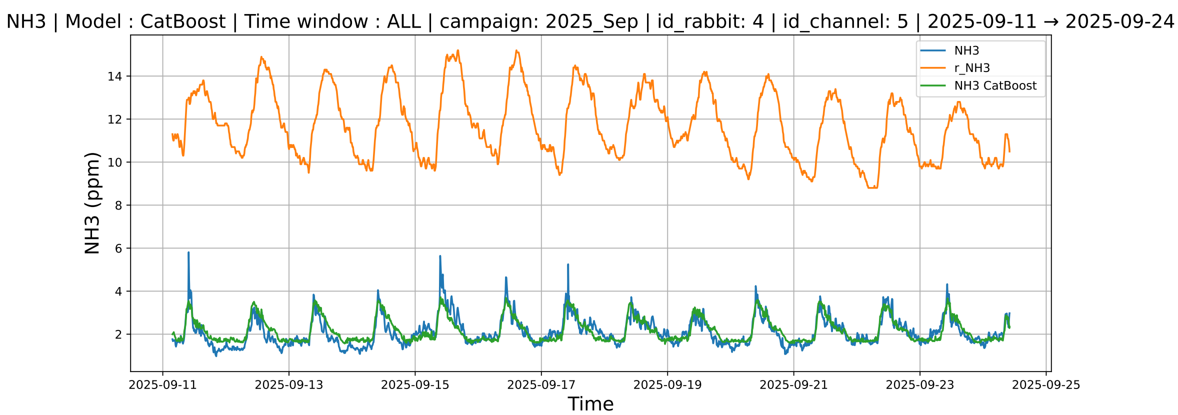




4.1.7.3 Time window : ALL

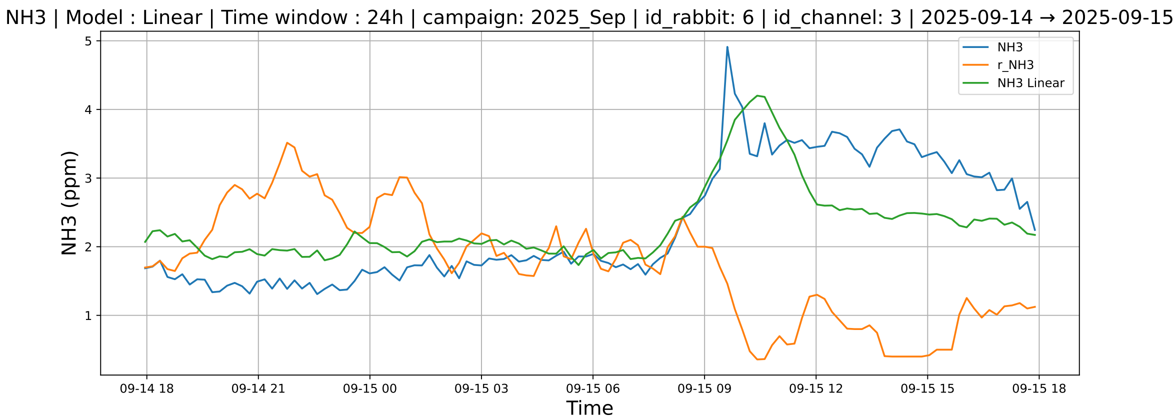


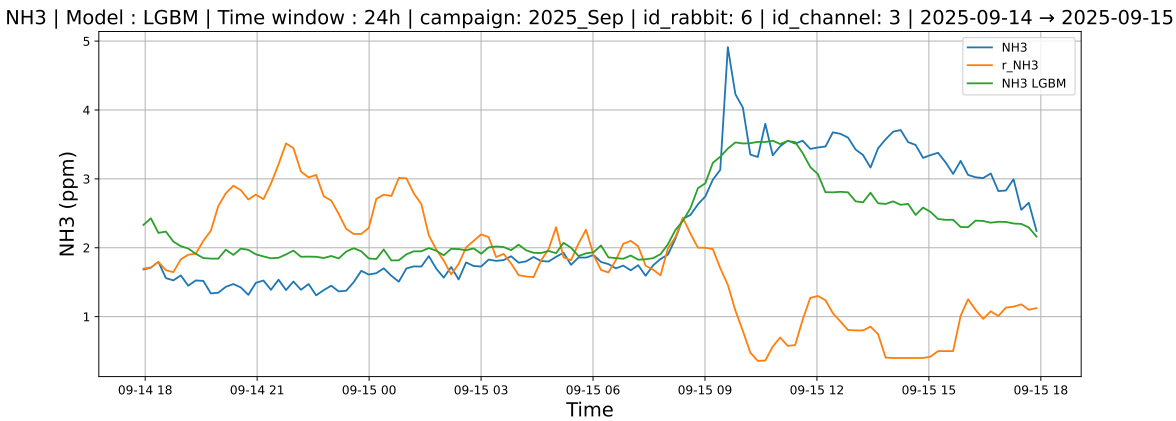


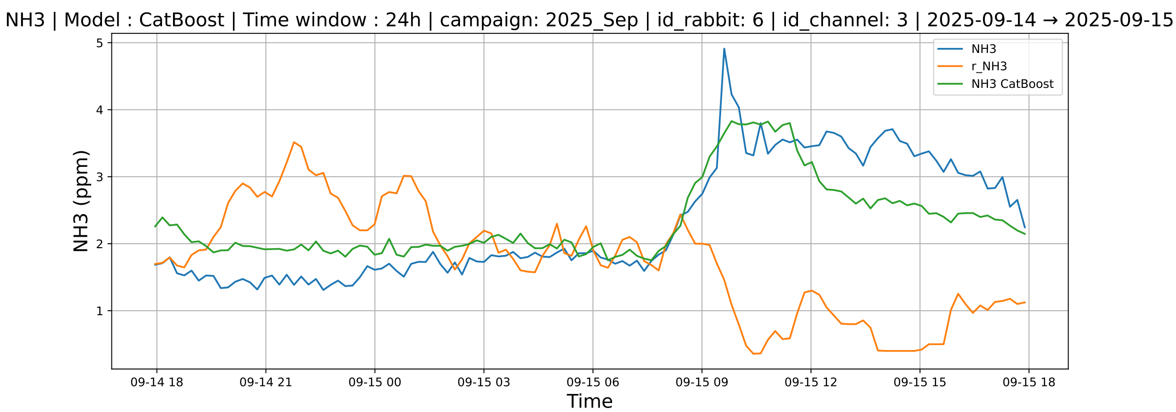


4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

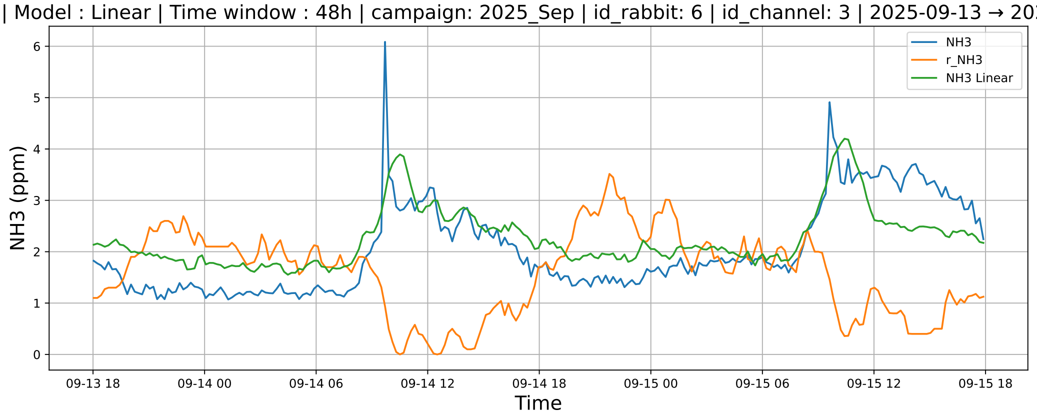




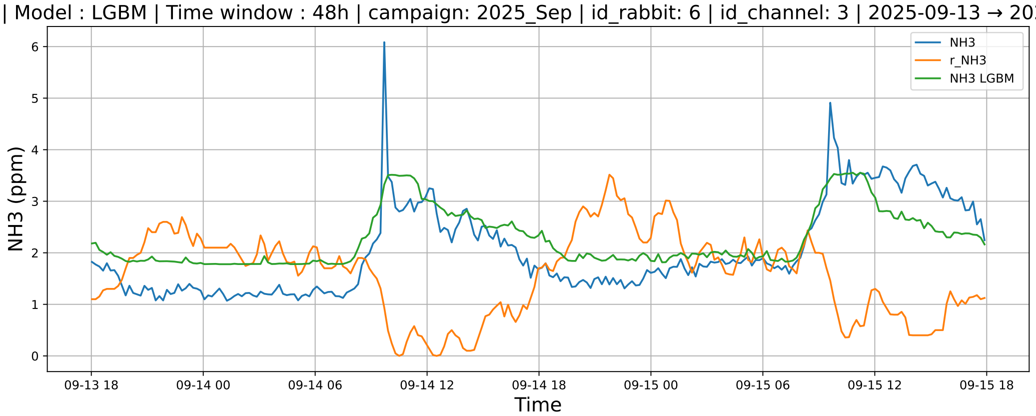


4.1.8.2 Time window : 48

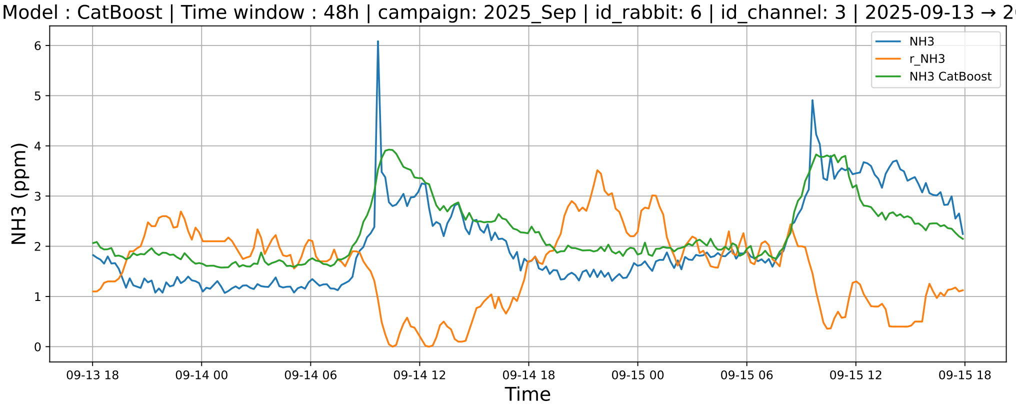
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



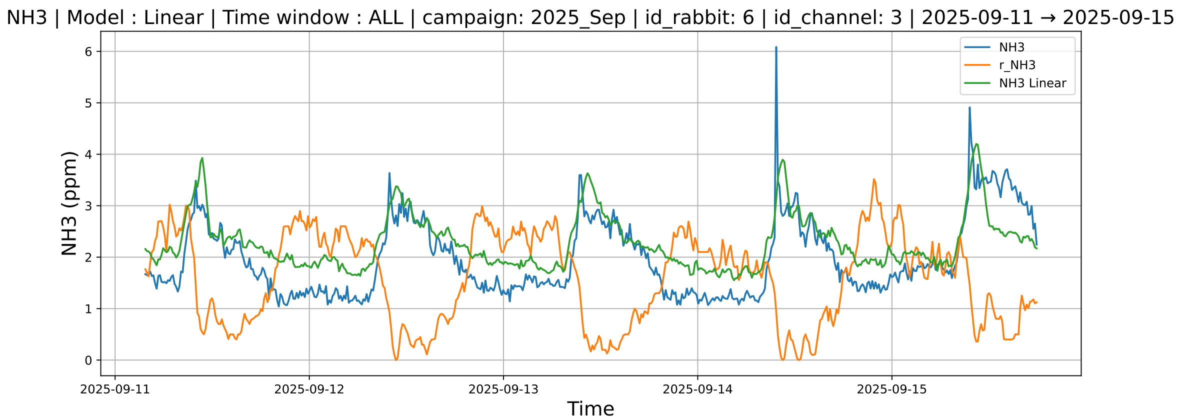
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

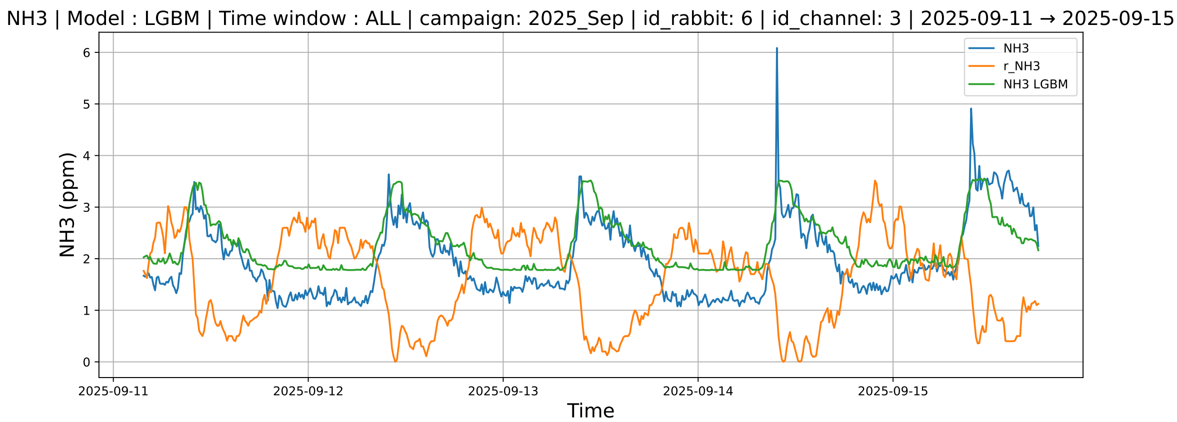


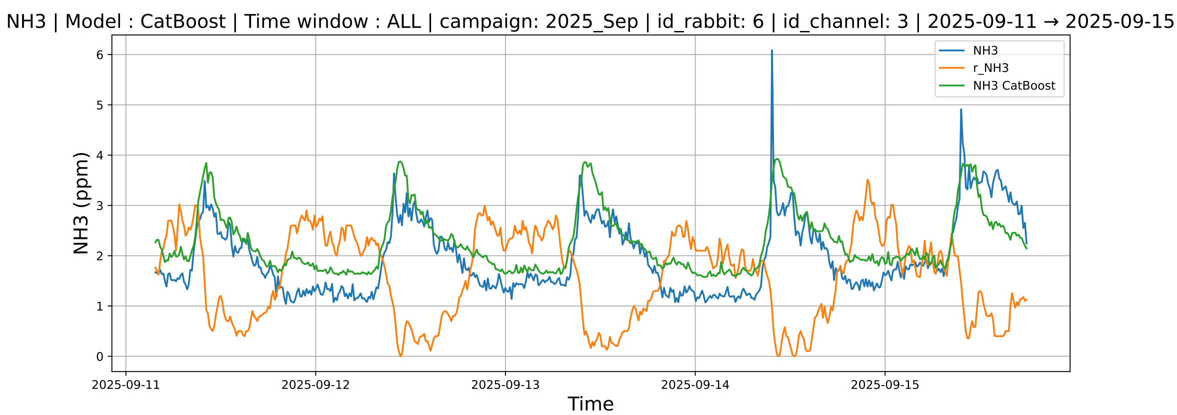
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.1.8.3 Time window : ALL





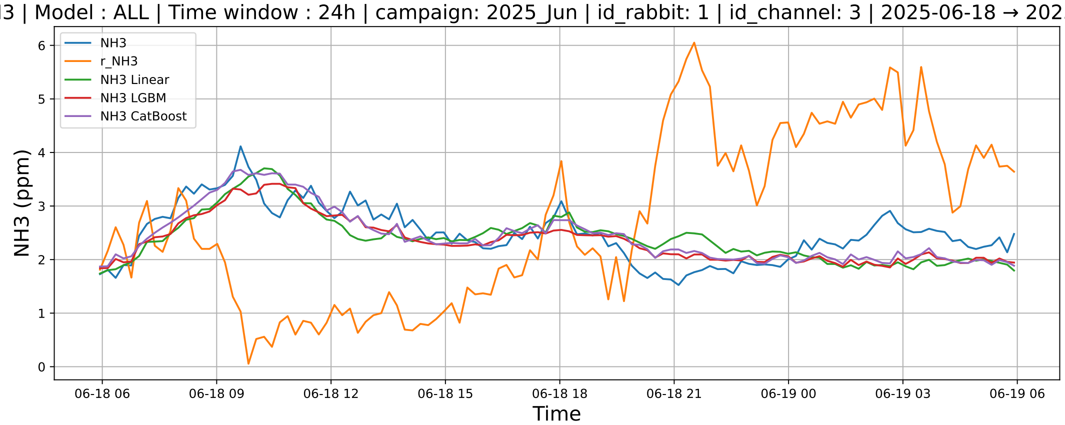


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

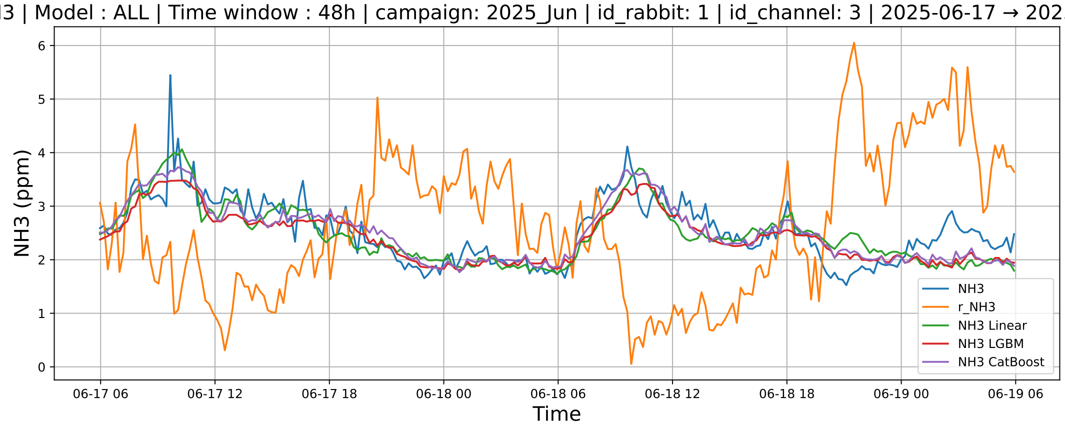
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



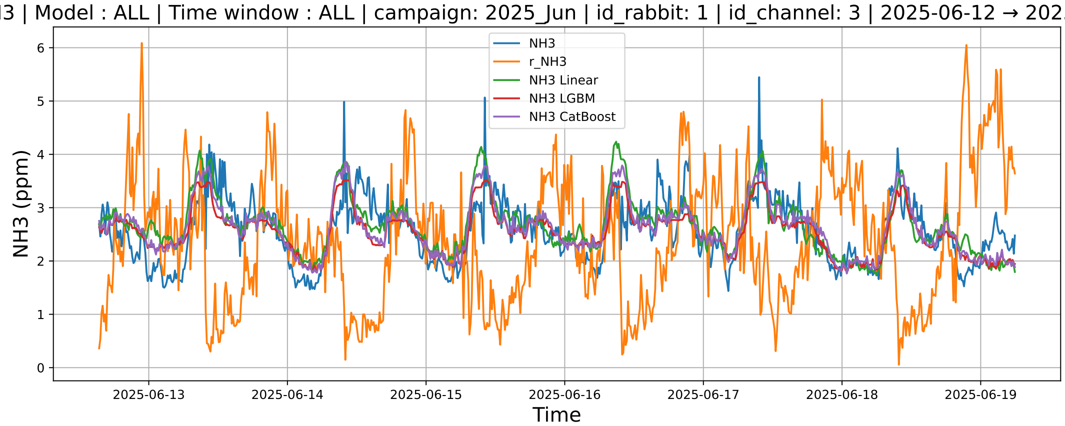
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

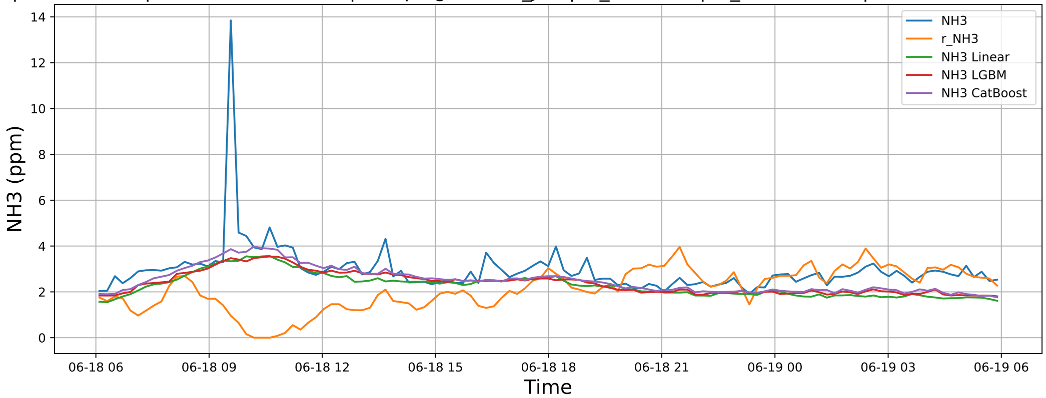
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

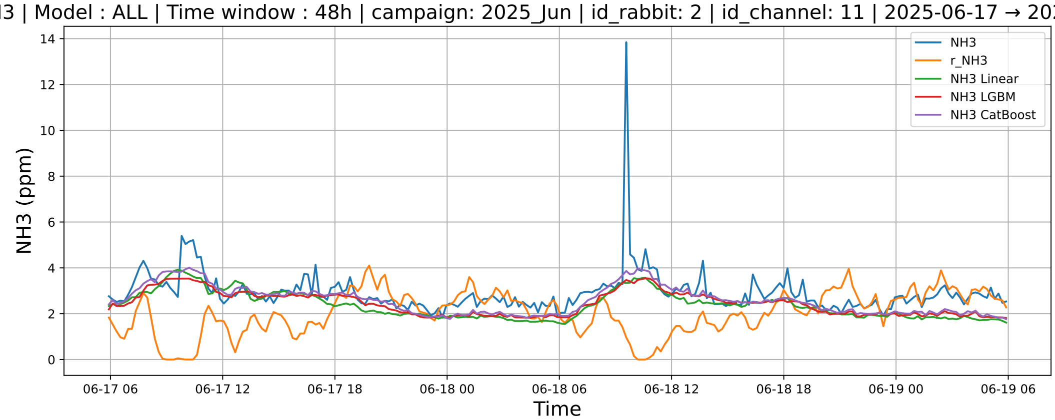
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

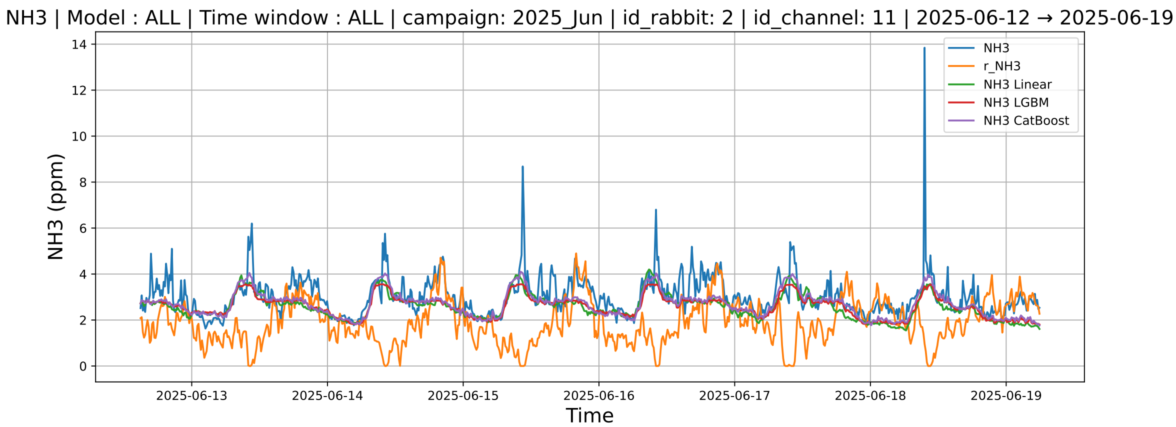


4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

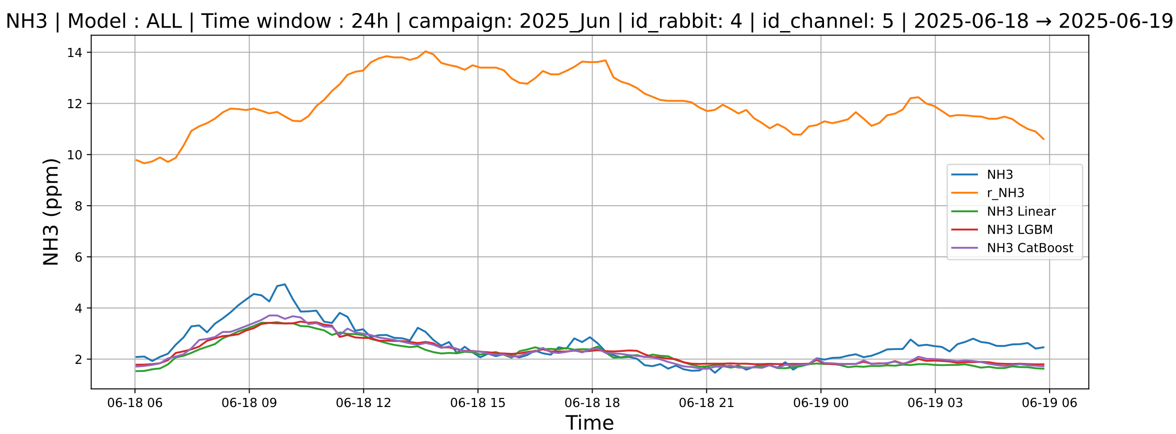


4.2.2.3 Time window : ALL

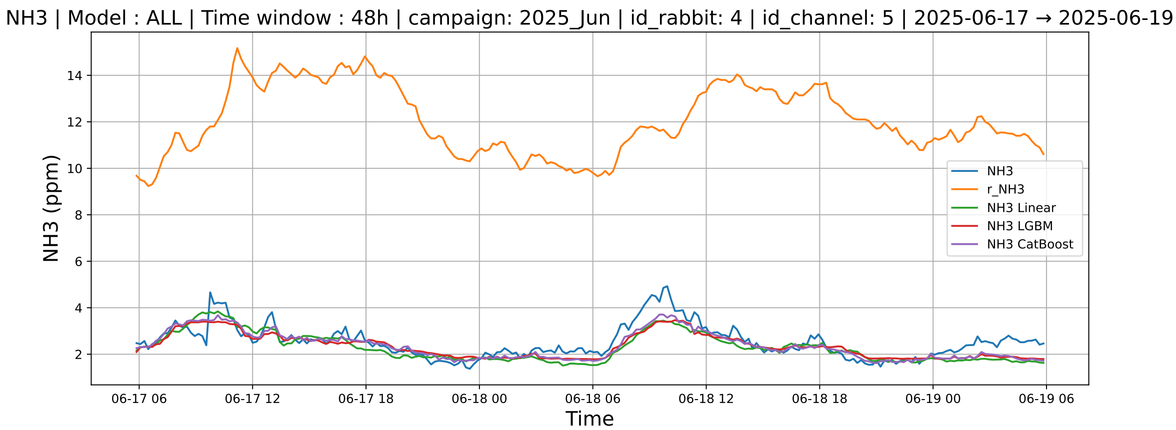


4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

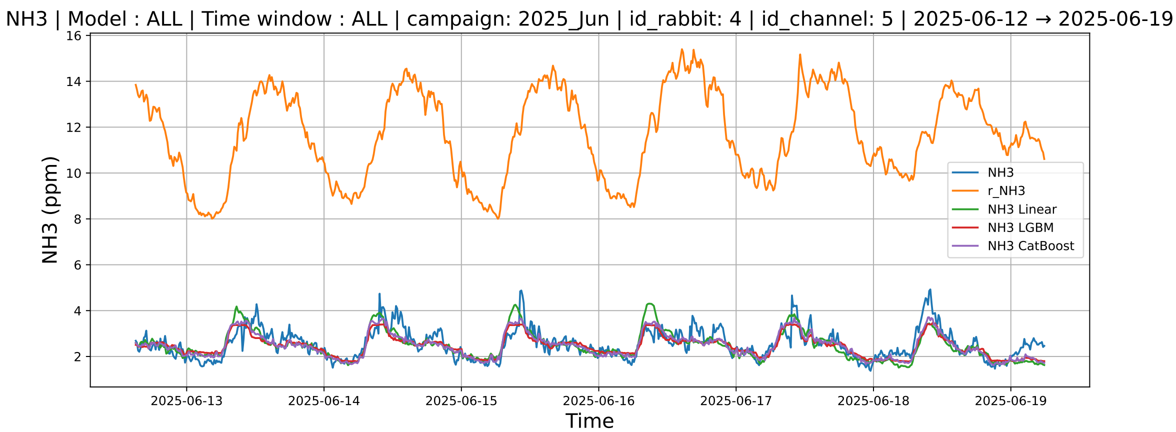
4.2.3.1 Time window : 24



4.2.3.2 Time window : 48

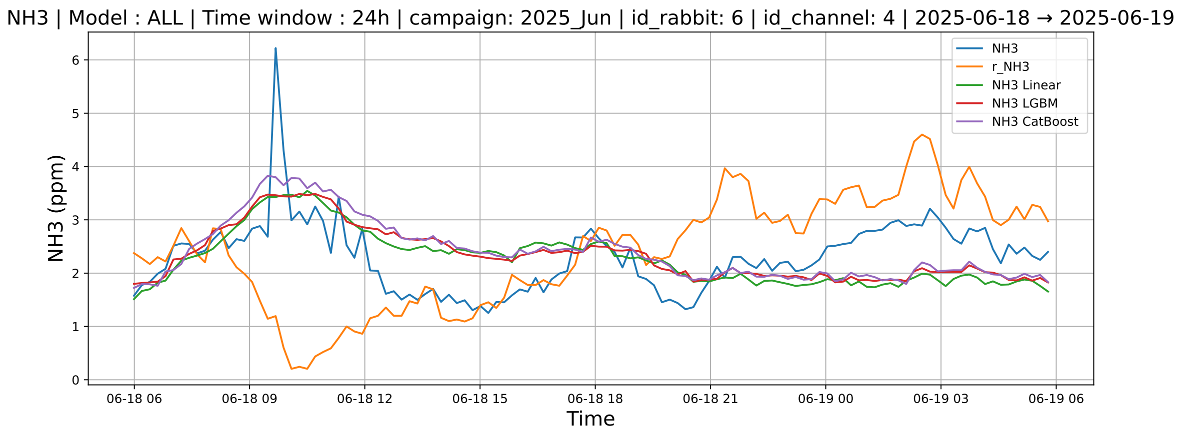


4.2.3.3 Time window : ALL

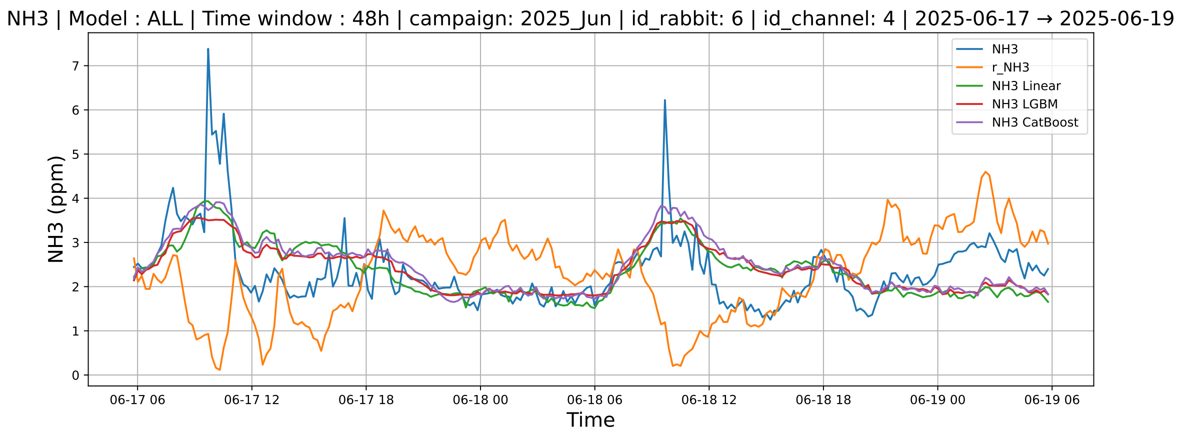


4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

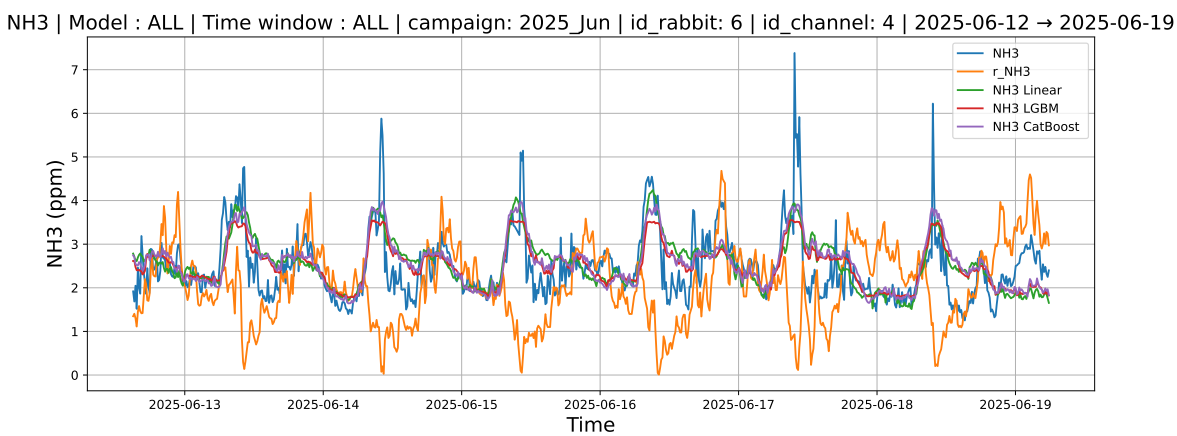
4.2.4.1 Time window : 24



4.2.4.2 Time window : 48

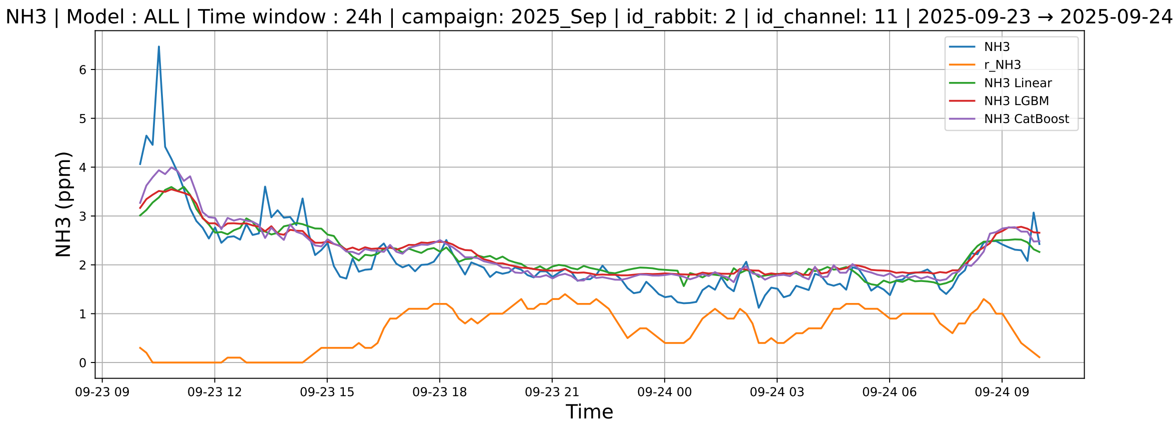


4.2.4.3 Time window : ALL

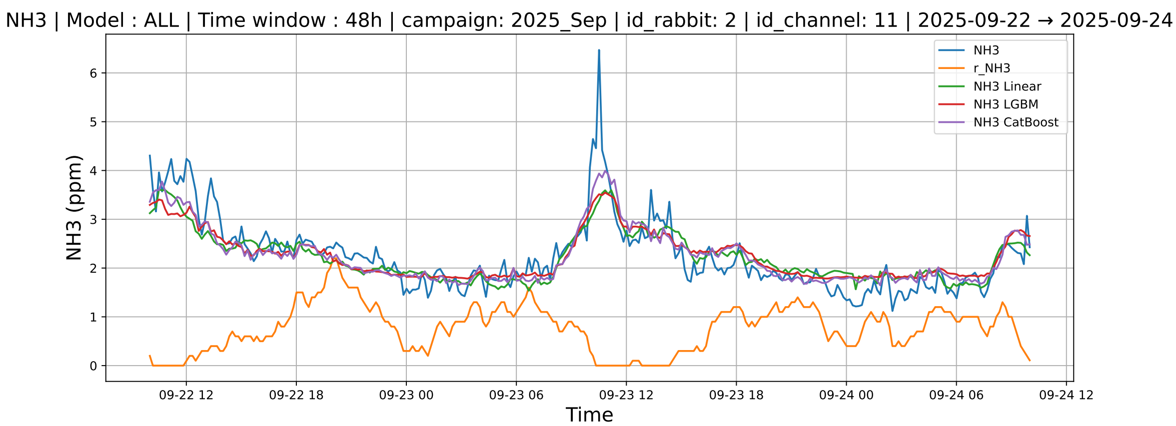


4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

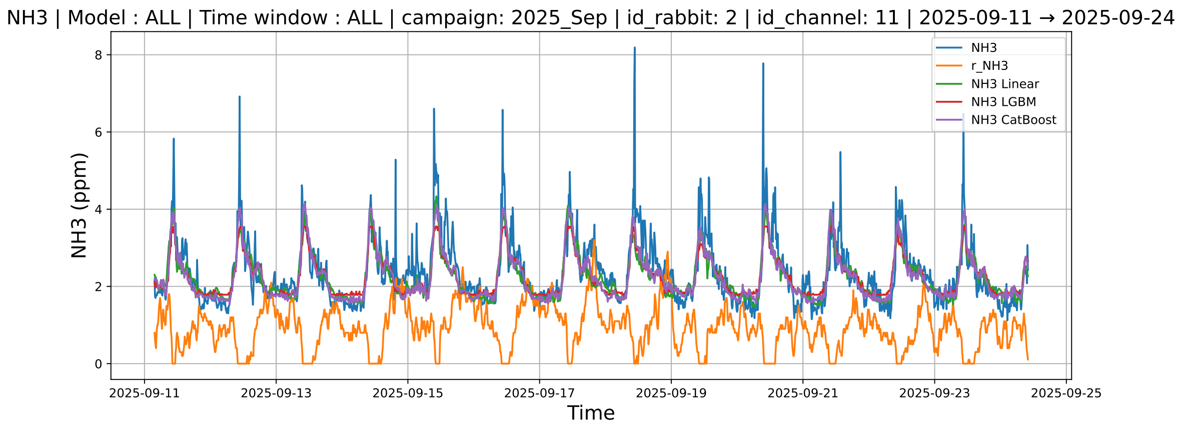
4.2.5.1 Time window : 24



4.2.5.2 Time window : 48

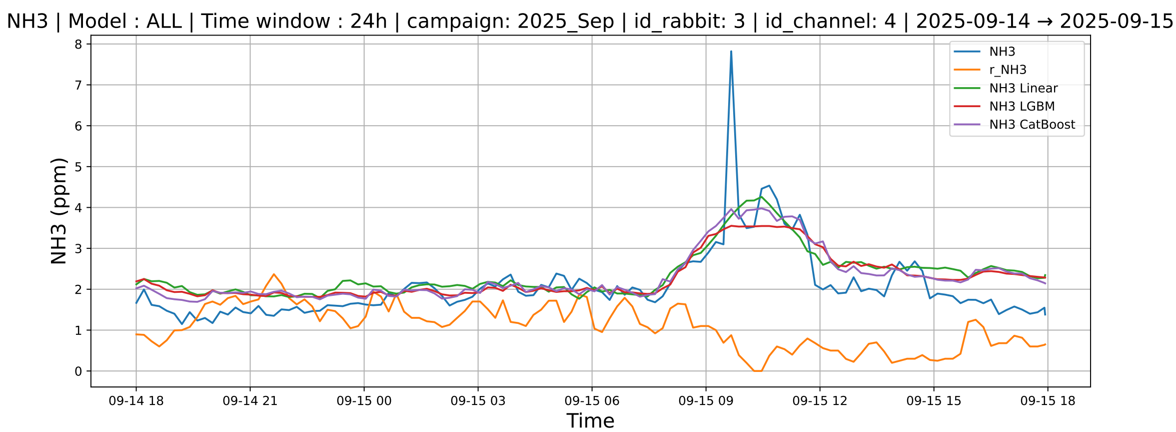


4.2.5.3 Time window : ALL

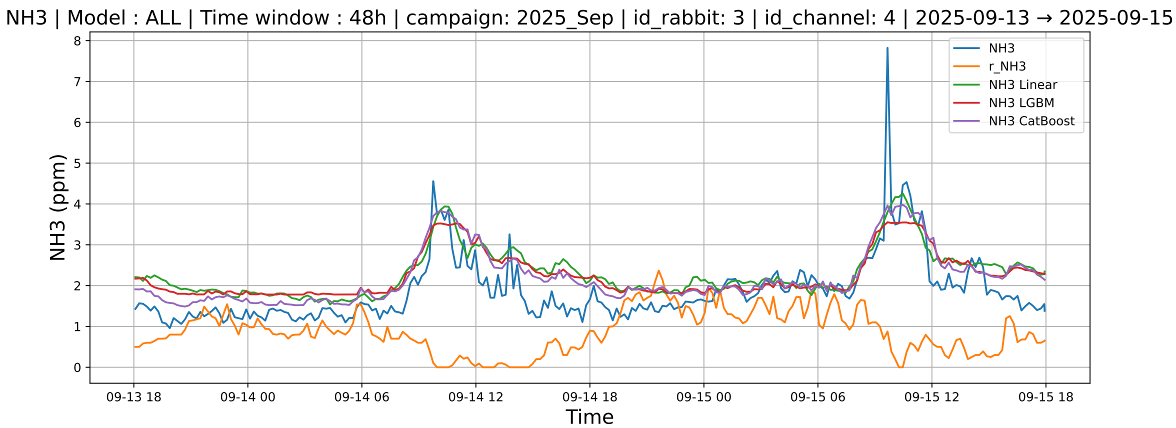


4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

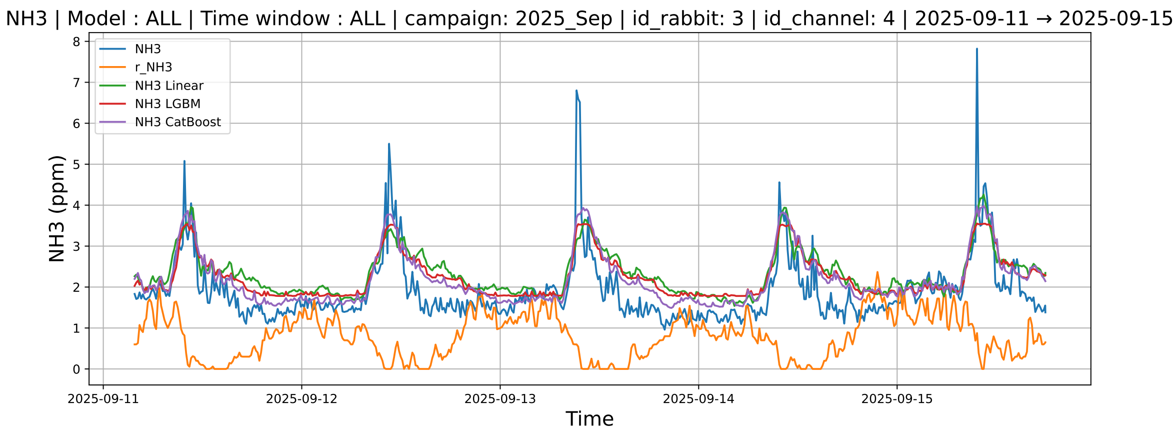
4.2.6.1 Time window : 24



4.2.6.2 Time window : 48



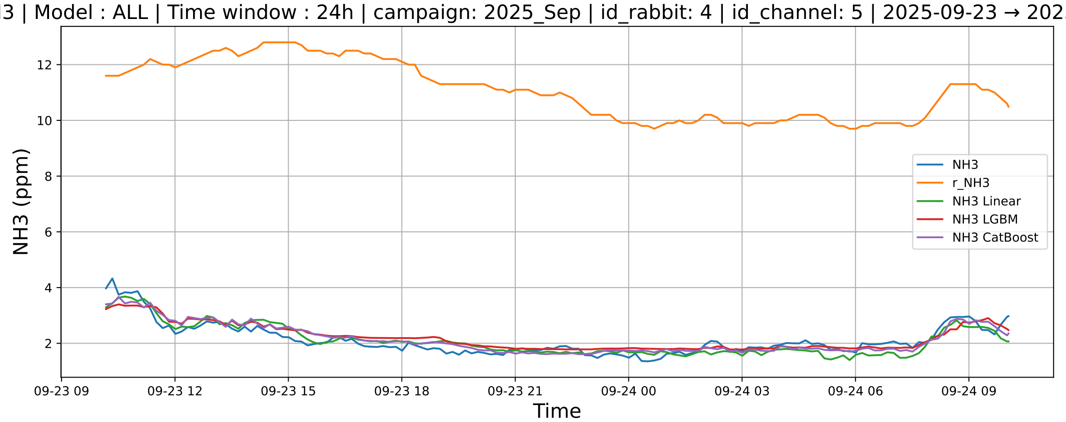
4.2.6.3 Time window : ALL



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

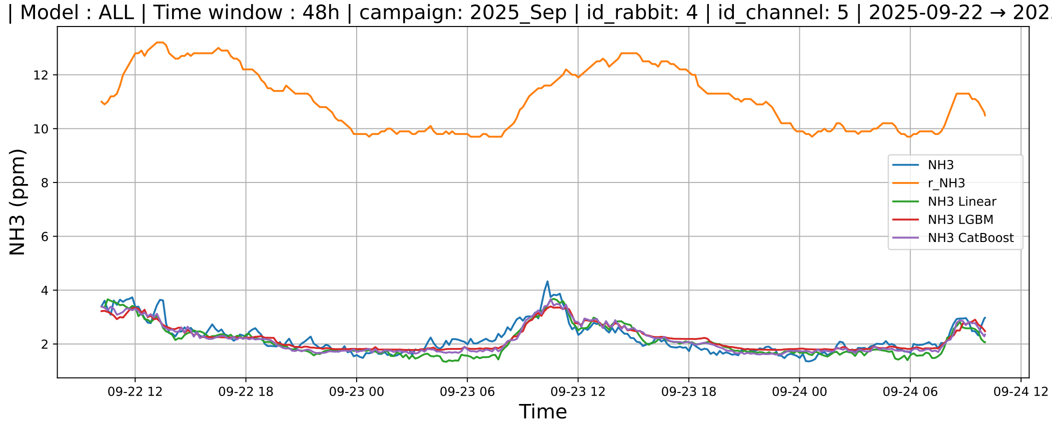
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



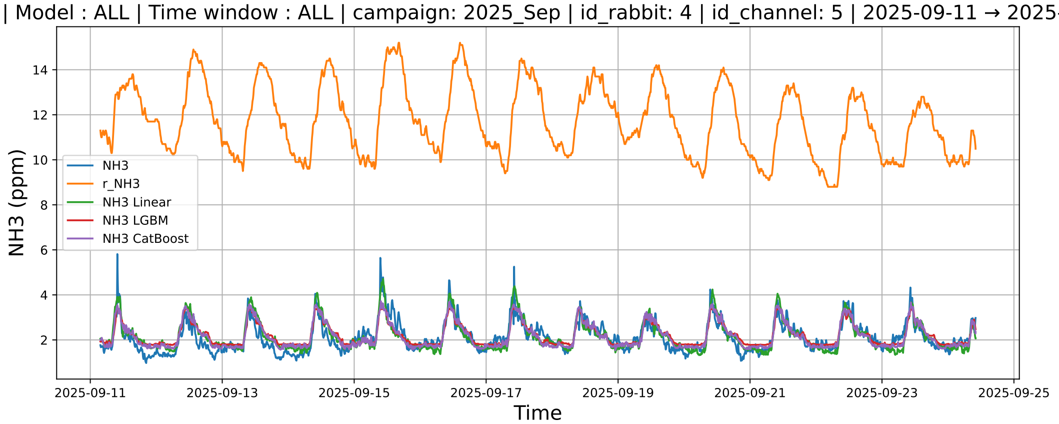
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



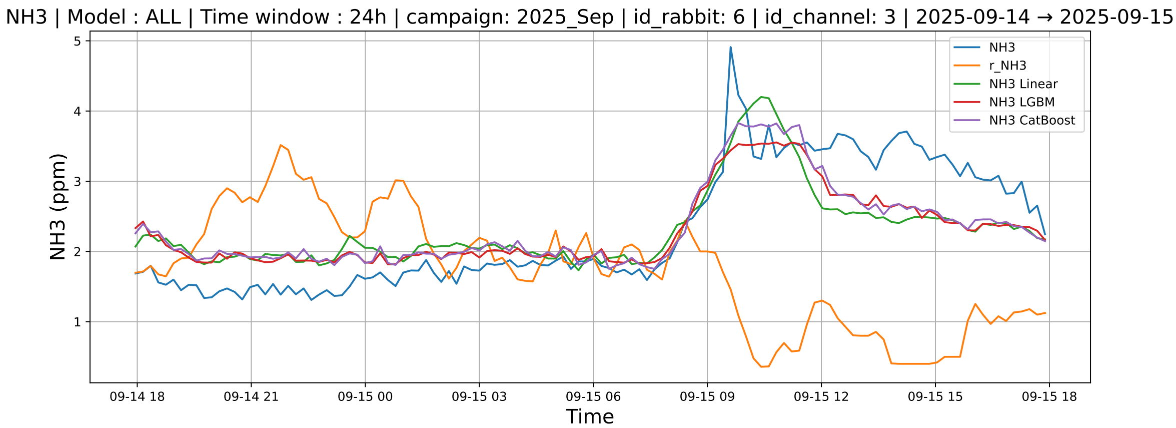
4.2.7.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

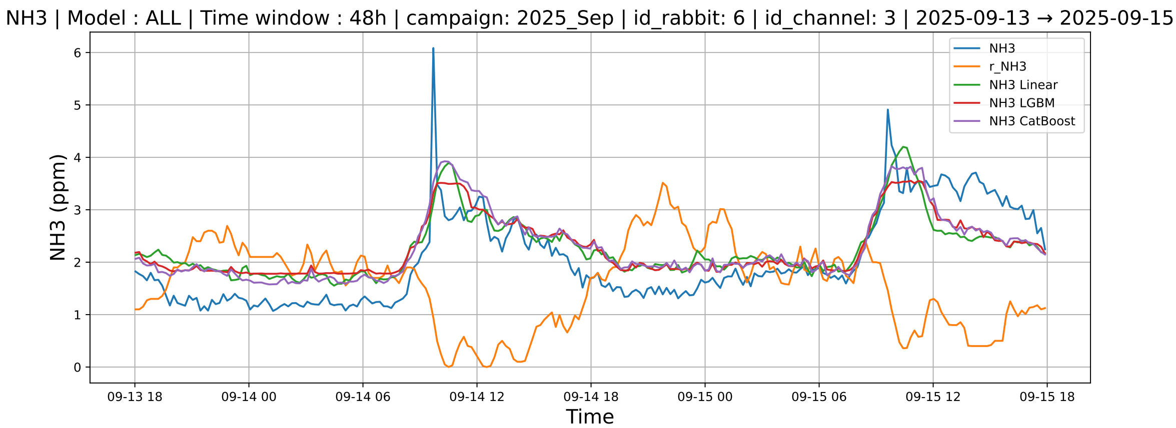


4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

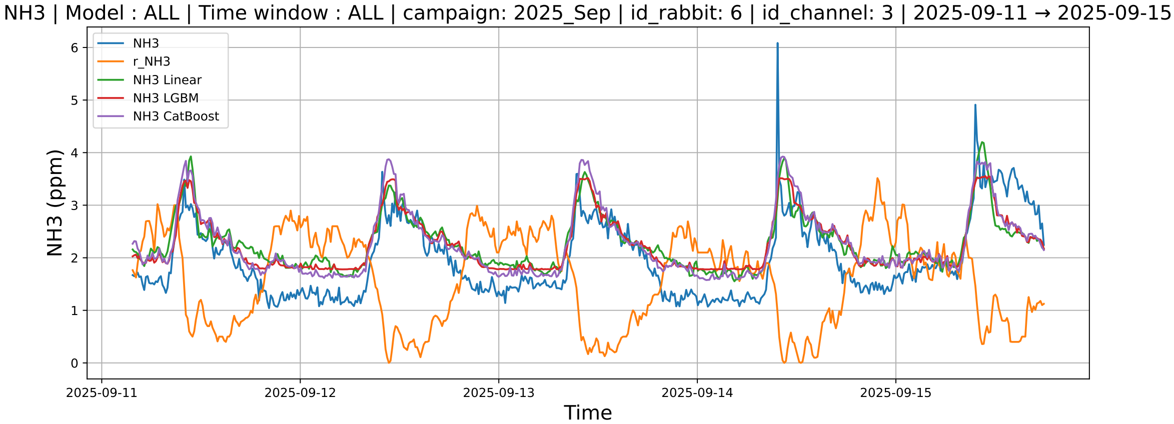
4.2.8.1 Time window : 24



4.2.8.2 Time window : 48



4.2.8.3 Time window : ALL


NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
0
1
2
3
4
5
6
2025-09-11
2025-09-12
2025-09-13
2025-09-14
2025-09-15
Time